Applied Microeconomics — Master Notes

by Jakob V. Stangel

> One running document for the whole course, organised **by lecture (= textbook chapter)**. Format per lecture: **Résumé → ★ Heart of it → Key concepts → Section-by-section → Cases → Key terms → Exam pointers.** Textbook: **Richard Friberg, *A Free Introduction to Microeconomics*** · Lecturer: **Luigi Butera** (lbu.eco@cbs.dk). *Exam: written sit-in, **open book, No AI**, Week 50 (9 Dec 2026).*

Overview — where everything is

(Everything below is clickable.)

- **Lecture 1: Principles of Economics (Ch. 1)** — what is economics? · the three principles · opportunity cost & marginal thinking · models: positive vs normative · ★ Fun facts
- **Lecture 2: Supply and Demand (Ch. 2)** — demand & the demand function · supply & the supply function · ★ solving market equilibrium · shifts & comparative statics · price ceilings & floors · ★ Fun facts
- **Lecture 3: Elasticities & Taxes (Ch. 2 recap · §6.4)** — why elasticity? · ★ price elasticity of demand · computing & the point-elasticity trick · elastic / inelastic / unit-elastic · cross-price · income elasticity & Engel curves · supply elasticity · elasticity over time · ★ taxes: the wedge, incidence & equivalence · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 4: Consumer Choice (Ch. 5)** — modelling choice: the 3 ingredients · preferences & rationality axioms · utility & utility functions · ★ marginal utility · indifference curves & their 3 types · ★ marginal rate of substitution (MRS) · budget constraint & MRT · ★ the optimum: MRS = MRT · comparative statics · interior vs corner · Cobb-Douglas shortcut · behavioral economics · cases · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 5: Applying Consumer Theory (Ch. 5 · §6.5)** — ★ deriving the demand curve · solving it analytically: substitution & Lagrangian · the dual: expenditure minimization · ★ income & substitution effects · normal good · inferior good · Giffen goods · ★ application: labour supply · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 6: Firms & Production (Ch. 3–4)** — the firm & perfect competition · inputs & the production function · short run vs long run · ★ marginal product & diminishing returns · returns to scale · ★ short-run costs (MC · ATC · AVC · AFC) · revenue & profit · ★ profit maximisation: MR = MC = p · the shutdown rule · long-run entry & zero profit · ★ optimal inputs: isoquants, isocost & cost minimisation · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 7: Competition & Long-Run Production (Ch. 4)** — market structure & perfect competition · ★ the three output rules · are short-run profits sustainable? · long-run output & shutdown · ★ the long-run price = min AC · ★ worked example (entry → how many firms?) · ★ the long-run market supply curve · is the firm's demand really flat? · when LR supply is *not* flat · formula sheet · ★ Fun facts

Lecture 1: Principles of Economics (Chapter 1)

Required reading: Friberg, ch. 1 (*Introduction*).

Theme of the lecture (Butera): the "**economic way of thinking**" — economics studies how people **allocate scarce resources** among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

> ★ **The heart of the chapter:** economics = **doing the best you can with scarce resources.** The toolkit is **three principles — Optimization · Equilibrium · Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

The red thread. Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an **opportunity cost** and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.
- **The three principles: Optimization · Equilibrium · Empiricism.**
- **Opportunity cost** — the value of the *best alternative* you give up.
- **Marginal thinking** — decide by the value of *one more* unit; **law of diminishing marginal returns**.
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

Section-by-section main points

What is economics?

Core: economics = **the study of how people allocate *scarce* resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition…) *and* the **means to get them** (money, **time, attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).
- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

The three principles of economics

Core: the whole course rests on three ideas.

Principle	Meaning
① **Optimization**	We choose the **best feasible option** given our limited resources (money, time, attention, willpower are all scarce).
② **Equilibrium**	A situation in which **no one would benefit** from changing their *own* behaviour, given the choices of everyone else.
③ **Empiricism**	We build **models** → models generate **predictions** → we use **data/experiments** to test (falsify) them.

Optimization — opportunity cost + marginal thinking

Core: to optimise, use two tools — **opportunity cost** (weigh each use against its best alternative — "apples to apples") and **marginal thinking** (what does *one more* unit give me?).

Opportunity cost = the value of the best alternative you give up — nothing "free" is free if it uses a scarce resource.

OPPORTUNITY COST

$$\text{OC}(\text{action}) = \text{benefit}(\text{action}) - \text{value of the best alternative given up}$$

Reads: the true cost of a choice is what it gives you **minus** what you gave up to get it. — **OC(action)** = the opportunity cost of doing it · **benefit(action)** = the gain the action delivers · **value of the best alternative given up** = the payoff of the *next-best* option you sacrificed by choosing this one.

- *Social media:* 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118\ \text{DKK/hr} \times 1\ \text{hr/day} \times 365\ \text{days} = \mathbf{43{,}070\ \text{DKK / year}}$$

Reads: price the time you *could* have been earning, added up over a year. — **118 DKK/hr** = Denmark's average hourly wage (i.e. the value of one hour of your time) · **1 hr/day** = the time you spend · **365 days** = a full year → **43,070 DKK** of wages forgone.

- *Trip to Berlin (worked example):* drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$\text{OC}(\text{drive}) = (1{,}930 - 1{,}280) - (118 \times 11) = 650 - 1{,}298 = \mathbf{-648\ \text{DKK}} \implies \mathbf{\text{fly}}$$

Reads: the money you save by driving, minus the value of the time driving wastes. — **(1,930 − 1,280)** = the extra *ticket* cost of flying (650 DKK) · **(118 × 11)** = the value of the **11 hours** flying saves you, priced at the 118 DKK/hr wage (1,298 DKK) · a **negative** result means driving is **648 DKK worse** than flying → **fly**.

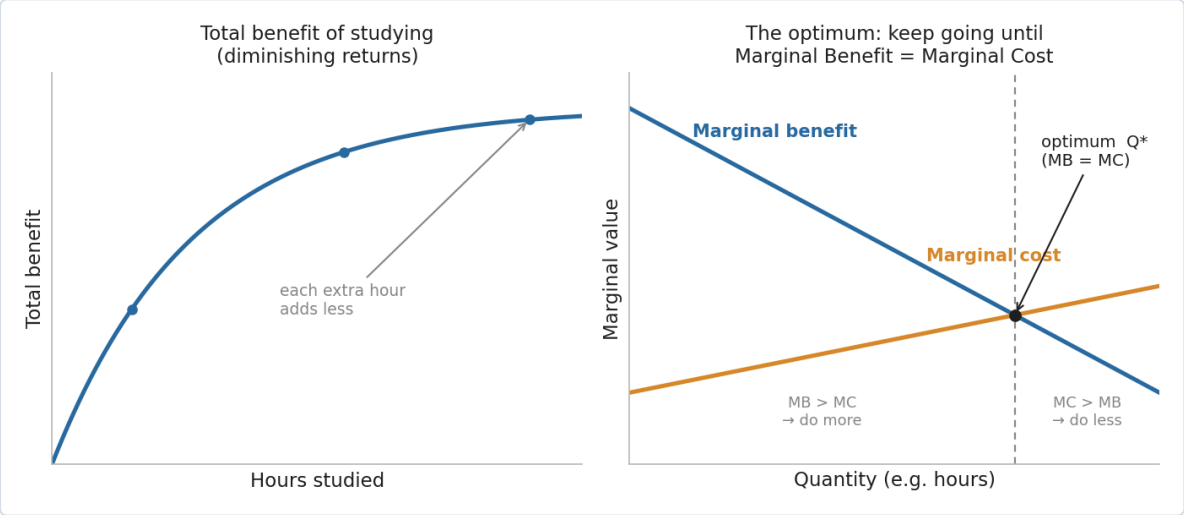
(Still a model — what if you enjoy the drive?)

Marginal thinking: decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

> OPTIMUM CONDITION
>
> choose the quantity where **Marginal Benefit = Marginal Cost** (MB = MC)

Reads: keep going until one more unit's benefit exactly equals its cost. — **Marginal Benefit (MB)** = the *extra* benefit from **one more** unit · **Marginal Cost (MC)** = the *extra* cost of that unit · at the **optimum** MB = MC (while **MB > MC** do more; once **MC > MB** you've gone too far).

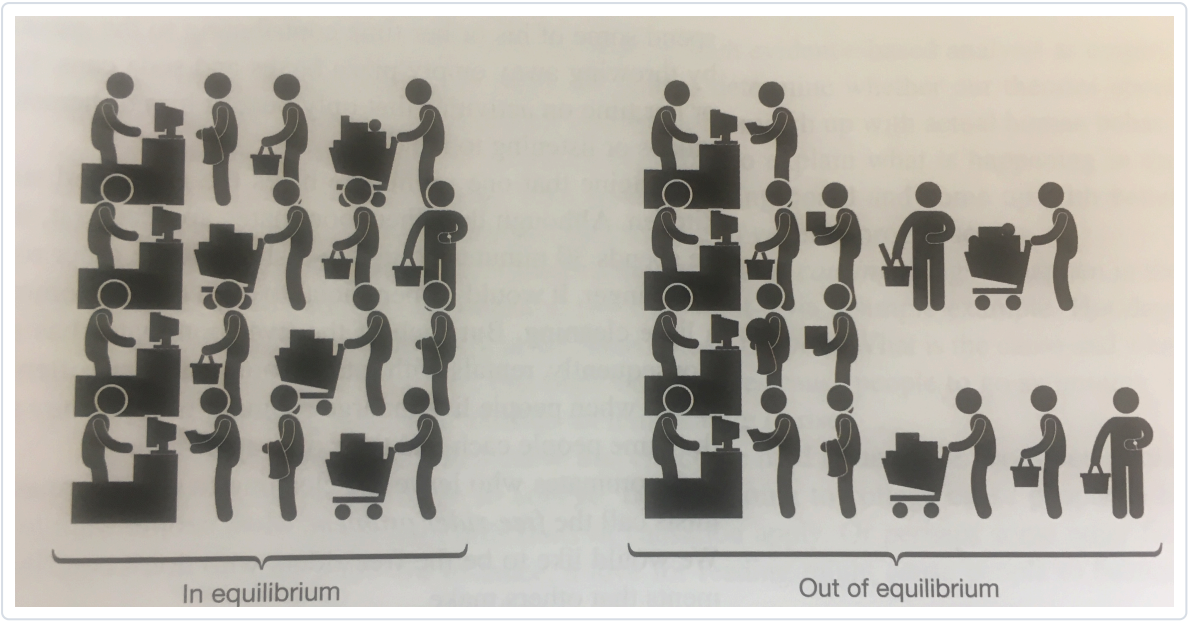
- *Night before the exam:* total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (**law of diminishing marginal returns**). Optimal hours = where MB of one more hour just equals its MC.



How to read it (illustrative diagram — the graph the book draws in a later chapter). *Left:* **total benefit** rises but each extra hour adds less — the curve **flattens** (diminishing returns). *Right:* that flattening means **marginal benefit slopes down**; **marginal cost** slopes up. Left of the crossing, **MB > MC** so one more unit is worth it (*do more*); right of it, **MC > MB** (*do less*). The best you can do is the quantity where they meet — **the optimum Q*, where MB = MC.**

Equilibrium

Core: an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour.** Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.

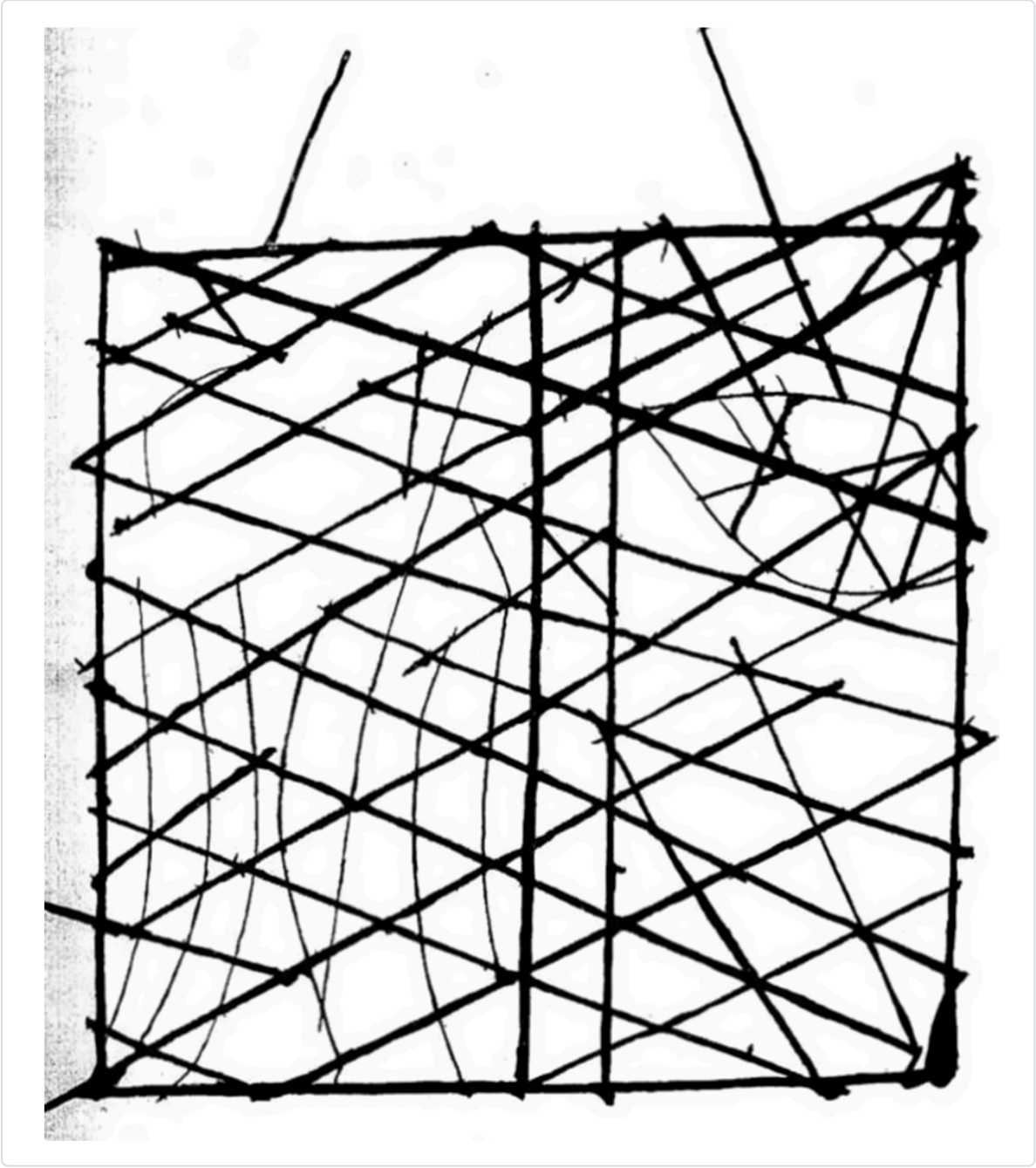


How to read it (equilibrium = supermarket queues, from the lecture). Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

Empiricism — models, and positive vs normative

Core: economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- **A model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



How to read it (Friberg Fig. 1.1 — what a "model" is). This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate.** That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by.**

- **Positive vs normative:**
  - **Positive** = a **testable/falsifiable** cause-effect claim — "*each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices*" (Glaeser et al. 2018).
  - **Normative** = a **value judgment** — "*gentrification is bad because it forces the poor out.*"
  - **Economics is about positive statements**, not normative ones.
- **Correlation ≠ causation:** in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.

- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

Friberg's framing — allocation mechanisms (Ch. 1's example)

Core: a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews · secondary-school grades · open entry + hard exams · **waiting lists** · **queues** ("camp on the sidewalk") · **lotteries** · social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) · **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

Cases & examples

Each: *the example → the concept it teaches.*

- **1 hr/day of TikTok** → **opportunity cost** (≈43,070 DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → **marginal thinking** + **diminishing returns**; optimise where MB = MC.
- **Ice-cream & drowning** → **correlation ≠ causation.**
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification "is bad"** → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

Key terms

Term	Meaning
Scarcity	Limited resources vs unlimited wants — the basis of economics
Optimization	Choosing the best feasible option given constraints
Equilibrium	No one gains from changing their own behaviour, given others'
Empiricism	Build models → predict → test with data
Opportunity cost	Value of the best alternative given up
Marginal thinking	Deciding by the value of one more unit
Law of diminishing marginal returns	Each extra unit adds less than the last
Model	Simplified representation of relationships between variables
Positive statement	A testable/falsifiable cause-effect claim
Normative statement	A value judgment (good/bad, should/ought)
Allocation mechanism	Rule that decides who gets a scarce good (price/lottery/queue…)

Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking** + **MB = MC** to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation ≠ causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

★ Fun facts & memorable details

> Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "***All models are useful, but all models are wrong.***"
- **1 hour of TikTok a day ≈ 43,070 DKK a year** in forgone wages (at Denmark's ~118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
- "*During the summer people eat more ice-cream and also drown more*" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
- "**Joy of giving**" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
- Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.

Lecture 2: Supply and Demand (Chapter 2)

Required reading: Friberg, ch. 2 (*Supply and Demand*). · Lecturer: **Luigi Butera.**

Theme of the lecture (Butera): build **tractable models of demand and supply**, then use them to find the **market equilibrium** — the price at which "*no one would benefit from changing their own behaviour given the choices of others.*" The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

> ★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium (p*, q*)**, where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for p* and q***; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

The red thread. For *every* question ask the one diagnostic that trips people up: **is this a *movement ALONG* a curve (only the good's *own price* changed) or a *SHIFT of* the whole curve (something *else* changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

Key concepts / "modes" to use

- **Demand function** `Q_D = D(p, pₛ, p_c, Y, τ)` → its 2-D **demand curve**; **inverse demand** (p as a function of q).
- **Supply function** `Q_S = S(p, p_o)` → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium** `Q_D = Q_S` → solve for **p*, q***; **excess demand (shortage) / excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

Section-by-section main points

2.1 Demand — the demand function

Core: demand answers "*how much do buyers want at each price?*" The **law of demand:** all else equal, **lower price → higher quantity demanded** (the curve slopes **down**).

- *In plain terms — why it slopes down:* two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded `Q_D` as a function of price **and everything else**:

DEMAND FUNCTION

$$Q_D = D(\ p\ ,\ p_s\ ,\ p_c\ ,\ Y\ ,\ \tau\ )$$

Reads: how much buyers want depends on the good's own price **and** a handful of other factors. — **Q_D** = quantity demanded (what we're solving for) · **D(·)** = "is a function of" · **p** = the good's **own price** · $\mathbf{p_s}$ = price of a **substitute** · **p_c** = price of a **complement** · **Y** = consumers' **income** · **τ** = **taxes**.

- **How the 2-D curve appears — "flush" the other variables into a constant.** A demand *curve* plots `Q_D` against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature T=20°, substitute price p_r=2, income I=5:

COLLAPSING TO A LINEAR DEMAND CURVE

$$Q_D = 1 - p + 0.25{\cdot}T + 0.75{\cdot}p_r + 0.5{\cdot}I = 1 - p + \mathbf{5 + 1.5 + 2.5 = 10 - p}$$

$$(\text{intercept } a = 10 \text{ bundles all the non-price factors})$$

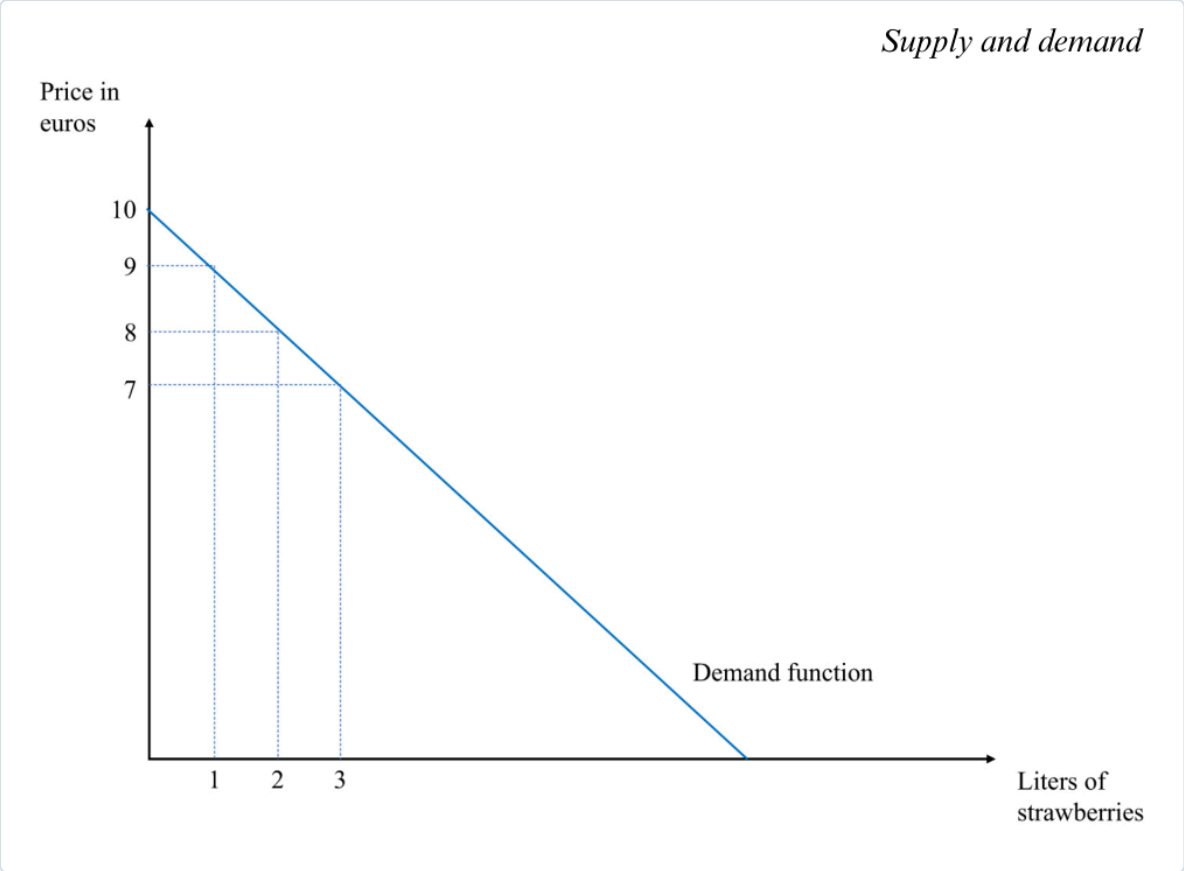
Reads: plug fixed numbers into every non-price factor and they merge into one constant. — **T** = temperature (=20) · $\mathbf{p_r}$ = price of raspberries (=2) · **I** = income (=5); the **coefficients** (0.25, 0.75, 0.5) say how strongly each shifts demand · adding the constant pieces (1 + 5 + 1.5 + 2.5) gives the **intercept a = 10**, so only **p** is left as a variable → **Q_D = 10 − p**.

LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

$Q_D = a - b{\cdot}p \rightarrow$ e.g. $\mathbf{Q_D = 10 - p}$ $(a = 10,\ b = 1)$. Solve for $p \Longrightarrow$ **inverse demand: p = 10 − Q_D** · *(deck's version: $Q_D = 20 - 2p \Longrightarrow p = 10 - ½{\cdot}Q_D$)*

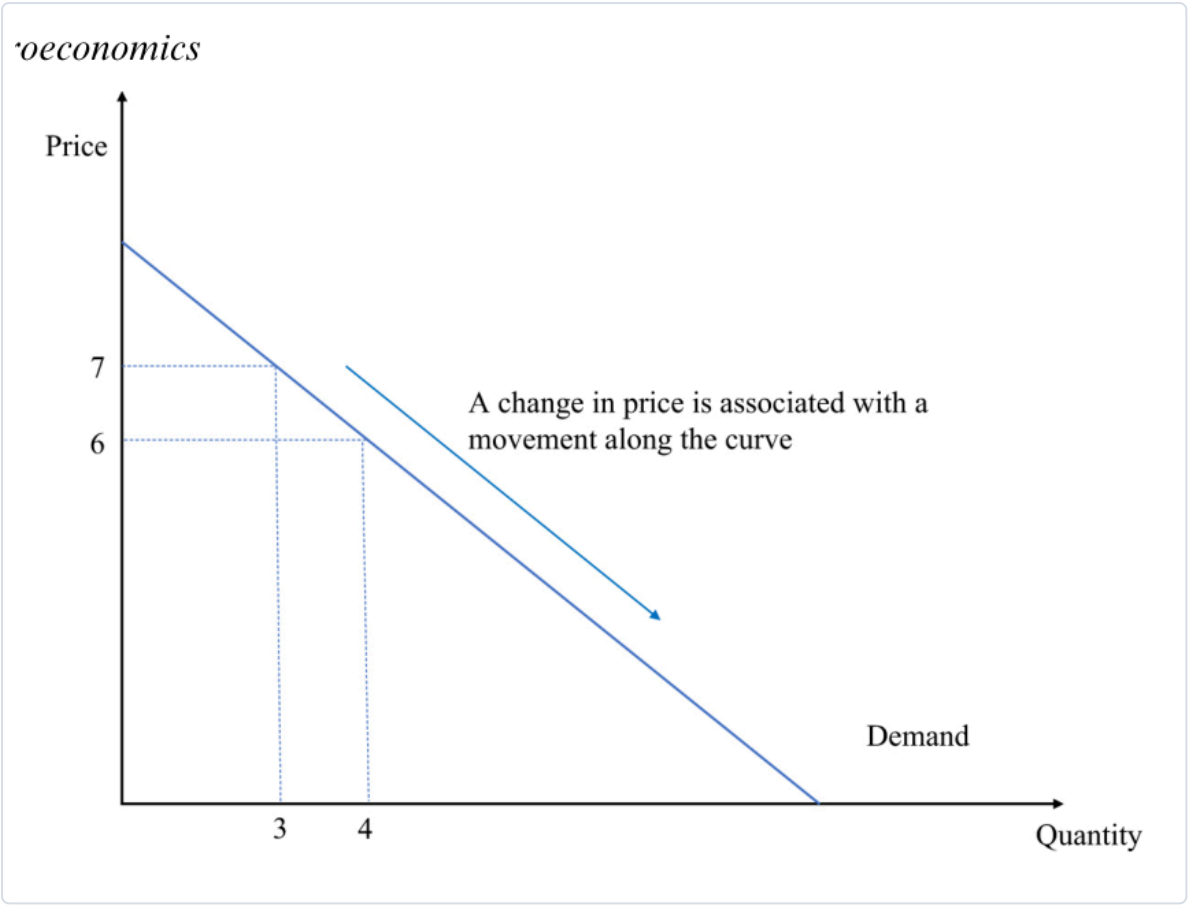
Reads: a straight-line demand — quantity falls as price rises. — **a** = the **intercept** (quantity demanded if price were 0; it bundles income, other prices, tastes — *shift it and the whole curve moves*) · **b** = the **slope** (how many units demand drops per +1 in price) · **p** = price. The **inverse** just rearranges it to put **p** alone on the left, because graphs put price on the vertical axis.

Check: p = 7 → Q_D = 3; p = 6 → Q_D = 4. **a** = the intercept (shift it and the whole curve moves); **b** = how many units demand falls per +1 in price.



How to read it (the demand curve). By convention **price is on the vertical axis, quantity on the horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price p, how much is bought?" *and* "for quantity q, what's the most a buyer will pay?" Here the line hits **p = 10 when q = 0** (nobody buys above €10) and slides down to more litres as price falls.

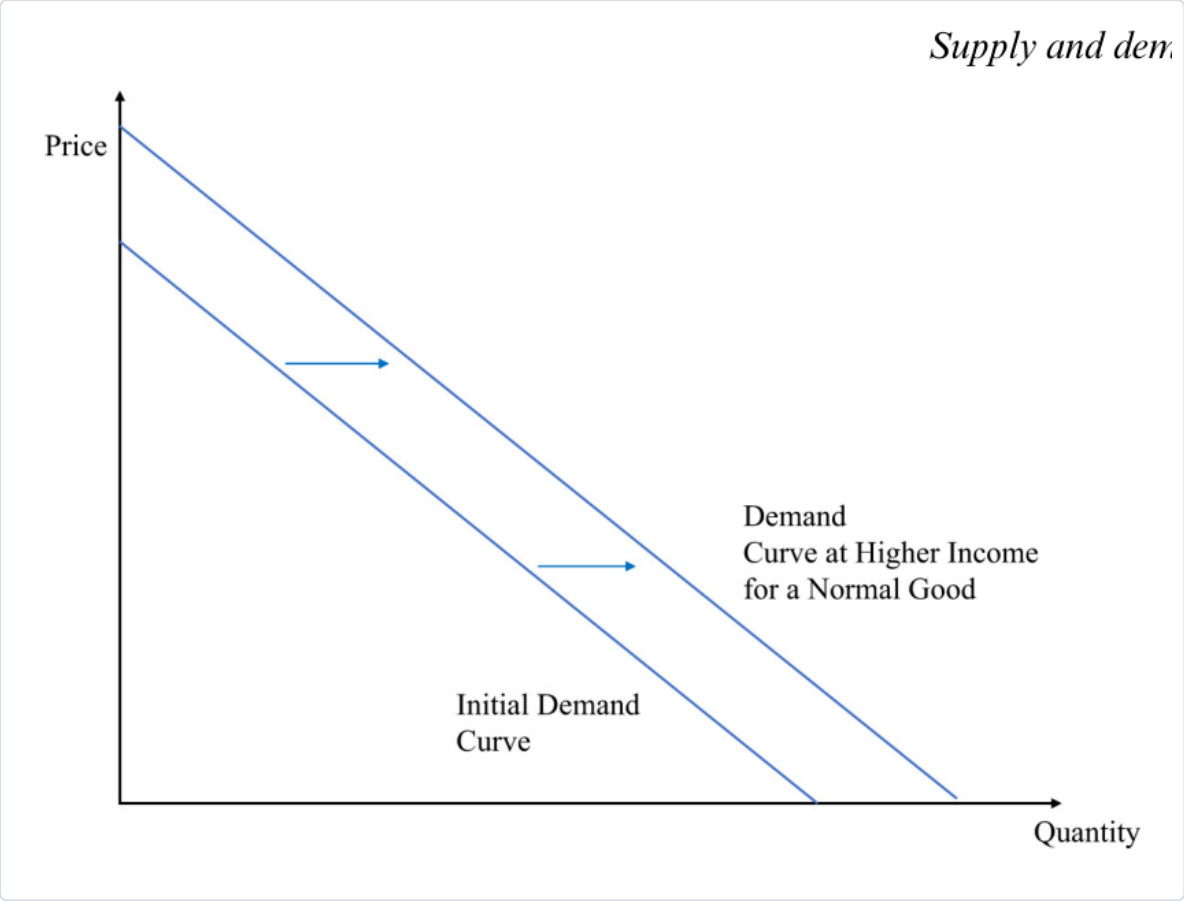
- **★ Movement ALONG vs SHIFT — the make-or-break distinction:**
  - **Own price changes → MOVE ALONG** the existing curve (you slide to a new point).
  - **Anything else changes → the whole curve SHIFTS** (a *new* curve).

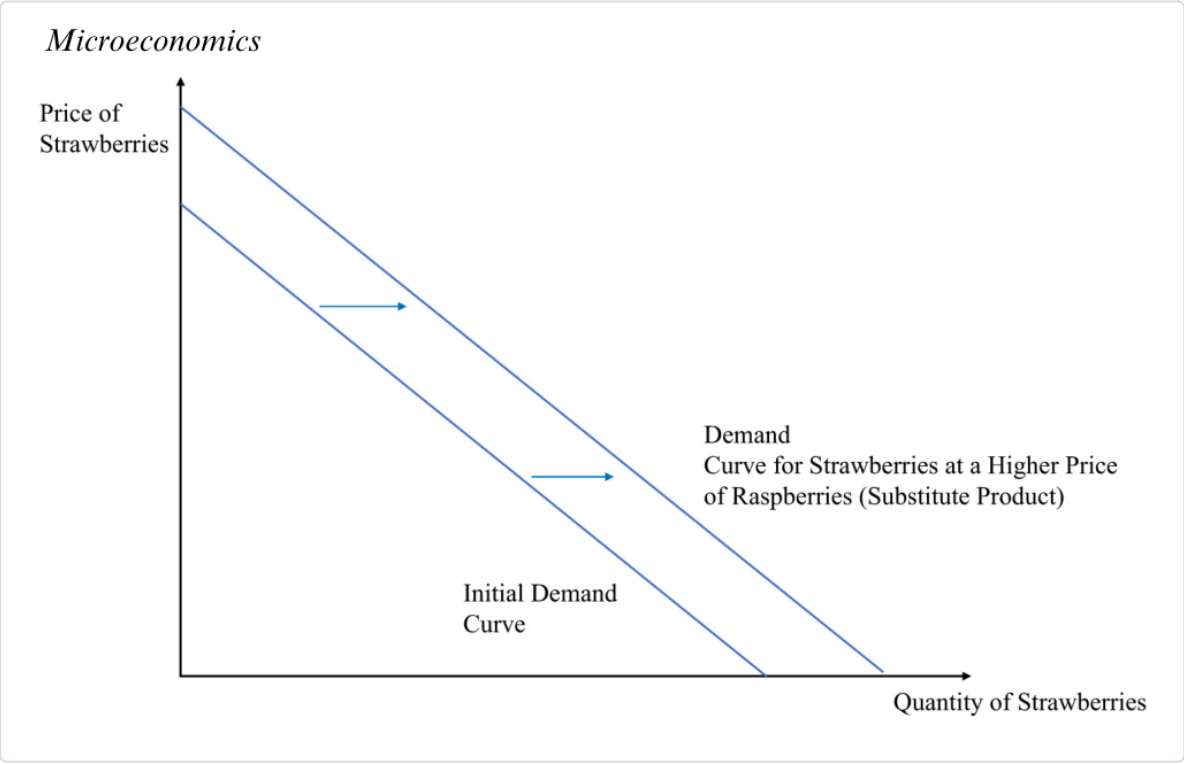


How to read it (movement along). Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters (each moves the *whole* curve = changes the intercept a):**

Shifter	Direction	Why
Income ↑ (normal good)	shift **out** (right)	richer → buy more at every price
Income ↑ (inferior good, e.g. instant noodles)	shift **in** (left)	richer → buy *less* of it
Price of a substitute ↑ (raspberries ↑ → strawberry demand)	shift **out**	switch toward the now-relatively-cheaper good
Price of a complement ↑ (cereal ↑ → milk demand)	shift **in**	the pair is consumed together
Tastes / expectations (heatwave → ice-cream; COVID → toilet paper)	out or in	common-sense direction





How to read them (shifts). The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions): substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's

price rises (milk ↔ cereal).

- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

> AGGREGATE (MARKET) DEMAND
>
> $$Q(p) = q_1 + q_2 + \ldots = D_1(p) + D_2(p) + \ldots \textit{ (add quantities across buyers at the same price)}$$

Reads: market demand = every individual's demand added up **at each price**. — **Q(p)** = total quantity the whole market wants at price p · **q_1, q_2, …** = the quantities buyer 1, buyer 2, … each want at *that same* price · **$D_1(p)$, $D_2(p)$** = each buyer's own demand function. (It's a **horizontal** sum — fix the price, add the quantities.)

Deck example: Mike's inverse demand p = 100 − 2q_M, Linda's p = 100 − ½q_L → invert each to q(p), then add: at each price, total q = q_M + q_L. *(Watch for a **kink** where one buyer drops out at high prices.)*

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good **to itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

2.2 Supply — the supply function

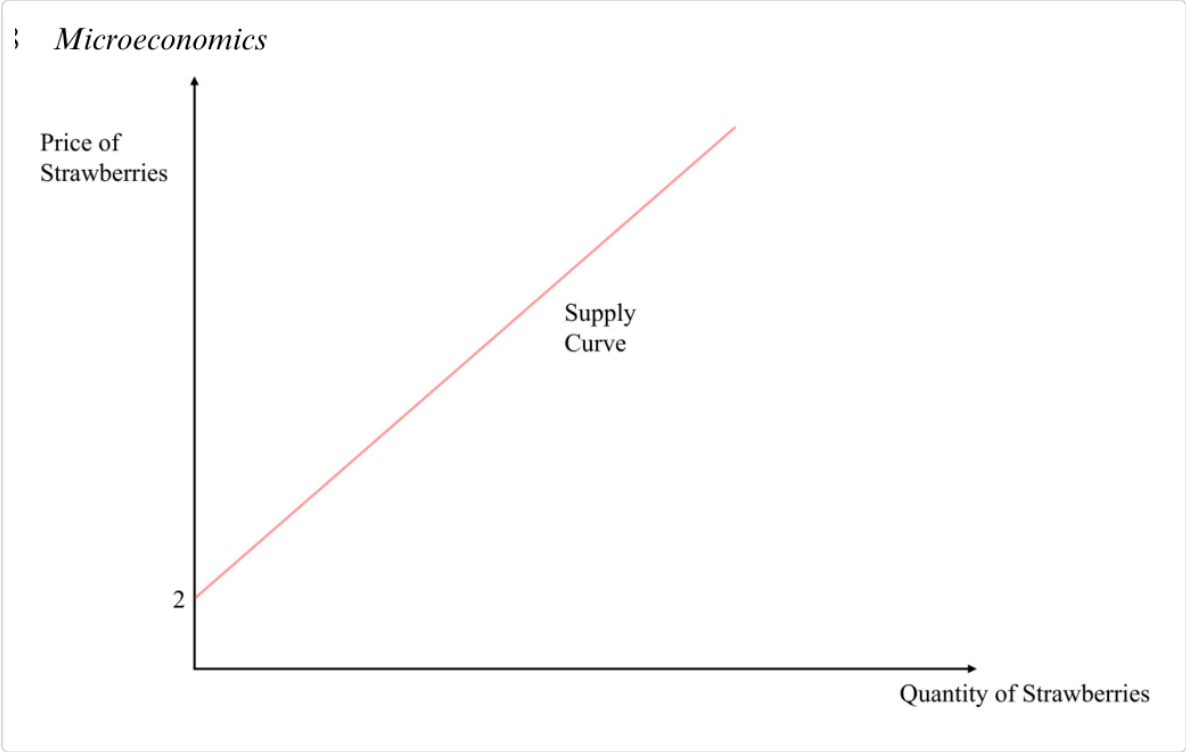
Core: supply answers "*how much do firms bring to market at each price?*" Firms are **price-takers** (they choose *quantity*, not the price). The curve usually slopes **up** — higher price → more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

> SUPPLY FUNCTION & ITS INVERSE
>
> $Q_S = S(\ p\ ,\ p_o\ )$ · linear example: $\mathbf{Q_S = -2 + p} \Longrightarrow$ inverse supply $\mathbf{p = 2 + Q_S}$

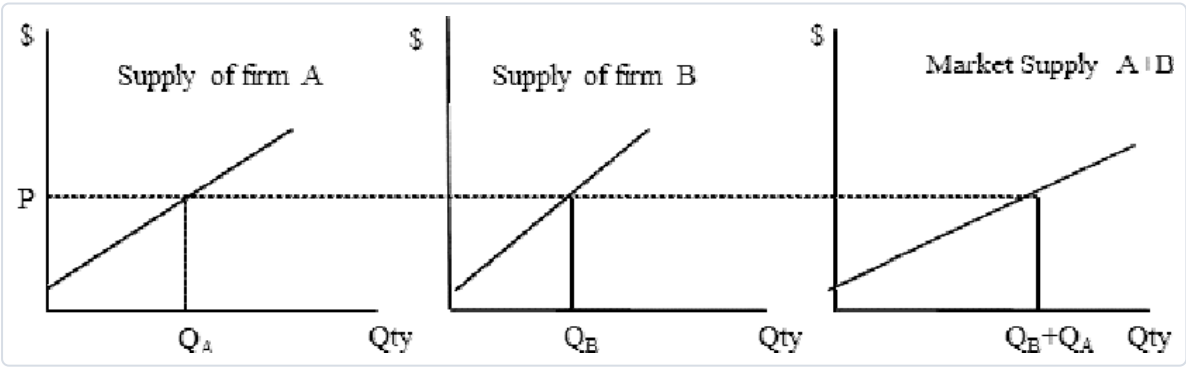
Reads: how much firms bring to market depends on the price they get and their costs. — **Q_S** = quantity supplied · **S(·)** = "is a function of" · **p** = the good's **price** · **p_o** = price of **inputs** (wages, materials — higher costs shift supply in). In the example the "**−2**" is the price-intercept read backwards: **price must exceed 2 before any is supplied** (at p = 2, Q_S = 0).

- **Why the "−2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before *any* is supplied** (at p = 2, Q_S = 0). *Check:* p = 3 → Q_S = 1; p = 4 → Q_S = 2.



How to read it (the supply curve). Start at the **€2 price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve): input/cost ↑** (wages, transport) → shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers** → shift **out**; **disruptions** (strikes, extreme weather, mine closures) → shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



How to read it (aggregate supply). At any price **P**, read each firm's quantity off its own curve (Q_A, Q_B) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

★ 2.3 Market equilibrium — solving for p* and q*

Core: equilibrium = the **one price where quantity demanded = quantity supplied** (`Q_D = Q_S`), the **market-clearing** price — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- **The recipe (memorise these three steps):**

STEP 1 — SET DEMAND = SUPPLY

$$Q_D = Q_S \implies 10 - p = -2 + p$$

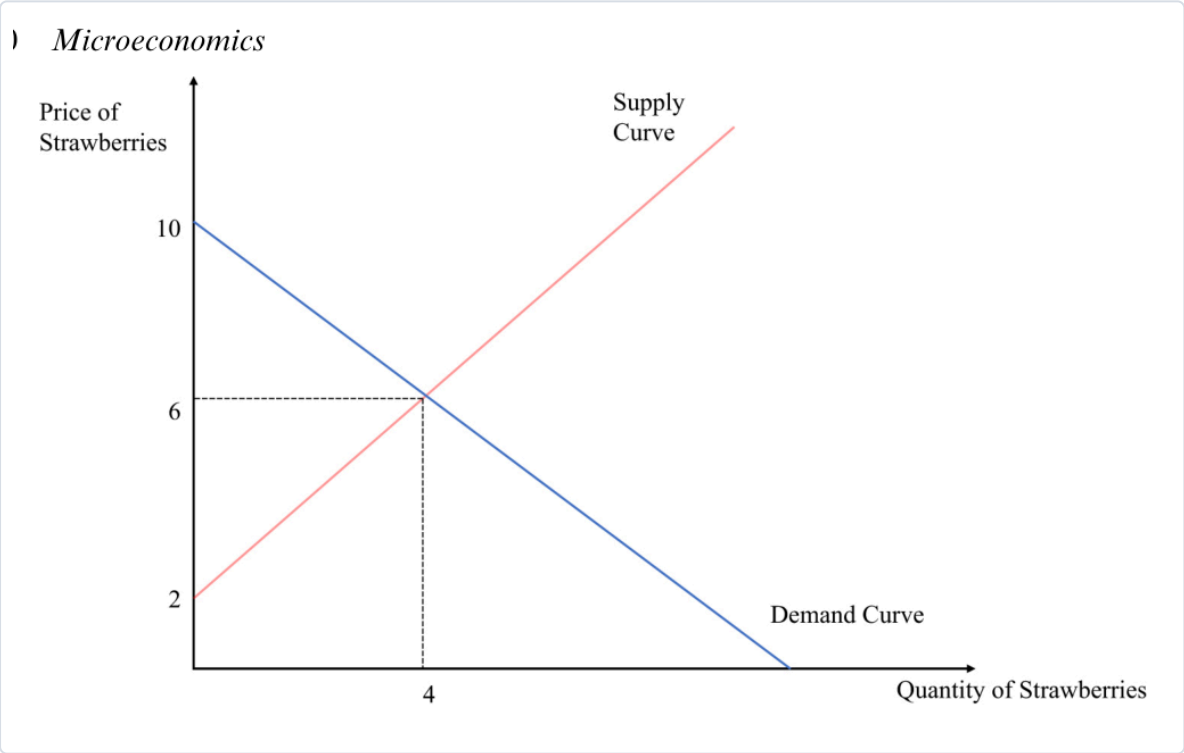
STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE P*

$$10 + 2 = p + p \implies 12 = 2p \implies \mathbf{p^* = 6}$$

STEP 3 — PUT P* BACK INTO EITHER CURVE FOR Q*

$$Q^* = 10 - 6 = \mathbf{4}\ (\text{check: } Q_S = -2 + 6 = 4\ \checkmark) \implies \textbf{equilibrium } (p^*, q^*) = (6, 4)$$

Reads: find the price where the two plans agree, then the quantity that trades there. — **Q_D** = quantity demanded · **Q_S** = quantity supplied · setting **Q_D = Q_S** solves for **p*** = the market-clearing **price**; substituting p* back into either curve gives **q*** = the **quantity** traded. The star (*) just marks "equilibrium value." (Both curves must give the *same* q* — that's the ✓ check.)



How to read it (equilibrium). The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

EXCESS DEMAND (SHORTAGE) — PRICE BELOW P*

$$\text{at } p = 4\text{: } Q_D = 10-4 = \mathbf{6},\ Q_S = -2+4 = \mathbf{2} \Longrightarrow \text{shortage} = Q_D - Q_S = \mathbf{4} \rightarrow \text{upward pressure on price}$$

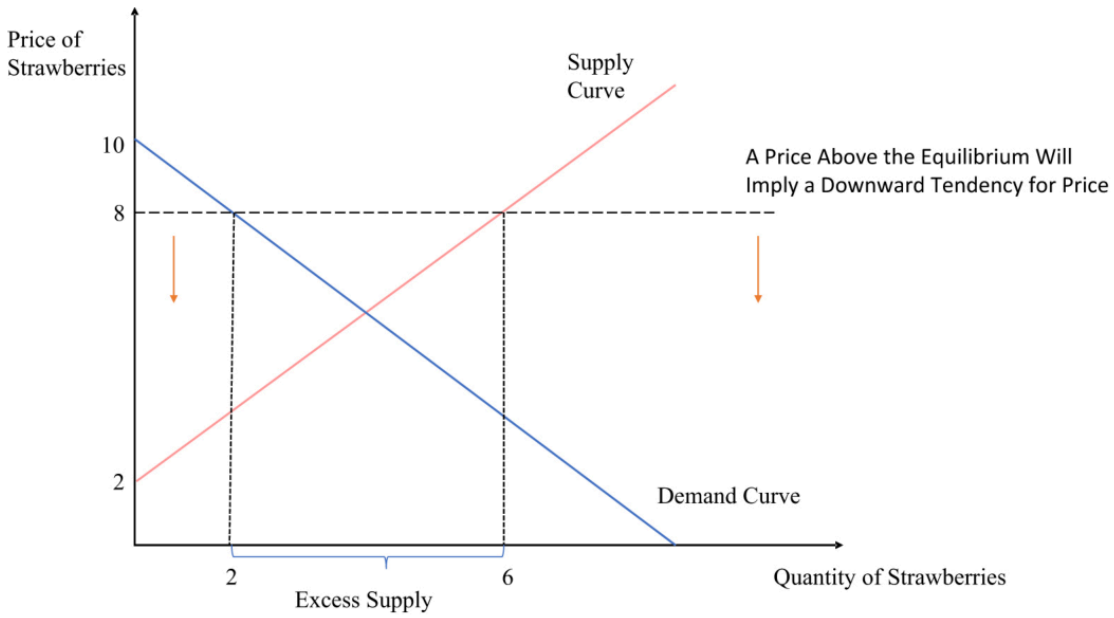
Reads: below equilibrium, buyers want more than sellers will supply. — plug the low price (**p = 4**) into **both** curves → **Q_D = 6** wanted, **Q_S = 2** offered; the gap **Q_D − Q_S = 4** is the **shortage**, which pushes the price **up** toward p*.

EXCESS SUPPLY (SURPLUS) — PRICE ABOVE P*

$$\text{at } p = 8\text{: } Q_D = 10-8 = \mathbf{2},\ Q_S = -2+8 = \mathbf{6} \Longrightarrow \text{surplus} = Q_S - Q_D = \mathbf{4} \rightarrow \text{downward pressure on price}$$

Reads: above equilibrium, sellers offer more than buyers want. — plug the high price (**p = 8**) into **both** curves → **Q_S = 6** offered, **Q_D = 2** wanted; the gap **Q_S − Q_D = 4** is the **surplus**, which pushes the price **down** toward p*.





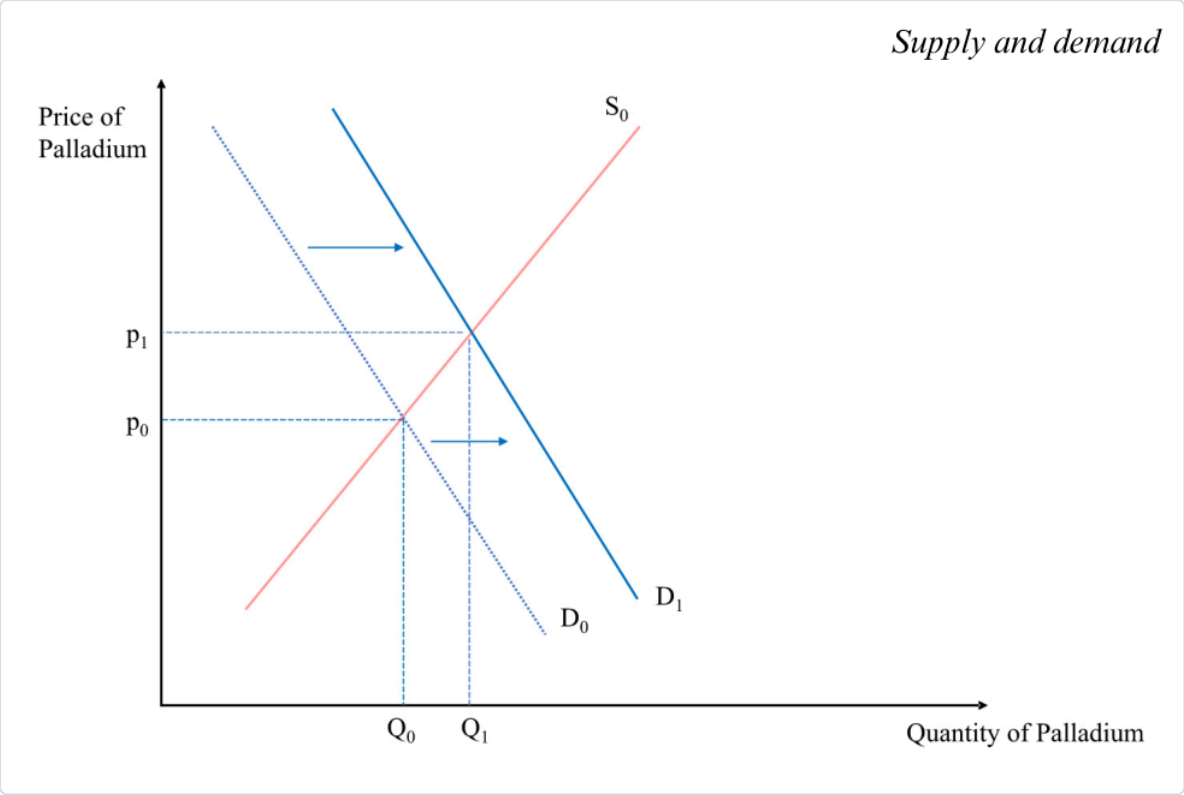
How to read them. Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward p*.

- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do.** This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- *(Who actually "sets" the price if everyone's a price-taker? The theoretical **Walrasian auctioneer** — and **Vernon Smith's 1962 experiments** showed real people reach the equilibrium price fast.)*

2.4 Shifts in the equilibrium — comparative statics

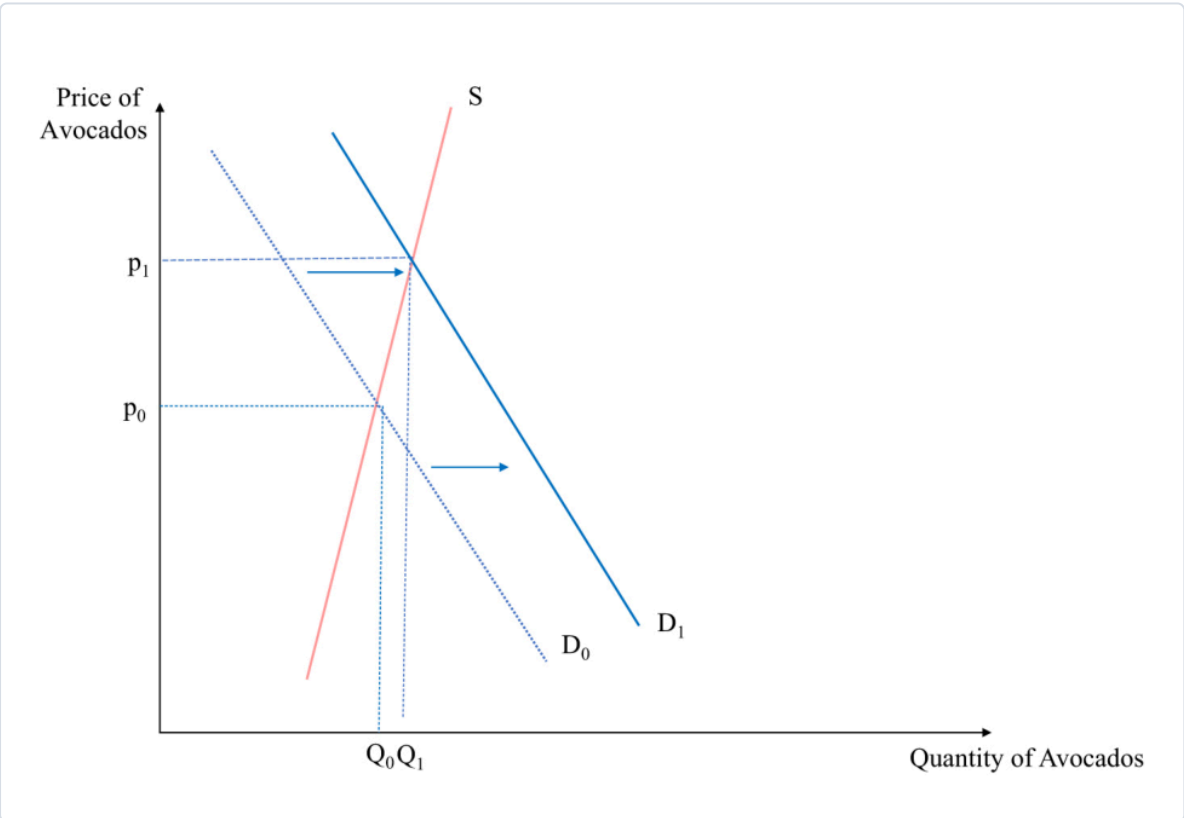
Core: to analyse any shock, **don't jump to price/quantity.** First figure out **which curve moves and which way**; the effect on p* and q* follows.

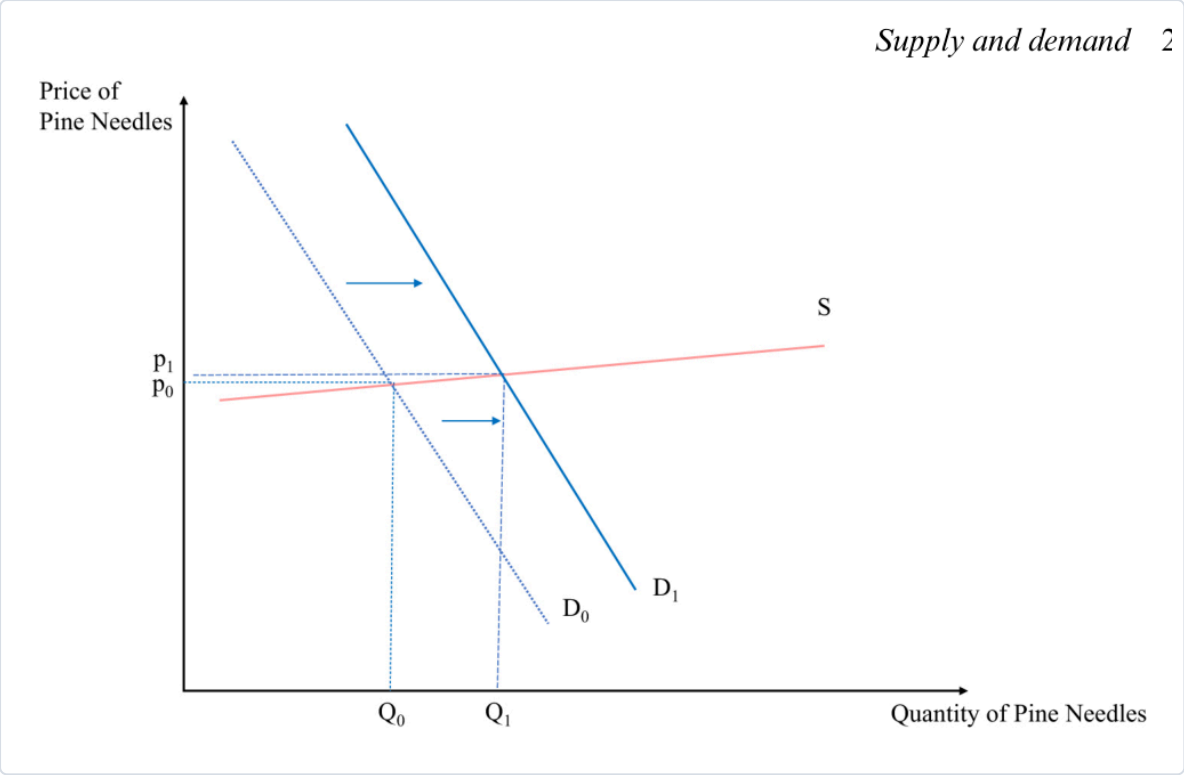
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*): ① Demand or supply?** (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ *Then* read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.



How to read it (a demand shock). Demand shifts **out** ($D_0 \rightarrow D_1$); supply is unchanged, so producers **move along** supply to the new cross — **both p and q rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- ★ **Slope decides the *size* of the effect.** A **demand** shock's impact depends on the **slope of the *supply* curve**:





How to read them. Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as **quantity**; the steeper, the more it shows up as **price**.*

Price ceilings & floors — when the government fixes the price (deck)

Core: a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set p to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below p***, so **Q_D > Q_S → a shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped p into both curves; shortage = Q_D − Q_S.
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above p***, so **Q_S > Q_D → a surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus = Q_S − Q_D.
- **The trade-offs (positive, not normative):** *ceilings* — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). *floors* — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "*Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice.*"

Is the model realistic? — perfect competition (deck)

Core: the supply-demand model assumes **perfect competition** — "*every model is wrong, but useful.*"

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

Cases & examples

Each: *the example → the concept it teaches.*

- **Strawberries (the running example)** → the whole toolkit: `Q_D = 10 − p`, `Q_S = −2 + p`, equilibrium **(6, 4)**, and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply *and* demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles → inferior good**; **raspberries → substitute**; **cereal & milk → complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea → steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela → price ceiling → shortage**; **minimum wage → price floor → surplus (unemployment)**.

Formula sheet — quick reference

Object	Formula	Note
Demand function	$Q_D = D(p, p_s, p_c, Y, \tau)$	many variables; curve fixes all but p
Linear demand / inverse	$Q_D = a - b \cdot p$ / $p = (a - Q_D)/b$	a = intercept (shifters), b = slope
Supply function / inverse	$Q_S = -2 + p$ / $p = 2 + Q_S$	negative intercept = min price to supply
Aggregate (market)	$Q(p) = \Sigma D_i(p)$; $\Sigma S_i(p)$	**horizontal** sum (add quantities)
Equilibrium	**set $Q_D = Q_S$ → p^*, then q^***	the market-clearing point
Excess demand (shortage)	$Q_D - Q_S$ at $p < p^*$	→ upward price pressure; price ceiling
Excess supply (surplus)	$Q_S - Q_D$ at $p > p^*$	→ downward pressure; price floor

Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get **p^***, back-substitute for **q^*** — the strawberry case ($p^*=6$, $q^*=4$) is the model answer.

- **Compute a shortage/surplus:** plug a below- or above-equilibrium price into *both* curves and take the gap (**$Q_D - Q_S$** or **$Q_S - Q_D$**) — and know a **ceiling→shortage**, **floor→surplus**.
- **Diagnose movement vs shift**, and for a shock say **which curve moves, which way**, and the effect on p* & q* (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

★ Fun facts & memorable details (Lecture 2)

> Sticky bits from Supply & Demand.

- **Marshall's scissors:** asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
- The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
- **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
- A negative supply intercept (**$Q_S = -2 + p$**) isn't a mistake — it just means **price must top €2 before anyone supplies a drop.**
- **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.
- Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
- Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.

Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

Required reading: Friberg, **§6.4 Elasticities** (own-price, cross-price, income; supply elasticity) — building on the **Chapter 2** supply-and-demand model (see Lecture 2). *The lecture deck also teaches **taxes on goods & services** — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

Theme of the lecture (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"*"shape" of a demand or supply curve — its elasticity — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity.* The deck's

three goals: **(1)** understand what demand & supply **elasticities** mean, **(2) derive** them from demand/supply curves, and **(3)** use them to analyse the **effects of taxes**.

> ★ **The heart of the lecture: elasticity = % responsiveness** — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed.**

The red thread. For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax.** Elasticity is the "shape" number that ties all three together.

Key concepts / "modes" to use

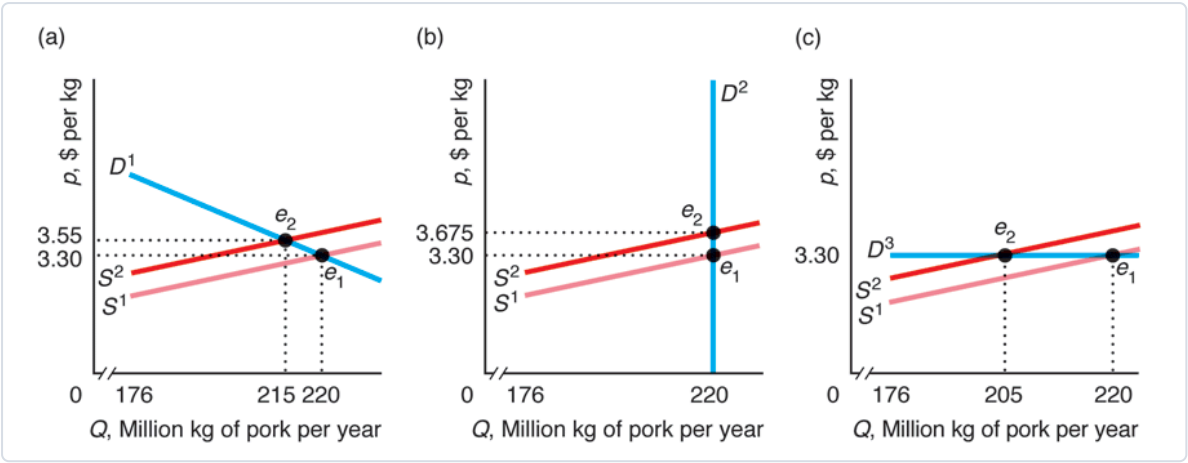
- **Elasticity** — the **% change in one variable ÷ % change in another** (a unit-free number).
- **Price elasticity of demand (E_D)** — % change in quantity demanded ÷ % change in own price (**negative**).
- **Elastic / inelastic / unit-elastic** — $|E| > 1$ / $|E| < 1$ / $|E| = 1$.
- **Point-elasticity trick** — `E_D = (dQ/dp)·(p/Q)`; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (−).
- **Income elasticity** — normal (+) vs inferior (−); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** — `p_b = p_s + t`; who bears the burden depends on **relative elasticities**; **legal ≠ economic** incidence; unit vs ad-valorem tax; **deadweight loss**.

Section-by-section main points

Why do we care about elasticity? (deck)

Core: the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity.** A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "**sin taxes**" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers** "**passed on**" to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



How to read it (why elasticity matters). All three panels apply the **same supply shift** `S¹→S²`. The **only** thing that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** (3.30→3.675). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** (220→205).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity.** *This one picture is the whole reason we measure elasticity.*

★ Price elasticity of demand — definition

Core: the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity ÷ % change in price.** It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up → quantity down).

PRICE ELASTICITY OF DEMAND

$$E_D = (\Delta Q/Q \times 100) \div (\Delta p/p \times 100) = \%\Delta Q \div \%\Delta p$$

Reads: **how responsive quantity demanded is to a price change** — "a **1% rise in price** p leads to an **E_D % change in quantity** q." — **ΔQ/Q×100** = the **percentage change in quantity demanded** · **Δp/p×100** = the **percentage change in price** · **E_D** = the ratio (a **unit-free** number, **negative** for a normal good). The ×100s cancel, so it's just **%ΔQ over %Δp**.

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A **percentage** measure is **comparable across markets, currencies and time.**
- **Why negative:** demand curves slope **down**, so ΔQ and Δp have opposite signs → the ratio is negative. *(When people say a "higher" elasticity they usually mean a bigger **absolute value** $|E_D|$.)*
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups → `E_D = −8% ÷ +10% = −0.8`. Since $|-0.8| < 1$, demand is **inelastic** — revenue actually *rises* (price up 10%, quantity down only 8%). *(Full step-by-step computations are in the next section.)*

Computing elasticity — and the point-elasticity trick

Core: plug the two percentage changes into the formula. **On a linear demand curve the elasticity is *not* constant** — it depends on *where* you evaluate it, via `E_D = (dQ/dp)·(p/Q)`.

- **Simple example (textbook Table 6.2):** quantity falls **6 → 4** while price rises **1 → 2**. Then

WORKED — SIMPLE EXAMPLE

$$E_D = (\ (4-6)/6\ ) \div (\ (2-1)/1\ ) = (\ -2/6\ ) \div (\ 1\ ) = \mathbf{-1/3}$$

Reads: a demand that is **inelastic** — **(4−6)/6 = −0.33** = the **% fall in quantity** (−33%) **· (2−1)/1 = 1** = the **% rise in price** (+100%) **· −1/3** = quantity falls **only ⅓ as fast** as price rises **→ inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda ≈ **−1.37.** *If price rises 5%,* quantity changes by `E_D × 5% = −1.37 × 5 = −6.85%` (a ~7% fall). *To cut demand 15%,* price must rise `−15 ÷ −1.37 ≈ 11%`. *→ this is exactly the sum a policymaker designing a **soda tax** needs.*
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q/\Delta p) \cdot (p/Q) = (\mathrm{d}Q/\mathrm{d}p) \cdot (p/Q)$$

Reads: elasticity = **slope of the demand function × the price-to-quantity ratio at your chosen point** — **dQ/dp** = the **slope** (how many units Q changes per €1 of price; constant for a straight line) **· p/Q** = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked (Q = 10 − 2p):** here `dQ/dp = −2` (always). But the **elasticity varies**:
  - at **p = 2.5** (so Q = 5): `E_D = −2 × 2.5/5 = −1` (**unit elastic**).
  - at **p = 4** (so Q = 2): `E_D = −2 × (p/Q) = −2 × 4/2 = −4` **→ elastic.** *(⚠ The textbook misprints this as `−2 × 2/2 = −2` — it accidentally puts Q=2 in the numerator instead of p=4; the correct p/Q is 4/2 = 2, so **E_D = −4**.)*
  - **The pattern: demand is more elastic at higher prices** — at p=2.5 it's unit-elastic (−1), at p=4 it's elastic (−4). A fixed unit change is a bigger *percentage* change when Q is small and p is large.
- **Constant-elasticity demand (contrast):** the form `Q = a·p^ϵ` has the **same** elasticity **ϵ** at *every* price (used in empirical estimation via `ln Q = ln a + ϵ·ln p + error`). *So: **linear → elasticity changes along the curve; constant-elastic → it doesn't.***
  - **Worked example:** with `Q = 100·p^(−1.2)`, the elasticity is **−1.2** whether you evaluate at **p = €1** (Q = 100), **p = €5** (Q ≈ 15), or **p = €50** — it never changes. *(Contrast the linear `Q = 10 − 2p` above, where it swung from −1 to −4.)*

Elastic, inelastic, unit-elastic — and why it matters

Core: we **name** the ranges of elasticity, and each has a sharp economic meaning for **how supply shocks split into price vs quantity.**

Name	Level (of E_D)	Meaning	Curve looks…
Inelastic	−1 < E_D < 0 (|E| < 1)	%ΔQ **smaller** than %Δp	relatively **steep**
Unit elastic	E_D = −1	%ΔQ **equals** %Δp	—
Elastic	E_D < −1 (|E| > 1)	%ΔQ **bigger** than %Δp	relatively **flat**

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big *price* changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity ≈ −0.19 to −0.96; policy point estimate −0.34). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic — substitutes.** The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity ≈ **−4.74**, but beer **overall** ≈ **−0.60.**

Cross-price elasticity — substitutes vs complements

Core: the **cross-price elasticity** measures how demand for good **A** responds to a change in the price of a *different* good **B**. Its **sign** tells you the *relationship* between the two goods; its **size** tells you *how strong* that relationship is.

CROSS-PRICE ELASTICITY

$$E_{cross} = \%\Delta Q_A \div \%\Delta p_B \quad \rightarrow \quad > 0 = \text{substitutes} \cdot < 0 = \text{complements} \cdot \approx 0 = \text{unrelated}$$

Reads: how the quantity of **A** responds to a price change in **B** — **%ΔQ_A** = percentage change in demand for good A · **%Δp_B** = percentage change in the price of good B · **E_cross** = the ratio (**can be + or −**, unlike own-price which is always −).

- **The sign = the *relationship*:**
  - **Positive → substitutes.** B gets dearer → buyers switch *to* A → A's demand **rises** (butter ↑ → margarine demand ↑; Heineken ↑ → Carlsberg ↑).
  - **Negative → complements.** B gets dearer → buyers use less B *and* less of its partner A → A's demand **falls** (petrol ↑ → big-car demand ↓; smartphones ↑ → phone-case demand ↓).
  - **≈ 0 → unrelated** goods (the price of bread barely touches demand for haircuts).
- **The size = the *strength*:** the larger the **absolute value** |E_cross|, the **closer** the substitutes (or the stronger the complementarity). Two brands of the *same* soda have a **big** positive cross-price elasticity; soda and bottled water a **small** one; soda and umbrellas ≈ 0.
- **Worked example: Danish** spirits get **10% dearer** → Swedish demand for spirits changes by `E_cross × 10% = +0.3 × 10 = +3%` (Swedes buy **more at home** instead of driving to Denmark) → **positive → substitutes** (Asplund 2007). *Flip it: if smartphones get **10% dearer** and phone-case demand falls **6%**, `E_cross = −6 ÷ 10 = −0.6` → **negative → complements**.*

- ★ **Its killer application — defining markets & judging mergers.** This is *the* reason cross-price elasticity matters in practice. Competition authorities use it to decide **whether two products are in the same market**: a **high positive** cross-price elasticity means A and B are **close substitutes**, so a firm selling both has real market power (a merger between them removes a competitor). This is the logic of the **SSNIP / "hypothetical monopolist" test** — *would a **S**mall but **S**ignificant **N**on-transitory **I**ncrease in **P**rice (say 5%) be profitable, or would enough buyers flee to substitutes to make it a loss?* If they flee, the substitute is in the same market.
- **Asymmetry (watch out):** cross-price elasticity is **not symmetric** — A's response to B's price ≠ B's response to A's price. A cheap **complement** (phone case) reacts strongly to the price of the expensive main good (phone), but the phone barely reacts to the case's price.
- **Real numbers (Asplund et al. 2007):** Swedish **spirits +0.3**, **beer +0.5**, **wine +0.2** with respect to their **Danish** prices — all **positive**, i.e. all **substitutes** for cross-border shopping (beer the closest). *(Deck aside: cigarettes & alcohol ≈ −1 — strong **complements**, Krauss et al. 2014.)*

Income elasticity — normal vs inferior goods, Engel curves

Core: the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

INCOME ELASTICITY

$$E_{income} = \%\Delta Q \div \%\Delta I \quad \rightarrow \quad > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

Reads: how quantity demanded responds to an income change — **%ΔQ** = percentage change in quantity · **%ΔI** = percentage change in income · **sign**: **positive → normal** (demand rises with income), **negative → inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Worked example:** a household's **income rises 10%** and its spirits purchases rise **14%** → `E_income = +14 ÷ +10 = +1.4` → **normal** *and* **income-elastic** (a "luxury," >1) (Asplund 2007). *Flip it: income **+10%** but instant-noodle demand **−5%** →* `E_income = −5 ÷ 10 = −0.5` *→ **inferior**.*
- **Nuance:** a good can be **normal at low incomes and inferior at higher** ones (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive → "**defensive stock**").
- **Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior. Engel's law:** the **share of income spent on food *falls* as income rises** → the income elasticity of food is **less than 1.**

Supply elasticity — same principle

Core: the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

PRICE ELASTICITY OF SUPPLY

$$E_S = \%\Delta Q_S \div \%\Delta p \quad \textbf{(positive)}$$

Reads: how responsive quantity supplied is to a price change — **%ΔQ_S** = percentage change in quantity supplied · **%Δp** = percentage change in price · **E_S** = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year → `E_S = +4 ÷ +20 = +0.2` → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

Elasticity over time — short run vs long run

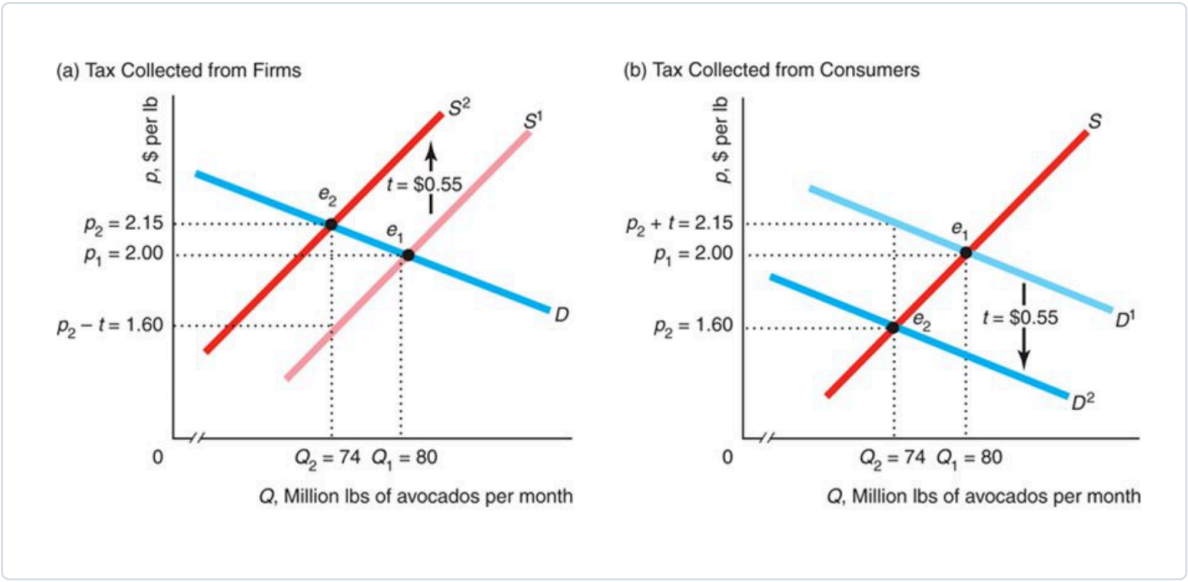
Core: demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes.**

- **Electricity (Labandeira et al. 2017):** ≈ **−0.24 short-run** vs **−0.6 long-run** — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas** prices does little in the short run but more in the long run; likewise **computers**. *Exam framing: "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."*

★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

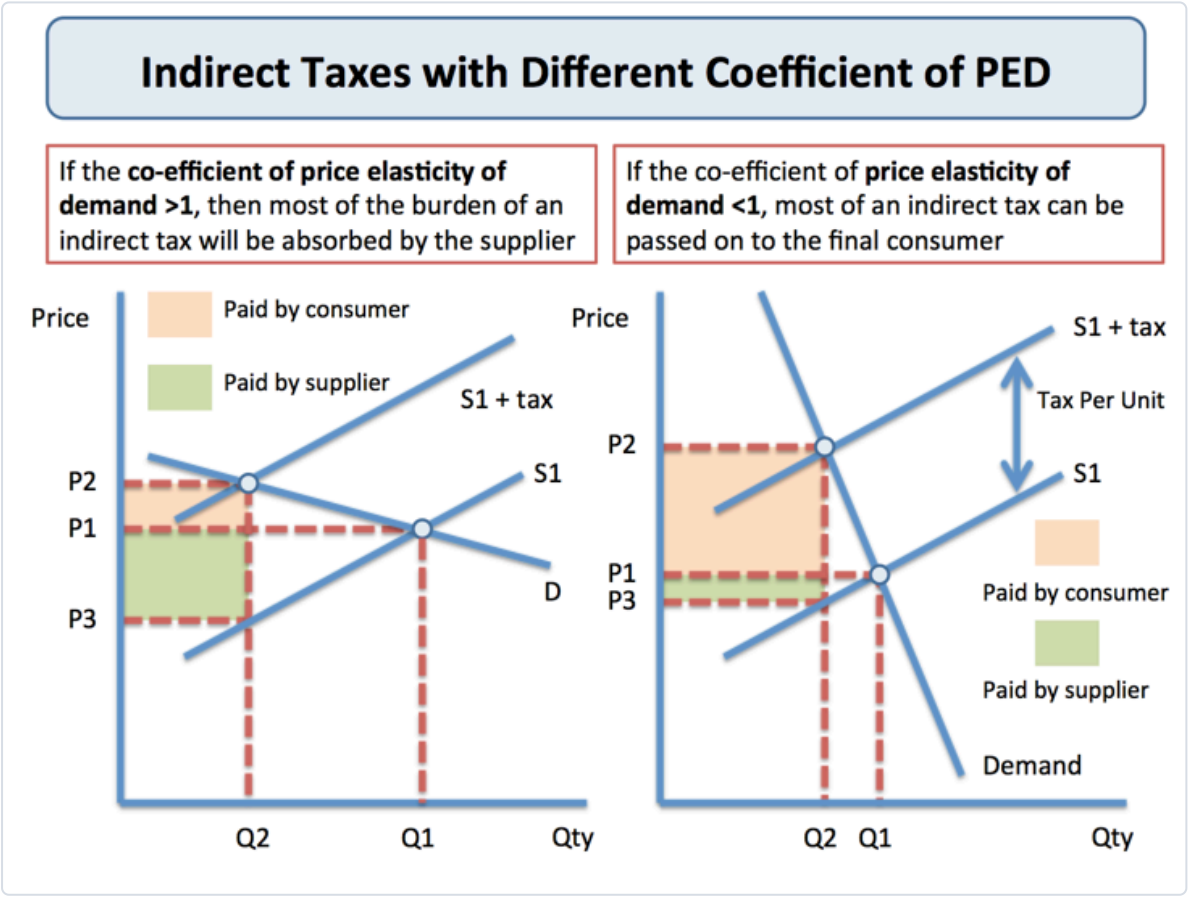
Core: a tax **drives a wedge** between the price **buyers pay** (`p_b`) and the price **sellers receive** (`p_s`): **`p_b = p_s + t`**. It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss.** The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
  - **Unit (specific) tax** — a fixed **amount *t* per unit** (per litre of petrol, per cigarette).
  - **Ad valorem tax** ("sales tax"/VAT) — a **fraction α of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example, t = €3):** with `Q_D = 23 − p_b`, `Q_S = −1 + p_s/2`, and `p_b = p_s + 3`, solving gives **p_s = 14, p_b = 17, Q = 6.** Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
  - **Tax revenue** = `t × Q = 3 × 6 = €18`. The rest of the surplus lost is **deadweight loss** (here `3 × (7−6)/2 = €1.5`) — trades that *would* have happened but now don't.
- **Legal incidence ≠ economic incidence (the key result).** It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by *t*) or **consumers** (demand shifts down by *t*) — the quantity, the prices actually paid/received, and the DWL are **identical.** *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



How to read it (tax equivalence). Both panels impose the **same $0.55 tax** on avocados. **(a)** collects it from **firms** → the supply curve shifts **up** by t; **(b)** collects it from **consumers** → the demand curve shifts **down** by t. In **both** cases the result is identical: **Q falls to 74**, **buyers pay 2.15**, **sellers receive 1.60**, and the wedge between them is exactly **$0.55.** → *The economics doesn't care who writes the cheque.*

- **What *does* decide the split: relative elasticities.** The **more inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The **more price-sensitive (elastic)** side is affected **less** on price, **more** on quantity.



How to read it (incidence & elasticity). The tax raises the supply curve to `S + tax`. The **new higher price P2** is what **consumers** pay; **P3** is what **suppliers** keep. The split of the rectangle between them is

the **incidence**:

- **Left — elastic demand (PED > 1):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand (PED < 1):** consumers keep buying regardless, so the firm **passes most of the tax through** → the **consumer bears most** of it.
- **Policy take-aways:** (1) "**sin taxes**" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Putting a number on the split — the pass-through formula.** The diagrams say *which* side pays more; this formula says *how much*. Each side's share of the tax is the **other side's** elasticity over the total:

> TAX INCIDENCE (PASS-THROUGH)
>
> $$\text{consumer share} = E_S \div \left(E_S + |E_D| \right) \quad \cdot \quad \text{producer share} = |E_D| \div \left(E_S + |E_D| \right)$$

- **Read it as "the other side's elasticity on top":** the **consumer** share carries **E_S** (supply's elasticity); the **producer** share carries **|E_D|** (demand's elasticity). Same denominator; the two shares **add to 1** (they split the whole tax — a handy check). - **Why the *other* side's elasticity?** A side pays a big share when it **can't escape** — i.e. when it's inelastic *and the other side is elastic*. If supply is very elastic (big E_S), sellers walk away rather than absorb the tax, so it lands on **consumers** → E_S on top of the consumer share. This is the same "inelastic side bears more" rule from the diagrams, now as an equation. *(Worked in [practice problem 11](#micro3-practice).)* - **Subsidies are the mirror image:** `p_s = p_b + s` — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — → small DWL). - **Worked example (subsidy s = €3):** same market (`Q_D = 23 − p_b`, `Q_S = −1 + p_s/2`) but now `p_s = p_b + 3` → solving gives **p_b = 15, p_s = 18, Q = 8** — buyers pay **less** and sellers get **more**, and quantity *rises* above the no-subsidy level (6 → 8). The subsidy costs the government `s × Q = 3 × 8 = €24`, part of which is **deadweight loss**.

Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019).** Own-price elasticity ≈ **−1.37** (elastic). A 5% price rise → **~7% fall** in quantity; to cut demand 15% you'd raise price ~11%. → *Illustrates: elasticity as the **design tool for a corrective ("sin") tax**.*
- **World cocoa demand.** Inelastic (≈ −0.19 to −0.96; policy estimate −0.34). → *Illustrates: **inelastic demand → supply shocks hit price**, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021).** Product-level elasticity ≈ **−4.74**, category ≈ **−0.60.** → *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*

- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007).** Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price −1.3** (elastic), **cross-price (Danish price) +0.3** (substitute), **income +1.4** (normal good). → *Illustrates: **all three elasticities at once** — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017).** ≈ −0.24 short-run vs −0.6 long-run. → *Illustrates: demand is **more elastic in the long run** (time to adjust equipment/habits).*

Formula sheet — quick reference

Quantity	Formula	Sign / meaning
Price elasticity of demand	`E_D = %ΔQ_D ÷ %Δp`	**negative**; |E|>1 elastic, |E|<1 inelastic
Point elasticity (linear)	`E_D = (dQ/dp)·(p/Q)`	varies along the curve; more elastic at higher p
Constant-elastic demand	`Q = a·p^ϵ` → `ln Q = ln a + ϵ ln p`	elasticity = **ϵ** everywhere
Cross-price elasticity	`E_cross = %ΔQ_A ÷ %Δp_B`	**+ substitutes**, **– complements**
Income elasticity	`E_income = %ΔQ ÷ %ΔI`	**+ normal**, **– inferior**; food < 1 (Engel)
Supply elasticity	`E_S = %ΔQ_S ÷ %Δp`	**positive**
Tax wedge	`p_b = p_s + t`	buyers pay more, sellers get less; DWL
Tax revenue	`t × Q_taxed`	rectangle; rest of loss = deadweight loss
Subsidy wedge	`p_s = p_b + s`	expands Q above efficient; also a DWL

Exam pointers

- **Compute an elasticity** from two (p, Q) points and **name** it (elastic/inelastic/unit) — the **−1/3 example**.
- Use the **point-elasticity trick** `E_D=(dQ/dp)(p/Q)` on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for **cross-price** (substitutes/complements) and **income** (normal/inferior) elasticities, with an example each.
- **Solve a unit-tax** market (`p_b = p_s + t`) for `p_b, p_s, Q`, compute **revenue** and **deadweight loss** — the **t=€3 example**.
- State and **explain** the two big tax results: **legal ≠ economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the **sin-tax** application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for **narrower product categories**.

Bridge. Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on E_D), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade.**

★ Practice problems — the whole chapter (with worked solutions)

> A set covering **every calculation** in this lecture. **Try each closed-book first**, then check against the worked solution. *(Exam is open-book, no AI — so the skill being tested is doing these fast and correctly, not memorising.)*

What you must be able to calculate (each maps to a problem below):

Skill	Formula	Problem
Own-price elasticity from two points + name it	`E_D = %ΔQ ÷ %Δp`	**1**
Predict %ΔQ from %Δp (and reverse)	`%ΔQ = E_D · %Δp`	**2**
Point elasticity on a linear curve (+ find the unit-elastic point)	`E_D = (dQ/dp)·(p/Q)`	**3**
Elasticity → total revenue (raise or cut price?)	`TR = p·Q`	**4**
Constant-elastic demand	`Q = a·p^ϵ`	**5**
Cross-price elasticity + classify	`E_cross = %ΔQ_A ÷ %Δp_B`	**6**
Income elasticity + classify	`E_income = %ΔQ ÷ %ΔI`	**7**
Supply elasticity	`E_S = %ΔQ_S ÷ %Δp`	**8**
Unit tax: solve `p_b, p_s, Q`, revenue, DWL, incidence	`p_b = p_s + t`	**9**
Subsidy: solve `p_b, p_s, Q`, cost, DWL	`p_s = p_b + s`	**10**
Incidence from elasticities (pass-through)	`consumer share = E_S ÷ (E_S + |E_D|)` · `producer share = |E_D| ÷ (E_S + |E_D|)`	**11**
Equilibrium + a price control *(Ch. 2 recap)*	set `Q_D = Q_S`	**12**

Problems

1. A bakery raises bread from **€4 → €5**; weekly sales fall from **100 → 80** loaves. Find the price elasticity of demand and say whether demand is elastic, inelastic, or unit-elastic.

2. (a) The own-price elasticity of petrol is **−0.3**. If the price rises **10%**, by how much does quantity demanded change? (b) A cinema wants ticket sales to rise **9%**; demand elasticity is **−1.5**. What price change is needed?

3. Demand is `Q = 120 − 4p`. (a) Find E_D at **p = 10**. (b) Find E_D at **p = 20**. (c) At what price is demand **unit-elastic**?

4. Using the same demand `Q = 120 − 4p`: at **p = 10** and at **p = 20**, should the firm **raise or cut** price to increase **total revenue**? At what price is revenue **maximised**?

5. Demand is `Q = 500·p^(−2)`. (a) What is the elasticity of demand? (b) Find quantity at **p = 5** and **p = 10**. (c) If price rises **1%** from €5, roughly what is %ΔQ?

6. Classify each pair as substitutes, complements, or unrelated, and give E_cross: (a) price of **tea +8%** → demand for **coffee +4%**; (b) price of **printers +10%** → demand for **ink +6% fall**; (c) price of **umbrellas +10%** → demand for **coffee 0%**.

7. Classify the good and give E_income: (a) income **+5%** → **restaurant meals +15%**; (b) income **+10%** → **bus tickets −4%**; (c) income **+8%** → **bread +2%**.

8. When the price rises from **€8 → €10**, quantity supplied rises from **60 → 69**. Find the price elasticity of supply and classify it.

9. A market has `Q_D = 100 − 4p` and `Q_S = −20 + 2p`. (a) Find the pre-tax equilibrium. (b) A **unit tax t = €6** is imposed. Solve for **p_b, p_s, and Q**. (c) Find **tax revenue** and the **deadweight loss**. (d) How is the burden split between buyers and sellers, and **why**?

10. Same market (`Q_D = 100 − 4p`, `Q_S = −20 + 2p`). Now the government pays a **subsidy s = €6** per unit. Solve for **p_b, p_s, Q**, the **cost to government**, and the **deadweight loss**.

11. At the pre-tax equilibrium of problem 9, the elasticities are **E_D = −4** and **E_S = +2**. Use the pass-through formula to find the **share of the tax borne by consumers vs producers**, and check it against your answer to 9(d).

12. *(Ch. 2 recap.)* `Q_D = 80 − 2p`, `Q_S = −10 + p`. (a) Find equilibrium **p*** and **q***. (b) The government sets a **price ceiling at €25**. Compute the resulting **shortage**.

Solutions

1. %ΔQ = (80−100)/100 = **−20%**; %Δp = (5−4)/4 = **+25%**.

$$E_D = -20\% \div +25\% = \mathbf{-0.8}$$

|−0.8| < 1 → **inelastic** (quantity moves less than price). *(Note: revenue actually rises — see problem 4's logic.)*

2. (a) `%ΔQ = E_D · %Δp = −0.3 × 10% =` **−3%** (quantity falls 3%). (b) rearrange: `%Δp = %ΔQ ÷ E_D = 9% ÷ (−1.5) =` **−6%** → the cinema must **cut** price 6%.

3. Slope `dQ/dp = −4` (constant). Use `E_D = (dQ/dp)·(p/Q)`.

- (a) p = 10 → Q = 120 − 40 = 80 → `E_D = −4 × 10/80 =` **−0.5** (inelastic).

- (b) p = 20 → Q = 120 − 80 = 40 → `E_D = −4 × 20/40 =` **−2** (elastic).
- (c) unit-elastic: set `−4·(p/Q) = −1` → `Q = 4p` → `120 − 4p = 4p` → `8p = 120` → **p = 15** (Q = 60). *(On any straight-line demand, the unit-elastic point is the **midpoint**.)*

4. Revenue `TR = p·Q`.

- At **p = 10** (inelastic, |E|<1): raising price **raises** revenue. Check: p=10 → TR = 10×80 = 800; p=11 → Q = 76, TR = **836**. → **raise price.**
- At **p = 20** (elastic, |E|>1): cutting price **raises** revenue. Check: p=20 → TR = 800; p=19 → Q = 44, TR = **836**. → **cut price.**
- **Revenue is maximised at the unit-elastic point** p = 15: TR = 15 × 60 = **900**. *(Rule: |E|<1 → raise price; |E|>1 → cut price; max TR where |E| = 1.)*

5. (a) For `Q = a·p^ε`, the elasticity **is the exponent**: `E_D =` **−2** (constant at every price). (b) p = 5 → Q = 500/25 = **20**; p = 10 → Q = 500/100 = **5**. (c) `%ΔQ ≈ E_D · %Δp = −2 × 1% =` **−2%**.

6. `E_cross = %ΔQ_A ÷ %Δp_B`. (a) `+4 ÷ +8 =` **+0.5 → substitutes** (tea dearer → more coffee). (b) `−6 ÷ +10 =` **−0.6 → complements** (printers dearer → less ink). (c) `0 ÷ +10 =` **0 → unrelated.**

7. `E_income = %ΔQ ÷ %ΔI`. (a) `+15 ÷ +5 =` **+3 → normal, and a luxury** (E>1). (b) `−4 ÷ +10 =` **−0.4 → inferior.** (c) `+2 ÷ +8 =` **+0.25 → normal necessity** (0<E<1; cf. Engel's law for food).

8. %ΔQ_S = (69−60)/60 = **+15%**; %Δp = (10−8)/8 = **+25%**.

$$E_S = +15\% \div +25\% = \mathbf{+0.6}$$

0 < E_S < 1 → **inelastic supply** (output hard to expand quickly).

9. (a) **Pre-tax:** set `Q_D = Q_S`: `100 − 4p = −20 + 2p` → `120 = 6p` → **p* = 20, Q* = 20**.

- (b) **With t = 6**, use `p_b = p_s + 6`, `Q_D = 100 − 4p_b`, `Q_S = −20 + 2p_s`. Set equal: `100 − 4(p_s + 6) = −20 + 2p_s` → `76 − 4p_s = −20 + 2p_s` → `96 = 6p_s` → **p_s = 16, p_b = 22, Q = 100 − 4(22) = 12**. *(check Q_S = −20 + 32 = 12 ✓)*
- (c) **Tax revenue** = `t × Q = 6 × 12 =` **€72**. **DWL** = `½ × t × (Q\* − Q_tax) = ½ × 6 × (20 − 12) =` **€24**.
- (d) Buyers bear `p_b − p\* = 22 − 20 =` **€2**; sellers bear `p\* − p_s = 20 − 16 =` **€4** (they sum to t = 6). **Sellers bear more because supply is the *more inelastic* side** (slope 2 vs demand's 4 → demand is more elastic, so it escapes more of the tax onto quantity).

10. With s = 6, use `p_s = p_b + 6`: `100 − 4p_b = −20 + 2(p_b + 6)` → `100 − 4p_b = −8 + 2p_b` → `108 = 6p_b` → **p_b = 18, p_s = 24, Q = 100 − 4(18) = 28**. *(check Q_S = −20 + 48 = 28 ✓)* — buyers pay **less** (18<20), sellers get **more** (24>20), and **Q rises** (20→28). **Cost to government** = `s × Q = 6 × 28 =` **€168**. **DWL** = `½ × s × (Q_sub − Q\*) = ½ × 6 × (28 − 20) =` **€24** (over-production: units that cost more than they're worth).

11. Pass-through shares:

INCIDENCE FROM ELASTICITIES

$$\text{consumer share} = E_S \div (E_S + |E_D|) \;\cdot\; \text{producer share} = |E_D| \div (E_S + |E_D|)$$

With E_S = 2 and |E_D| = 4: **consumer share = 2 ÷ (2+4) = 1/3**, **producer share = 4 ÷ 6 = 2/3.** Of the €6 tax that's **€2 on buyers, €4 on sellers** — exactly matching 9(d). ✔ *(Rule: the **more inelastic** side pays the bigger share.)*

12. (a) `80 − 2p = −10 + p` → `90 = 3p` → **p* = 30, q* = 20**. (b) At the ceiling **p = 25**: `Q_D = 80 − 50 = 30`, `Q_S = −10 + 25 = 15` → **shortage = Q_D − Q_S = 30 − 15 = 15 units.**

★ Fun facts & memorable details (Lecture 3)

> Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
- **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
- **The soda-tax number:** the own-price elasticity of sugary drinks (≈ **−1.37**) is a real input to real policy — a 5% price rise cuts consumption ~7%.
- **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
- **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
- **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
- **Brand vs category:** demand for *a* beer is wildly elastic (≈ −4.7 — switch brands!), but demand for *beer* is inelastic (≈ −0.6) — same drink, opposite elasticity, just different zoom level.
- **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits (+0.3) from the data (Asplund et al. 2007).

Lecture 4: Consumer Choice (Chapter 5)

Required reading: Friberg, **Chapter 5 — Consumer choice** (§5.1 modelling choice · §5.2 preferences & utility · §5.3 budget constraints · §5.4 utility maximization · §5.5 discussion of the framework · §5.6 behavioral economics · §5.7 appendix: solving for the optimal basket with Lagrange). *Lecturer: Luigi Butera — deck "**Lecture 4: Consumer choice.**" (Numbering note: this is the course's **4th lecture** but the textbook's **Chapter 5**; the deck's own "4.1–4.5" roadmap just re-numbers the chapter's sections.)*

Theme of the lecture (Butera — from the deck). So far (Lectures 2–3) we took **demand curves as given** and pushed them around. Now we **build demand from the ground up**: *where do preferences come from, and how does a rational consumer choose under scarcity?* The deck's answer is a **three-stage machine** — **preferences → utility → budget constraint → constrained optimization** — that will (next chapter)

generate the demand curve. The deck's five stops: **4.1 preferences · 4.2 utility · 4.3 budget constraints · 4.4 constrained choice · 4.5 behavioral economics.**

> ★ **The heart of the lecture:** a consumer **maximizes utility subject to a budget constraint.** We (1) describe what she **likes** with well-behaved **preferences** (complete + transitive), (2) quantify them with a **utility function** whose slope-of-indifference-curve is the **MRS** = how much of one good she'll trade for the other *and stay equally happy*, (3) describe what she **can afford** with a **budget line** whose slope is the **MRT** = p_F/p_E = how the market lets her trade the goods. The **optimum** is where the highest reachable indifference curve is **tangent** to the budget line — i.e. where **MRS = MRT** ($MU_F/MU_E = p_F/p_E$): the rate she's *willing* to trade equals the rate the market *lets* her trade.

The red thread. Every piece answers one of three questions: *what do I want?* (preferences → utility → MRS), *what can I afford?* (budget line → MRT), and *so what do I pick?* (tangency: MRS = MRT). Get those three and you can predict how choices shift when **prices** or **income** change — which is exactly a **demand curve** (next chapter). §5.6 then asks: *when does this rational machine break down?* (behavioral economics).

Key concepts / "modes" to use

- **Bundle** — a list of quantities of each good (e.g. "4 Entertainment, 2 Food"). Choice = ranking bundles.
- **Preferences & the rationality axioms** — **completeness** (can rank any two bundles) · **transitivity** (rankings "hold in transit") · **monotonicity** / "**more is better**" (convenient add-on).
- **Utility function** `U(F,E)` — assigns a number to each bundle so that **higher = preferred**. **Ordinal**, not cardinal (only the *ranking* means anything).
- **Marginal utility (MU)** — extra utility from one more unit of a good, holding the other fixed = the **partial derivative**; typically **diminishing**.
- **Indifference curve** — all bundles giving the *same* utility; **5 properties**; three shapes (**perfect substitutes**, **imperfect substitutes**, **perfect complements**).
- **Marginal rate of substitution (MRS)** — slope of the indifference curve = **MU_F / MU_E** = how much E you'll give up for one more F at equal utility.
- **Budget constraint / line** — `I = p_F·F + p_E·E`; the **opportunity set** is everything on or under it.
- **Marginal rate of transformation (MRT)** — slope of the budget line = **p_F / p_E** = how the *market* lets you swap the goods.
- **Consumer optimum** — highest indifference curve tangent to the budget line: **MRS = MRT**, i.e. `MU_F/MU_E = p_F/p_E` (equivalently, equal marginal utility *per euro*: `MU_F/p_F = MU_E/p_E`).
- **Interior vs corner solution** — tangency inside the axes vs "spend everything on one good" (perfect substitutes).
- **Behavioral economics** — systematic deviations: **social preferences**, **lack of self-control**, **cognitive biases** (loss aversion, endowment effect, framing, overconfidence).

Section-by-section main points

§5.1 How can we model choice? — the three ingredients (deck)

Core: we model an individual who **faces constraints** (income, time) and **does as well as possible** given them. Sounds almost tautological — the whole chapter is about *putting structure* on "does as well as possible." The deck lists the **ingredients we need**:

- **A language** to say "I like this more than that" → **preferences** (§5.2).
- **Assumptions** that make those preferences *rational* → the **axioms** (completeness, transitivity).
- **A tool to quantify wellbeing** → the **utility function** (§5.2.1).
- **A tool to quantify "you can't have it all"** → the **budget constraint** (§5.3).
- **Bringing it together** → **constrained optimization** (§5.4).
- **Three premises of the whole build (deck + §5.1):** *(1)* we **know and have stable preferences**, *(2)* we **face constraints** (time, money), *(3)* we **try to do as well as possible** given them.

§5.2 Preferences and the rationality axioms (deck slides 6–7)

Core: choice is modelled as **ranking bundles.** For the ranking to give a consistent model we assume it obeys two **essential** axioms (plus one convenient one). The deck's notation for the ranking:

Symbol	Reads	Meaning
a ≻ b	"a is **strictly preferred** to b"	a is *better* than b
a ⪰ b	"a is **weakly preferred** to b"	a is *at least as good* as b
a ~ b	"**indifferent** between a and b"	a and b are *equally good*

- **Completeness (essential):** the consumer can **rank any two bundles** — for any A, B she can say A≻B, B≻A, or A~B. *(No "I can't compare these.")*
- **Transitivity (essential):** rankings "**hold in transit**" — if **A≻B** and **B≻C**, then **A≻C**. This is what lets us predict choice in a *chain* of comparisons (and stops preferences from cycling).
- **Monotonicity / "more is better" (convenient, not essential):** all else equal, **more of a good is better.** This is what makes bundles *farther from the origin* preferred. *(Things you want **less** of — pollution, noise — are relabelled "bads," and we redefine the good as their **absence**.)*
- **Why the axioms matter:** completeness + transitivity are *exactly* the conditions under which preferences can be represented by a **utility function** (next section). No axioms → no utility function → no model.

§5.2.1 Utility & utility functions (deck slides 8–9)

Core: if preferences are **complete and transitive**, we can represent them with a **utility function** `U(·)` that assigns each bundle a **number**, higher for more-preferred bundles. Neoclassical economics inherits this from the **utilitarian** tradition (J.S. Mill: "*actions are right in proportion as they tend to promote happiness*").

UTILITY REPRESENTS PREFERENCES

$$\mathrm{a} \succ \mathrm{b} \iff U(\mathrm{a}) > U(\mathrm{b})$$

Reads: the utility function just **translates the ranking into real numbers** — bundle *a* is preferred to *b* **if and only if** *a* scores a higher utility number. (This is the **von Neumann–Morgenstern** representation the deck cites.) The function *orders* preferences; it does not measure "happiness" in any absolute unit.

- **Ordinal, not cardinal (the crucial caveat):** only the **ranking** matters, **not** the size of the gap. `U = 8` vs `U = 4` means the first is **preferred** — it does **not** mean "twice as good." *(Deck: "9 > 1, but do '1' and '9' mean something meaningful?" — no. Utility is unobservable.)*
- **General vs specific form:** general `U = U(F,E)`; a specific example the textbook uses is `U = √(EF)`. **Worked:** at **E=2, F=8**, `U = √(2·8) = √16 = 4`. Bundles with the **same** product `E·F = 16` (e.g. E=4,F=4 or E=8,F=2) sit on the **same** indifference curve; `E=5,F=5` gives `U = 5` → a **higher** curve.
- **Common functional forms (deck slide 12):**
  - **Cobb-Douglas:** `U = F^a · E^b` — the workhorse; convex, imperfect substitutes.
  - **Perfect substitutes:** `U = a·F + b·E` — straight-line indifference curves.
  - *(Perfect complements: `U = min(F, E)` — L-shaped.)*

★ Marginal utility & diminishing marginal utility (deck slides 10, 16)

Core: **marginal utility (MU)** is the extra utility from **one more (infinitesimal) unit** of a good, holding the others fixed — i.e. the **partial derivative** of `U`. It is usually **diminishing** (the 1st slice of pizza beats the 7th).

MARGINAL UTILITY

$$MU_F = \partial U(\mathrm{F},\mathrm{E})/\partial F \quad \cdot \quad MU_E = \partial U(\mathrm{F},\mathrm{E})/\partial E$$

Reads: **how much total utility rises per extra unit of one good, other goods held constant** — **∂U/∂F** = the partial derivative w.r.t. Food · **diminishing MU** = this derivative *shrinks* as you get more of the good.

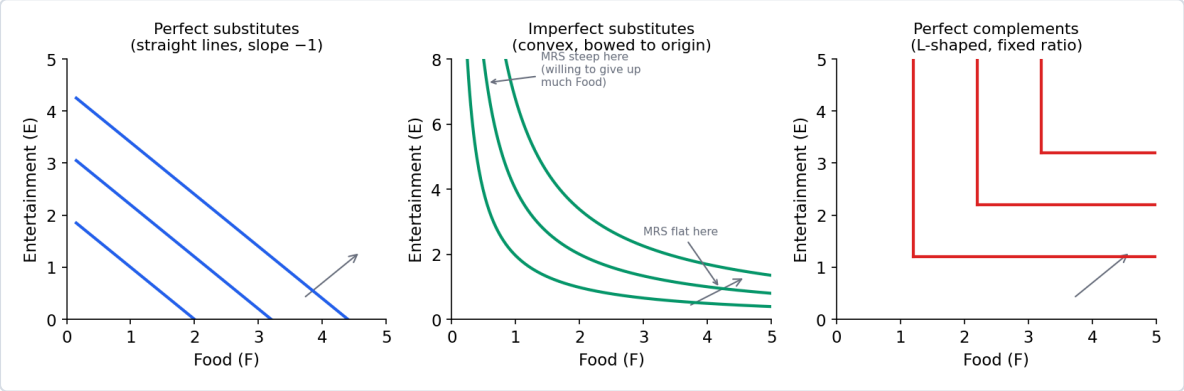
- **The deck's three consumers (utility over one good):** for each extra apple —
  - **Bob:** each apple adds the **same** utility → **constant** MU (linear U).
  - **Linda:** each apple adds a bit **more** → **increasing** MU (convex U). *(unusual)*
  - **Mike:** each apple adds a bit **less** → **diminishing** MU (concave U). *(the normal case)*
- **Worked example (`U = √(EF) = E^{0.5}F^{0.5}`):** `MU_F = ∂U/∂F = 0.5·E^{0.5}·F^{−0.5} = 0.5·√(E/F)`. At **E=2, F=8**: `MU_F = 0.5·√(2/8) = 0.5·0.5 = 0.25`. As **F rises** (holding E), `√(E/F)` falls → **MU_F falls** → diminishing marginal utility. ✓
- **Why it matters:** diminishing MU is *why* indifference curves are **bowed toward the origin** (convex) and *why* the **MRS falls** as you move down a curve — the next two ideas.

Indifference curves — 5 properties & the 3 types (deck slides 13–14)

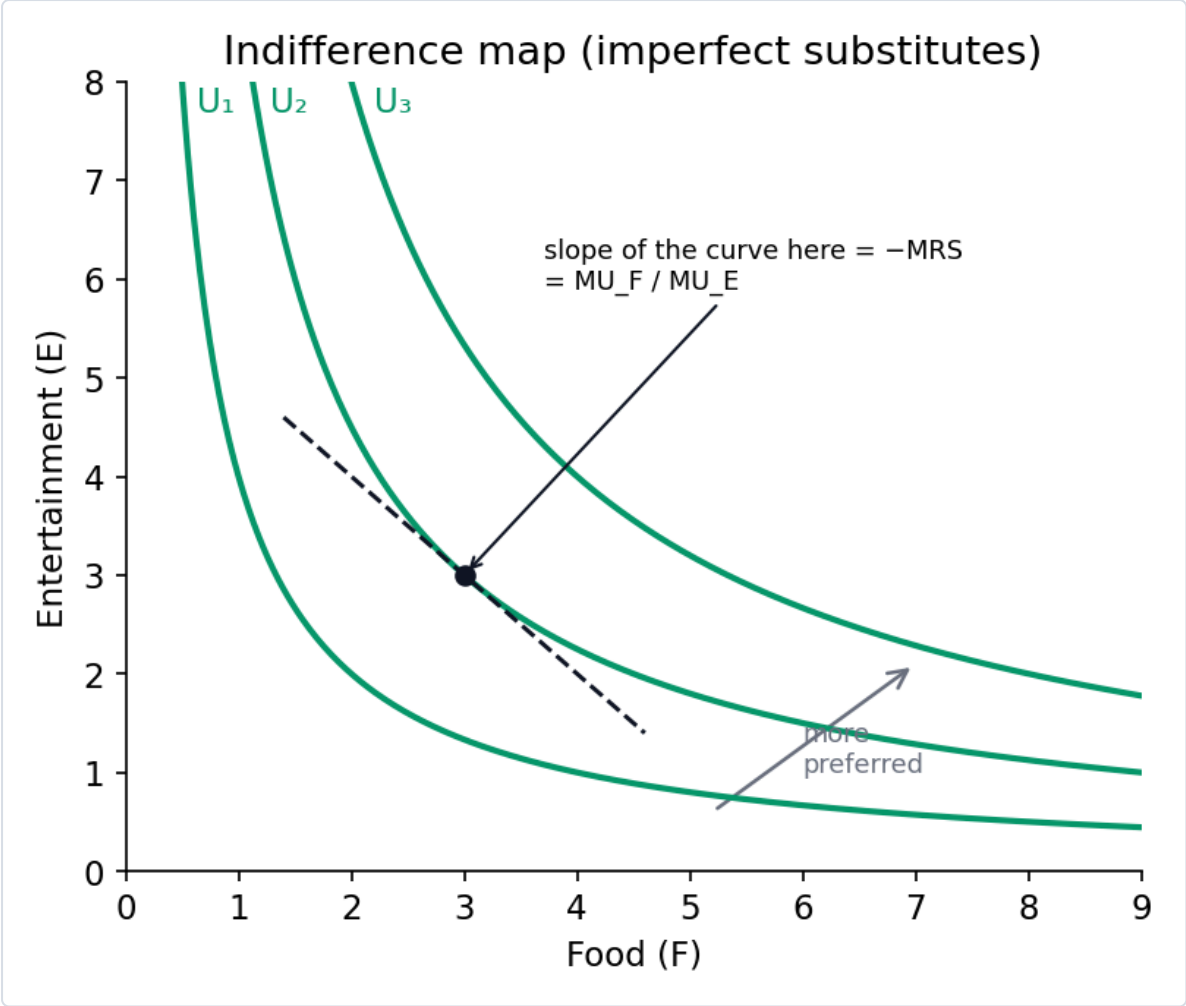
Core: an **indifference curve** connects **all bundles the consumer views as equally desirable.** It's the level-set of the utility function (fix `U` at one number). The deck's **five properties**:

#	Property	Why
1	Curves **farther from the origin** are preferred	"more is better" (monotonicity)
2	**Every** bundle lies on some indifference curve	completeness
3	Indifference curves **cannot cross**	transitivity (crossing → contradiction)
4	They **slope downward**	to keep utility fixed, more of one good ⇒ less of the other
5	They **cannot be "thick"**	a thick band would contain a strictly-better bundle (monotonicity)

- *(Utility is read **ordinally**: the labels on the curves rank them; the gaps between labels are meaningless.)*
- **Three shapes of preference (textbook §5.2):**



- **Perfect substitutes** — tradeable at a **constant rate** → **straight-line** curves (constant slope). *Example: one €20 note ~ four €5 notes, always.* `U = aF + bE`.
- **Imperfect substitutes (the normal case)** — willing to trade, but the **rate depends on how much of each you already have** → **convex** curves, "bowed toward the origin" (**strictly convex** preferences). *When you have lots of chips and no drink, you'll give up many chips for one drink; when you have few chips, you won't.* `U = √(EF)`, Cobb-Douglas.
- **Perfect complements** — consumed in a **fixed ratio** → **L-shaped** curves. *Left & right shoes: a 2nd right shoe alone adds nothing.* `U = min(F,E)`.



How to read it (indifference map). Each green curve is one utility level; **northeast = more preferred** (property 1). The dashed tangent at the marked bundle has slope **−MRS**: it gets **flatter** as you move right (more Food, less willing to give up Food) — that's diminishing MRS, the topic below.

★ Marginal rate of substitution (MRS) (deck slides 17–19)

Core: the **MRS** is the **slope of the indifference curve** — the **maximum amount of one good the consumer will give up to get one more unit of the other, staying equally happy.** Because utility is constant along the curve (`dU = 0`), it equals the **ratio of marginal utilities.**

MARGINAL RATE OF SUBSTITUTION

$$MRS = -\mathrm{d}E/\mathrm{d}F = MU_F \,/\, MU_E \quad \text{(good on the x-axis over good on the y-axis)}$$

Reads: **how much E you'll trade for one more F and stay on the same curve** — derive it by **totally differentiating** `U` and setting `dU = 0`: `0 = MU_F·dF + MU_E·dE` → `-dE/dF = MU_F/MU_E` . **MU_F** = marginal utility of the x-axis good (Food) · **MU_E** = marginal utility of the y-axis good (Entertainment) · the ratio = the curve's steepness at that point.

- **Diminishing MRS (why the curve bows):** as you move **down-right** along the curve (more F, less E), `MU_F` **falls** and `MU_E` **rises**, so the ratio `MU_F/MU_E` **shrinks** — the curve **flattens.** "*When I have relatively less pizza I'll give up more burritos for a slice than when I already have lots of pizza.*"

- **Worked example (`U = √(EF)`):** `MU_F = 0.5√(E/F)`, `MU_E = 0.5√(F/E)`, so `MRS = MU_F/MU_E = (0.5√(E/F))/(0.5√(F/E)) = E/F`. At bundle **E=2, F=8**: `MRS = 2/8 = 0.25` → she'll give up only **0.25 units of E** for one more F (she already has lots of F). At **E=8, F=2**: `MRS = 8/2 = 4` → now she'll give up **4 units of E** for one more F. *Same curve, opposite ends → diminishing MRS.* ✓

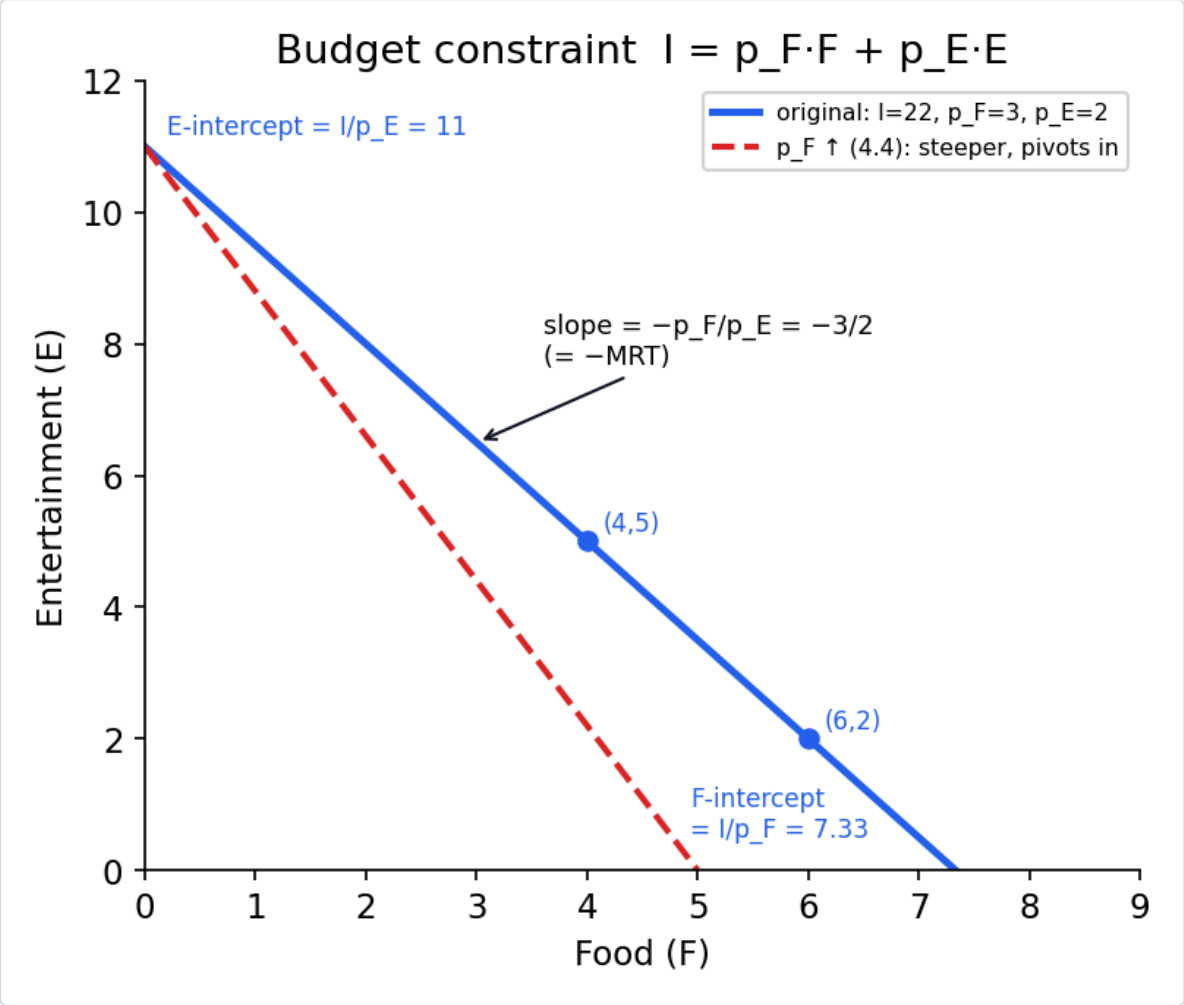
§5.3 Budget constraint & the MRT (deck slides 21–22)

Core: the **budget constraint** operationalizes "you can't have it all": given prices and income `I`, total spending can't exceed income. With two goods it's a **straight line** (the **budget line**); everything on or below it is the **opportunity set.**

BUDGET CONSTRAINT & LINE

$$I = p_F F + p_E E \quad \Longrightarrow \quad E = I/p_E - (p_F/p_E)\cdot F$$

Reads: **income = spending on Food + spending on Entertainment**; rearranged for graphing, **E-intercept = I/p_E** (all income on E), **slope = −p_F/p_E**, **F-intercept = I/p_F** (all income on F).



- **Worked example (deck: I=22, p_F=3, p_E=2):** E-intercept `= 22/2 = 11`; slope `= −3/2`; F-intercept `= 22/3 ≈ 7.33`. Bundle **(F=4, E=5)** is affordable: `3·4 + 2·5 = 22` ✓. So is **(F=6, E=2)**: `3·6 + 2·2 = 22` ✓.

- **MRT — marginal rate of transformation** = the **slope of the budget line** = how the **market** lets you swap goods:

MARGINAL RATE OF TRANSFORMATION

$$MRT = p_F \,/\, p_E \quad \text{(the relative price = the market "exchange rate" between the goods)}$$

Reads: **give up one F, the market gives you p_F/p_E units of E.** With p_F=3, p_E=2, `MRT = 1.5` : one less Food frees €3, which buys **1.5** Entertainment. *(MRS = what you're **willing** to trade; MRT = what the market **lets** you trade.)*

- **Comparative statics of the line (deck):**
  - **Income change → parallel shift** (slope unchanged; both intercepts move).
  - **Price change → pivot** (steepness changes). A **higher p_F ⇒ steeper**, pivoting **inward** around the E-intercept (see red dashed line); a **lower p_F ⇒** flatter.

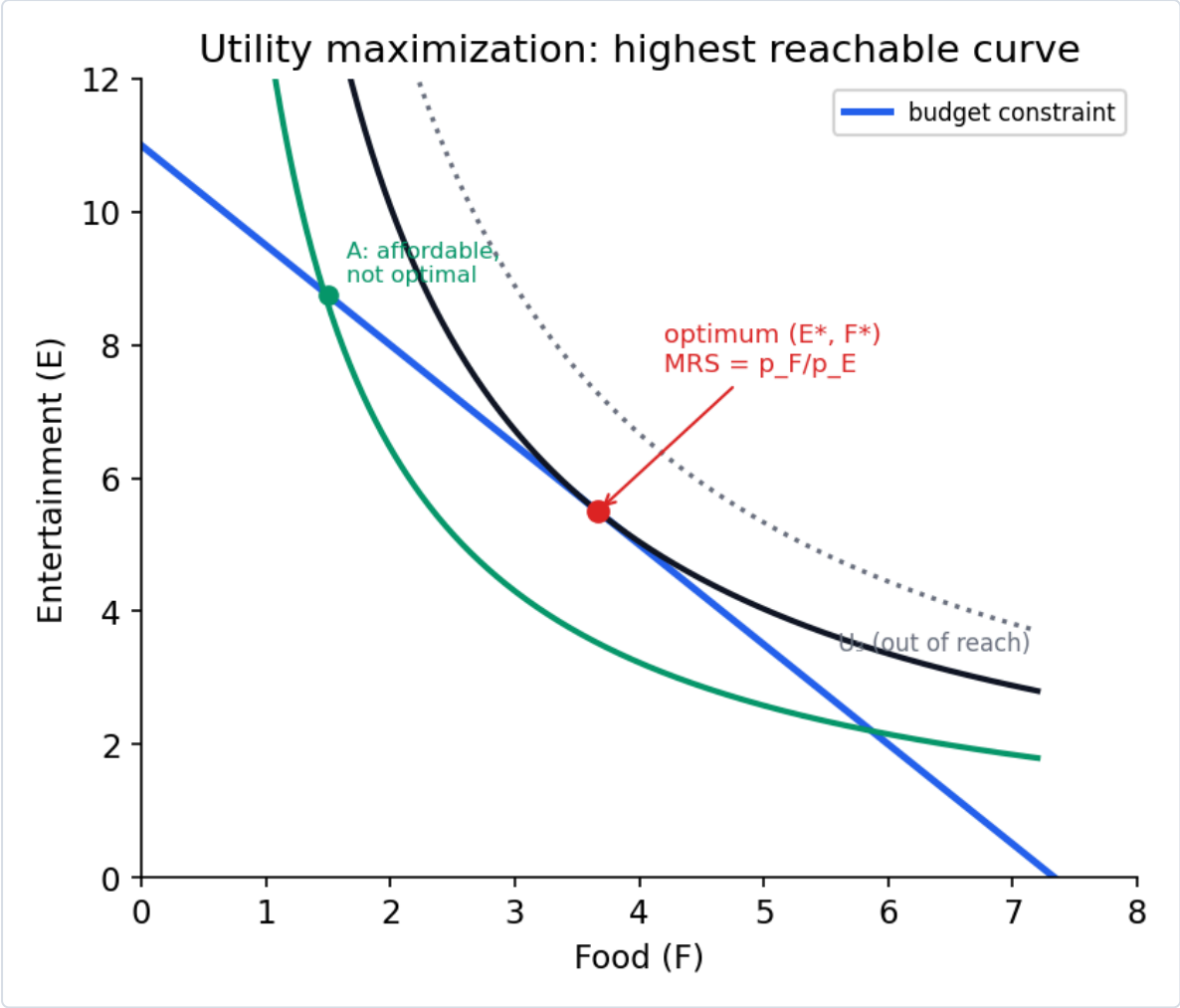
★ §5.4 The optimum: maximize utility subject to the budget (deck slide 23)

Core: the consumer solves `max U(F,E) subject to p_F·F + p_E·E ≤ I`. Graphically: find the **highest indifference curve** that still touches the budget line — that's the **tangency point.** At the tangency, the slope of the indifference curve (**MRS**) equals the slope of the budget line (**MRT**).

CONSUMER OPTIMUM (TANGENCY)

$$MRS = MRT \iff MU_F/MU_E = p_F/p_E \iff MU_F/p_F = MU_E/p_E$$

Reads: at the optimum, **the rate you're willing to trade = the rate the market lets you trade.** The last form is the "**equal marginal utility per euro**" (bang-for-the-buck) rule: **the last euro spent on Food yields the same extra utility as the last euro spent on Entertainment** — otherwise you'd reallocate toward the higher-MU-per-euro good.



How to read it (utility maximization). Point **A** is *affordable* (on the budget line) but **not optimal**: the curve through A is steeper than the line (MRS > MRT), so she values Food more than the market charges → she should **buy more Food, less Entertainment**, sliding down to the tangency. The **red optimum** is the best reachable bundle; the dotted curve U_3 would be better but her income can't reach it.

- **Why A isn't optimal (the deck's logic):** at A, `MU_F/MU_E > p_F/p_E` — food is "cheap relative to how much you value it." Reallocate toward food until the two ratios are equal.
- **Worked example (`U = √(EF)`, I=22, p_F=3, p_E=2):** MRS `= E/F`. Set `MRS = MRT`: `E/F = 3/2` → `E = 1.5F`. Plug into the budget `3F + 2E = 22`: `3F + 2(1.5F) = 22` → `6F = 22` → **F* = 3.67, E* = 5.5.** *(Check: `3·3.67 + 2·5.5 = 11 + 11 = 22` ✓ — matches the red dot in the figure.)*

Comparative statics · interior vs corner · the Cobb-Douglas shortcut (deck slide 24, §5.7)

Core: once you can find the optimum, you can ask **how it moves** when a price or income changes — and that movement, traced out, **is the demand curve** (Ch. 6). Watch for **corner solutions.**

- **Interior vs corner solution (deck):**
  - **Interior solution** — the tangency lies **strictly inside** the axes (buy *some* of both goods). Requires **convex** (imperfect-substitute) preferences.
  - **Corner solution** — the optimum is **on an axis** (buy *only one* good). Happens with **perfect substitutes**: if `MRS ≠ MRT` everywhere, you spend **everything** on whichever good gives more

utility per euro (`MU/p`). *Example: if* `U = 2F + E` *and* `p_F = p_E`*, Food gives twice the bang per euro → buy only Food.*

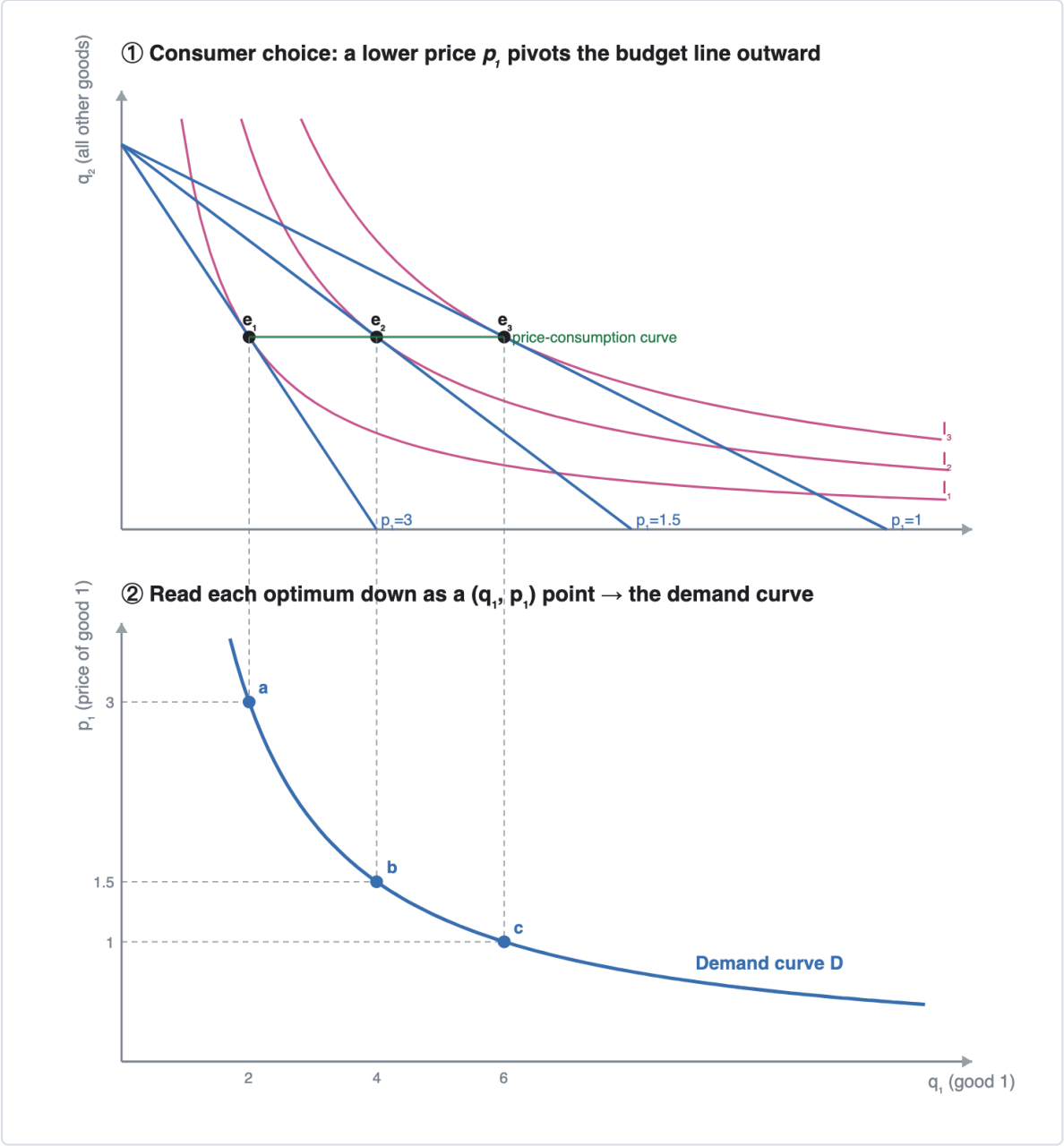
- **Perfect complements** — always a corner-of-the-L: buy the goods in their **fixed ratio** regardless of the price ratio (solve the ratio + the budget line together).

- **The Cobb-Douglas shortcut (worth memorising for exams):** for `U = F^a · E^b`, the tangency condition solves in closed form — the consumer **spends a *constant share* of income on each good**:

> COBB-DOUGLAS OPTIMAL DEMANDS
>
> $$F^* = [\ a/(a+b)\ ] \cdot I/p_F \quad \cdot \quad E^* = [\ b/(a+b)\ ] \cdot I/p_E$$

Reads: **spend the fraction a/(a+b) of income on Food and b/(a+b) on Entertainment**, then divide by each price to get quantities. The **exponents are the expenditure shares. Worked (`U = F^{0.5}E^{0.5}`, so a=b=0.5, I=22, p_F=3, p_E=2):** `F* = 0.5·22/3 = 3.67`, `E* = 0.5·22/2 = 5.5` — same answer as the tangency method above, in one line. *(Notice each good's demand **doesn't depend on the other good's price** — a special Cobb-Douglas feature.)*

- **The appendix method (§5.7) — Lagrange:** for a general `U`, combine objective + constraint into `ℒ = U(F,E) + λ(I − p_F F − p_E E)`, set the three partials to zero, and the first two give back `MU_F/MU_E = p_F/p_E` — the same tangency condition. `λ` = the **marginal utility of income** ("how much extra utility one more euro of income buys"). *You won't need Lagrange for the exam, but know that it reproduces MRS = MRT.*
- **The payoff (Ch. 6 foreshadow):** change `p_F`, re-solve the optimum, and record the new `F*` — repeat, and the `(p_F, F*)` pairs **trace out the demand curve.** *This is why we built all this machinery: choice → demand.*



How to read it (choice → demand, step by step). This is the picture behind "trace out the demand curve." ① **(top)** each time the price of the good **falls**, the budget line **pivots outward** and the tangency optimum slides to **more of the good** (`e₁→e₂→e₃`). ② **(bottom)** drop each optimum straight down to plot its **(quantity, price)** pair — `a, b, c` — and **join them: that's the demand curve.** The demand curve you took as given in Lecture 2 is nothing but the consumer's optimum **re-solved for every price.** *(Fully worked in Lecture 5 — Deriving the demand curve; and see practice Q12, which computes two of these points, (p=4, F=5) and (p=2, F=10).)*

§5.5–5.6 Discussion of the framework & behavioral economics

Core: the rational-choice model is powerful (Becker: "*maximizing behavior, market equilibrium, and stable preferences, used relentlessly and unflinchingly*") — but it's a **model.** Two clarifications, then the **deviations.**

- **Two common misconceptions (§5.5):**
  - "*Economics says be a selfish materialist.*" **No** — the framework maximizes **utility**, not *goods*. Giving to charity, choosing a low-paid academic job you love, buying organic — all perfectly rational if that's

your preference.
    - "*It only works for apples-vs-oranges.*" Actually **Gary Becker** extended it to fertility, marriage, crime, discrimination: assume **stable preferences** and let **costs/benefits differ** across contexts (e.g. why fertility differs across countries — Table 5.2 — need not be different *tastes*).
- **Behavioral economics (§5.6) — three deviations from the simple model:**

Deviation	What it is	Signature examples
Social preferences	We care about **others** — fairness, status, altruism, spite. *Not irrational* — just richer preferences.	**Ultimatum game**: proposers offer ~40% (not ~0%), responders reject ~16% of low offers to punish unfairness.
Lack of self-control	We over-weight the **present** — a myopic "*doer*" vs a "*planner*" (Thaler–Shefrin).	Smoking, over-spending, under-saving; the marshmallow test; policy responses = mandatory pensions, alcohol limits.
Cognitive limitations & biases	Systematic (not random) errors; **bounded rationality**, heuristics.	**Loss aversion / status-quo bias / endowment effect** (Cornell coffee-mug study: owners value the mug 2–4× more); **framing** (600-lives problem); **overconfidence** (68% of entrepreneurs think they'll beat the odds).

- **Random error vs bias (deck's dartboard):** *random* misses scatter around the bull's-eye (people are right *on average*); a **behavioral bias** is a **systematic** skew off-centre (Kahneman & Tversky, *Thinking, Fast and Slow*).
- **Final word (§5.6.5):** the rational model stays the **benchmark** — you need it *even to define* a "deviation." Behavioral economics tells you **when** the benchmark will mislead (e.g. demand for addictive painkillers vs baking soda).

Cases & examples

- **Why did shrimp consumption triple? (Freakonomics / Levitt) — §5.4.1.** Non-economists guessed **demand shifts** (health fads, movies, trendy chefs). The choice framework points first to **prices and income**: the **relative price of shrimp fell** — that alone moves the optimum along/across the budget line. *Lesson: shifts in the budget constraint, not just tastes, drive choices.*
- **Same preferences, different choices: US pizza vs French duck confit (Dubois et al. 2014) — Fig 5.8.** Give an American and a French consumer the **identical** indifference map. Pizza is **cheaper in the US**, confit **cheaper in France** → different **budget-line slopes** → the American picks more pizza, the French more confit. *Different choices need not mean different tastes — just different **relative prices.***
- **The ultimatum game — fairness is real (Güth 1982; Henrich 2001).** Split €10; responder can veto (then both get €0). Pure self-interest predicts "offer €0.10, accept." Instead proposers offer **~40%**, and low offers get **rejected** — even across 15 small-scale societies worldwide (offers never below ~25%). *Evidence for **social preferences**.*
- **The Cornell coffee-mug experiment — the endowment effect (Kahneman et al. 1990).** Mugs handed out at random; owners then valued them **2–4× higher** than non-owners would pay. Ownership shouldn't matter to a rational valuation — but a **reference point** does. *Loss aversion / status-quo bias in one image.*

- **Food stamps or cash? (deck slide 25).** ~50% of US kids use food stamps before age 20. In the standard model, **cash is weakly better** (it relaxes the budget line *more* — you can buy food *or* anything else), so an in-kind transfer can only tie or lose. *Yet* food stamps persist — because of **paternalism / self-control** concerns and political constraints. *A perfect budget-constraint application with a behavioral twist.*
- **Fertility rates: tastes or costs? (Becker; Table 5.2).** South Korea 0.97 kids/woman vs Mali 6.14. A Becker-style reading: **same preferences**, different **costs/benefits** of children (childcare compatibility with careers, old-age support, child mortality, pensions). *Rational choice reaches far beyond "goods."*

Formula sheet (Consumer Choice)

Concept	Formula	In words
Utility represents preferences	`a > b ⟺ U(a) > U(b)`	numbers rank bundles (**ordinal**)
Marginal utility	`MU_F = ∂U/∂F`	extra utility per extra unit; usually **diminishing**
MRS (slope of indifference curve)	`MRS = MU_F / MU_E`	how much E you'll trade for one more F, same utility
Budget line	`I = p_F·F + p_E·E → E = I/p_E − (p_F/p_E)F`	what you can afford; slope `−p_F/p_E`
MRT (slope of budget line)	`MRT = p_F / p_E`	how the **market** lets you trade F for E
Consumer optimum	`MRS = MRT ⟺ MU_F/MU_E = p_F/p_E ⟺ MU_F/p_F = MU_E/p_E`	tangency = equal MU per euro
Cobb-Douglas demands (`U=F^a E^b`)	`F* = a/(a+b) · I/p_F`, `E* = b/(a+b) · I/p_E`	spend constant income shares

Key terms

Term	One-line meaning
Bundle	a specific quantity of each good
Completeness	can rank *any* two bundles
Transitivity	A>B, B>C ⟹ A>C (rankings "hold in transit")
Monotonicity	"more is better" — farther-out curves preferred
Utility function	numbers on bundles preserving the ranking (**ordinal**)
Marginal utility (MU)	∂U/∂(good); extra utility from one more unit
Indifference curve	all bundles of equal utility
MRS	slope of indifference curve = MU_F/MU_E
Budget constraint / line	`I = p_F F + p_E E`; the opportunity set
MRT	slope of budget line = p_F/p_E
Consumer optimum	highest reachable curve tangent to the budget line
Interior / corner solution	some of both goods / all of one good
Behavioral economics	systematic deviations from the rational model
Loss aversion / endowment effect	losses hurt more than equal gains; owning raises valuation

Exam pointers

- **Draw the machine, then solve it.** For "find the optimal bundle": (1) write `MRS = MU_F/MU_E`, (2) set `MRS = p_F/p_E`, (3) combine with the **budget line** `I = p_F F + p_E E`, (4) solve the two equations. *Two equations, two unknowns — always.*
- **Cobb-Douglas → use the shortcut.** If `U = F^a E^b`, jump straight to `F* = a/(a+b)·I/p_F`. Saves time; derive it once so you trust it.
- **Check the second good.** After finding one quantity, plug back into the budget line for the other, and **verify it satisfies the constraint** (`p_F F* + p_E E* = I`).
- **MRS = MU of the x-axis good over MU of the y-axis good.** Getting the ratio upside-down is the #1 error. If Food is on the x-axis, `MRS = MU_F/MU_E` and the tangency is `MU_F/MU_E = p_F/p_E`.
- **Corner solutions:** perfect substitutes → compare `MU_F/p_F` vs `MU_E/p_E` and spend everything on the winner. Perfect complements → ignore the price ratio, use the **fixed ratio + budget line.**
- **Comparative statics language:** income change = **parallel shift**; own-price change = **pivot**. Re-solving the optimum after a price change is how you **derive a demand curve** (Ch. 6).
- **Ordinal, not cardinal:** never say "twice as happy." Utility numbers only **rank.**
- **Behavioral:** be able to name the bias from a scenario (endowment effect, loss aversion, framing, self-control, overconfidence) and say it's a **systematic** (not random) deviation.

Bridge to Lecture 5 (Ch. 6 — Demand curves & elasticities). We now have a consumer who picks the tangency bundle for *given* prices and income. **Change one price and re-solve**, over and over, and the optimal quantities trace out a **demand curve** — closing the loop back to the demand we assumed in Lecture 2, and connecting to the **elasticities** of Lecture 3.

★ Practice problems (with solutions)

> A full-chapter set: preferences & axioms → utility & MU → MRS → budget line & MRT → the optimum → Cobb-Douglas shortcut → corner solutions → comparative statics → behavioral. Try each before opening the solution. Use **Food (F)** on the x-axis, **Entertainment (E)** on the y-axis throughout.

1. A consumer has bundles **A = (E 4, F 2)**, **B = (E 8, F 2)**, **C = (E 3, F 3)**, **D = (E 6, F 5)**. Using only "**more is better,**" which rankings can you establish for certain, and which need the indifference map? (Compare A–B, C–D, and A–C.)

2. Preferences satisfy completeness, transitivity, monotonicity. A student says: "*I prefer coffee to tea, tea to cocoa, but cocoa to coffee.*" Which **axiom** does this violate, and why is it a problem for building a utility function?

3. `U = √(EF)`. (a) Find the utility of **(E 2, F 8)**, **(E 4, F 4)**, **(E 8, F 2)**, **(E 5, F 5)**. (b) Which lie on the **same** indifference curve? (c) Rank all four bundles.

4. `U = F^{0.5} E^{0.5}`. (a) Derive `MU_F` and `MU_E`. (b) Evaluate both at **(F 4, E 9)**. (c) Does `MU_F` rise or fall as F increases (E fixed)? Name the principle.

5. For `U = √(EF)`, show that **MRS = E/F**. Then compute the MRS at **(E 2, F 8)** and at **(E 8, F 2)** and interpret the difference in one sentence.

6. Income **I = 40**, `p_F = 4`, `p_E = 2`. (a) Write the budget line as `E = …`. (b) Give both intercepts and the slope. (c) Is **(F 6, E 8)** affordable? (d) What is the **MRT**, and what does it mean?

7. Same as 6 (`I = 40`, `p_F = 4`, `p_E = 2`). (a) The government gives a **€20 income top-up**. What happens to the budget line? (b) Instead, `p_F` **falls to 2**. What happens? (c) Sketch the difference in words (shift vs pivot).

8. `U = √(EF)`, **I = 40**, `p_F = 4`, `p_E = 2`. Find the **optimal bundle** using `MRS = MRT` + the budget line. Verify it satisfies the constraint.

9. `U = F^{0.25} E^{0.75}` (Cobb-Douglas), **I = 60**, `p_F = 3`, `p_E = 6`. Use the **expenditure-share shortcut** to find `F*` and `E*`. What share of income goes to each good?

10. *(Perfect substitutes.)* `U = 2F + E`, **I = 24**, `p_F = 3`, `p_E = 2`. (a) Compute `MU_F/p_F` and `MU_E/p_E`. (b) What is the optimal bundle, and why is it a **corner solution**?

11. *(Perfect complements.)* `U = min(F, E)`, **I = 30**, `p_F = 2`, `p_E = 3`. Find the optimal bundle. *(Hint: the optimum sits at the corner where `F = E`; combine with the budget line.)*

12. *(Comparative statics.)* Start from problem 8's optimum (`U = √(EF)`, I=40, p_F=4, p_E=2). Now `p_F` **falls to €2**. (a) Find the new optimum. (b) Did the quantity of Food rise or fall? (c) You now have **two `(p_F, F*)` points** — what curve are you tracing out?

13. *(Behavioral.)* Name the bias in each: (a) You keep a concert ticket you'd never pay today's resale price for. (b) "1 in 3 patients survive" feels better than "2 in 3 die." (c) You put your alarm clock across the

room so you can't hit snooze. (d) 80% of new founders rate their success odds ≥ 7/10.

Solutions

1. A vs B: B has more E (8 vs 4), same F → **B > A** (more is better). **C vs D:** D has more of *both* (E 6>3, F 5>3) → **D > C. A vs C:** A has more E (4>3) but less F (2<3) → **cannot rank** by "more is better" alone; you need the **indifference map** (depends on the MRS).

2. Violates **transitivity.** coffee > tea and tea > cocoa should imply coffee > cocoa, but she says cocoa > coffee — the preferences **cycle.** A cycling ranking **can't be represented by a utility function** (no set of numbers can have coffee > tea > cocoa > coffee), so the whole model of choice breaks down.

3. (a) `U = √(EF)`: (E2,F8) → `√16 = 4`; (E4,F4) → `√16 = 4`; (E8,F2) → `√16 = 4`; (E5,F5) → `√25 = 5`. (b) The **first three** (all `E·F = 16`) lie on the **same** indifference curve (U=4). (c) **(E5,F5) > {the other three, which are indifferent to each other}** — i.e. `(5,5) > (2,8) ~ (4,4) ~ (8,2)`.

4. (a) `MU_F = ∂U/∂F = 0.5 F^{−0.5} E^{0.5} = 0.5√(E/F)`; `MU_E = 0.5 F^{0.5} E^{−0.5} = 0.5√(F/E)`. (b) At (F4,E9): `MU_F = 0.5√(9/4) = 0.5·1.5 = 0.75`; `MU_E = 0.5√(4/9) = 0.5·0.667 = 0.333`. (c) As **F rises**, `√(E/F)` falls → `MU_F` **falls** = **diminishing marginal utility.**

5. `MRS = MU_F/MU_E = [0.5√(E/F)] / [0.5√(F/E)] = √(E/F)·√(E/F) = E/F`. At **(E2,F8):** `MRS = 2/8 = 0.25`; at **(E8,F2):** `MRS = 8/2 = 4`. **Interpretation:** with lots of F she'll give up only 0.25 E for one more F; with little F she'll give up 4 E — the MRS **falls** as F rises (diminishing MRS / convex curve).

6. (a) `40 = 4F + 2E` → **`E = 20 − 2F`.** (b) **E-intercept = 20, F-intercept = 10, slope = −2.** (c) (F6,E8): `4·6 + 2·8 = 24 + 16 = 40` → **exactly affordable** (on the line). (d) **MRT = p_F/p_E = 4/2 = 2** → giving up **1 Food** frees €4, which buys **2 Entertainment.**

7. (a) **Income +€20** (`I = 60`): **parallel outward shift** — new intercepts E `= 60/2 = 30`, F `= 60/4 = 15`; **slope unchanged (−2).** (b) **`p_F` falls to 2:** budget **pivots** — E-intercept **unchanged at 20**, F-intercept rises to `40/2 = 20`, slope flattens to `−p_F/p_E = −1`. (c) Income change = **parallel shift** (same slope, more of everything); own-price change = **pivot** (one intercept fixed, slope changes).

8. `MRS = E/F` (from problem 5). Set `MRS = MRT`: `E/F = 4/2 = 2` → **`E = 2F`.** Budget: `4F + 2E = 40` → `4F + 2(2F) = 40` → `8F = 40` → **F* = 5, E* = 10.** *(Check: `4·5 + 2·10 = 20 + 20 = 40` ✓.)*

9. Shortcut for `U = F^{0.25}E^{0.75}` (a=0.25, b=0.75, a+b=1): `F* = 0.25 · 60/3 = 0.25·20 = **5**`; `E* = 0.75 · 60/6 = 0.75·10 = **7.5**`. **Shares: 25%** of income (€15) on Food, **75%** (€45) on Entertainment. *(Check: `3·5 + 6·7.5 = 15 + 45 = 60` ✓.)*

10. `U = 2F + E`: `MU_F = 2`, `MU_E = 1`. (a) **`MU_F/p_F = 2/3 ≈ 0.67`**; **`MU_E/p_E = 1/2 = 0.5`.** (b) Food gives **more utility per euro** (0.67 > 0.5) and the rate is **constant** (perfect substitutes), so spend **everything on Food**: `F* = I/p_F = 24/3 = **8**, E* = 0`. **Corner solution** because `MU_F/p_F ≠ MU_E/p_E` *everywhere* — there's no interior tangency; the highest indifference (straight) line touches the budget line **on the F-axis.**

11. `U = min(F,E)`: no point in having more of one than the other, so the optimum has **`F = E`.** Budget: `2F + 3E = 30` with `E = F` → `2F + 3F = 30` → `5F = 30` → **F* = E* = 6.** *(Check: `2·6 + 3·6 = 12 + 18 = 30` ✓.) (Note: the price ratio is irrelevant to the ratio of goods — only to how much of the fixed bundle you can afford.)*

12. Now `p_F = 2`, `p_E = 2`, I = 40. `MRS = E/F = p_F/p_E = 2/2 = 1` → **`E = F`**. Budget: `2F + 2E = 40` with `E = F` → `4F = 40` → **F* = 10, E* = 10.** (b) Food **rose from 5 to 10** as its price fell 4→2 (a downward-sloping demand response). (c) The two points **(p_F=4, F=5)** and **(p_F=2, F=10)** are two points on the consumer's **demand curve for Food.**

13. (a) **Endowment effect / sunk-cost + status-quo bias** (you value it more because you own it). (b) **Framing effect** (same facts, "survive" vs "die"). (c) **Lack of self-control** — a commitment device (planner constrains the future doer). (d) **Overconfidence.**

★ Fun facts & memorable details (Lecture 4)

> Sticky bits from Consumer Choice.

- **"More is better" has a loophole:** economists just **redefine bads as the absence of the bad** — you don't demand pollution, you demand *clean air.* A neat trick that keeps monotonicity intact.
- **Utility is unobservable — so numbers lie.** `U = 8` is not "twice as good" as `U = 4`; utility is purely **ordinal.** Early economists agonised over this; the modern answer is "we only ever need the ranking."
- **The bang-for-your-buck rule is the whole optimum.** `MU_F/p_F = MU_E/p_E` — at the best bundle, the **last euro** spent on each good buys the **same** extra utility. If it didn't, you'd move money to the better-value good. That's *all* tangency means.
- **Cobb-Douglas people are creatures of habit:** with `U = F^a E^b` you always spend the **same share** of income on each good — no matter how rich you get or how prices move. That's why it's the field's favourite functional form.
- **Left shoes are worthless alone.** Perfect complements (L-shaped curves) are why a single earring, one glove, or a lone right shoe adds **zero** utility — you need the matching unit.
- **The ultimatum game breaks "Homo economicus."** Offered a lopsided split, people **pay real money to punish unfairness** — and this holds from Pittsburgh to Papua New Guinea. Fairness is not a Western quirk.
- **You'd sell your coffee mug for far more than you'd pay for it.** The **endowment effect**: Cornell students valued a randomly-given mug **2–4× higher** than students who didn't get one. Ownership rewires valuation.
- **The paradox of voting:** a strict cost-benefit reading says *don't vote* (your vote almost never decides the election). That people vote anyway is a clue that **narrow self-interest is too small a theory of us.**
- **Ulysses tied himself to the mast** to resist the Sirens — a 3,000-year-old **commitment device**, the same logic as not keeping cigarettes in the house. Self-control problems are ancient.

Lecture 5: Applying Consumer Theory (Chapter 5 · §6.5)

Required reading: Friberg, **Chapter 5 — Consumer choice** (applied) · **§6.5 — Income and substitution effects.** *Lecturer: L. Butera — deck "**Lecture 5: Applying consumer theory.**" (This lecture **uses** the machinery built in Lecture 4 — the optimum where **MRS = price ratio** — and puts it to work.)*

Résumé. Lecture 4 found the **best bundle for *given* prices and income.** Lecture 5 *applies* that result three ways: **(1)** it **derives the demand curve** — change one price, re-solve, repeat, and the optimal quantities trace out demand (closing the loop back to Lecture 2); **(2)** it **solves the optimum with calculus** (the substitution and Lagrangian methods) to get demand *functions* q(p, Y), and introduces the **dual** problem (expenditure minimization); **(3)** it opens the "black box" of a price change with **income and substitution effects** — the deeper decomposition that explains *inferior* and *Giffen* goods and the puzzle of **labour supply.**

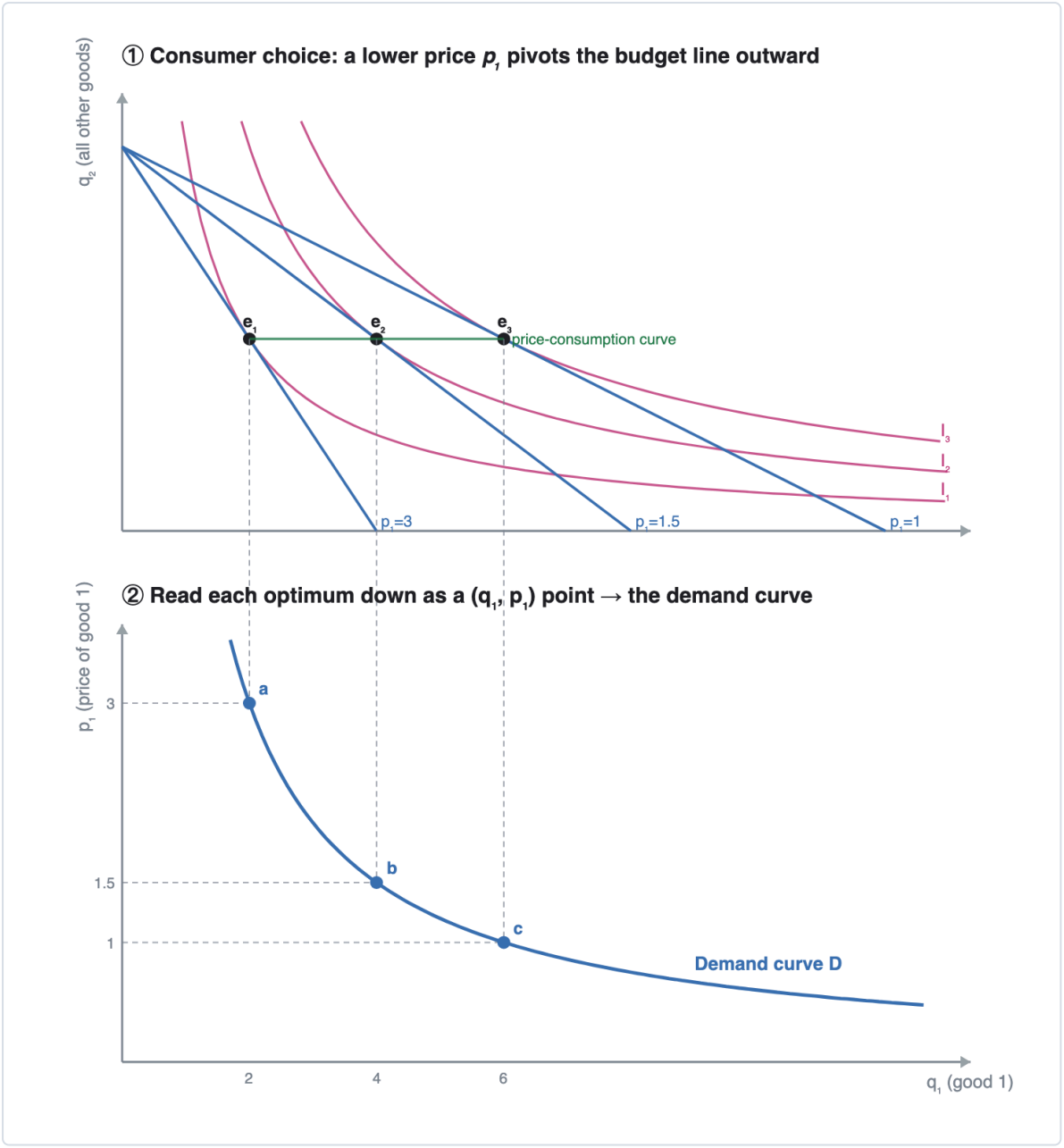
> ★ **The heart of the lecture:** a price change moves demand for **two** reasons, and separating them explains almost everything. The **substitution effect** (the good is now relatively dearer → switch away, **always** in the "less" direction) and the **income effect** (the price rise makes you effectively poorer → buy less if the good is **normal**, *more* if **inferior**). **Total effect = substitution effect + income effect.** When a strong, positive income effect on an inferior good *overwhelms* the substitution effect, demand slopes **up** — a **Giffen good**. The same tool cracks the wage puzzle: a raise makes you work *more* (substitution) but also lets you afford more leisure (income).

Key concepts / "modes" to use: *deriving demand from the optimum · the substitution method & the Lagrangian (`MU₁/p₁ = MU₂/p₂ = λ`) · demand functions $q(p_1,p_2,Y)$ · the dual / expenditure minimization / Hicksian (compensated) demand · substitution effect vs income effect (Slutsky/Hicks decomposition) · normal · inferior · Giffen · the labour–leisure model & backward-bending supply.*

★ Deriving the demand curve from consumer choice

Core: we never really *assumed* the demand curve of Lecture 2 — we can **derive** it. Hold **preferences, income and other prices constant**, then **change the good's own price and re-solve the optimum.** The resulting optimal quantities, plotted against price, **are** the demand curve.

- **The recipe:** at each price `p₁`, find the tangency bundle (`MRS = p₁/p₂`). Lower `p₁` → the budget line swings out → a **new, higher** optimal `q₁`. Collect the (`p₁`, `q₁`) pairs → the **individual demand curve** for good 1.



How to read it (building demand, step by step). ① **(top)** start from the consumer's optimum. Every time `p₁` **falls**, the budget line **pivots out** along the horizontal axis (same vertical intercept `Y/p₂`), and the tangency slides to a **new optimum** with **more** `q₁`: `e₁ → e₂ → e₃`. ② **(bottom)** each optimum hands you one **(quantity, price)** pair — drop a dashed line down from `e₁` to plot point `a = (q₁ 2, p₁ 3)`, from `e₂` to `b = (4, 1.5)`, from `e₃` to `c = (6, 1)`. **Join a, b, c → the demand curve `D`.** That's the whole trick: *the demand curve is just the consumer's optimum, re-solved and re-plotted for every price. (This example is Cobb–Douglas `U = q₁q₂`, so demand is `q₁* = Y/(2p₁)` — the smooth, unit-elastic curve shown.)*

- **Movement vs shift (ties back to Ch. 2):** changing the good's **own price** traces a **movement *along*** its demand curve; changing **income** or **another price** re-solves to a different curve — a **shift** of demand. *In plain terms:* the demand curve is just "the optimum, redrawn for every possible own-price."
- **Why this matters:** it makes demand a *result*, not an assumption — and it means the **shape** of demand (its elasticity, whether it can slope up) is inherited from the **shape of preferences** (the indifference curves).

Solving the optimum analytically — two methods

Core: the graphical tangency `MRS = p₁/p₂` can be solved with **calculus** to get the **demand functions** `q₁*(p₁,p₂,Y)` and `q₂*(p₁,p₂,Y)` explicitly. Two equivalent routes.

(a) The substitution method

Core: turn the constrained problem into an **unconstrained** one by substituting the budget line into utility.

1. **Budget:** `p₁q₁ + p₂q₂ = Y` → solve for one good: `q₂ = (Y − p₁q₁)/p₂`.
2. **Substitute** into `U(q₁, q₂)` → utility is now a function of `q₁` alone.
3. **Maximize:** set `dU/dq₁ = 0`. This reproduces the tangency condition **`MU₁/MU₂ = p₁/p₂`** and, with the budget line, yields `q₁*` and `q₂*` as functions of `(p₁, p₂, Y)`.

*(Requires the utility function to be **quasi-concave** — i.e. indifference curves **convex to the origin** — so the tangency is a maximum, not a minimum.)*

(b) The Lagrangian method

Core: the general tool for constrained optimization (met in L4's appendix). Build a function that bolts the constraint onto the objective with a multiplier **λ**.

$$\mathcal{L} = U(q_1, q_2) + \lambda\,(Y - p_1 q_1 - p_2 q_2)$$

- **First-order conditions** (set each partial derivative to 0):
  - `∂ℒ/∂q₁ = MU₁ − λp₁ = 0`
  - `∂ℒ/∂q₂ = MU₂ − λp₂ = 0`
  - `∂ℒ/∂λ = Y − p₁q₁ − p₂q₂ = 0` *(the budget, recovered)*
- **Rearrange the first two:** **`MU₁/p₁ = MU₂/p₂ = λ`** — the "*equal marginal utility per euro*" rule: at the optimum the **last euro spent on each good buys the same extra utility.** Dividing them gives back **`MU₁/MU₂ = p₁/p₂`** (MRS = price ratio).
- **What is λ?** the **marginal utility of income** — how much extra utility one more euro of budget would buy. *(A "shadow price" on the budget constraint.)*
- Solve the three equations together → the **demand functions.** *(Worked for Cobb–Douglas in the practice set — you get the tidy result that a Cobb–Douglas consumer spends **fixed budget shares** on each good.)*

The dual problem — expenditure minimization

Core: utility maximization has a **mirror image.** Instead of "*maximise utility for a given budget,*" ask "***minimise the spending** needed to reach a given utility* `Ū`." Same tangency, approached from the other side.

$$\min\ p_1 q_1 + p_2 q_2 \quad \text{subject to} \quad U(q_1, q_2) = \bar{U}$$

- **The solution is the *expenditure function*** **`E(p₁, p₂, Ū)`** — the **least money** needed to achieve utility `Ū` at prices `(p₁, p₂)`.

- **It gives the *Hicksian* (compensated) demands** — quantities as functions of prices holding **utility** constant (rather than income). *This is exactly the object the **substitution effect** uses below* — it asks how you'd re-shop if you had to stay on the **same indifference curve** but faced new relative prices.
- **Duality (the payoff):** max-utility-given-budget and min-budget-given-utility describe the **same tangency point** — two doors into the same room. The primal gives ordinary (Marshallian) demand `q(p,Y)`; the dual gives compensated (Hicksian) demand `q(p,Ū)`.

★ Income and substitution effects (§6.5)

Core: the deeper "behind-the-scenes" reading of a price change. When p_1 rises, quantity demanded moves for **two distinct reasons**, and the total is their sum:

Total effect = substitution effect + income effect

- **Substitution effect** — the change in quantity from the shift in **relative prices *alone*, holding utility constant.** If p_1 rises, good 1 is now **relatively dearer**, so you switch toward the (now cheaper) other good. **Always negative (or zero):** a price rise ⇒ substitution *away*. *(Formally, this is the move along the **original indifference curve** — the Hicksian/compensated response from the dual.)*
- **Income effect** — the change in quantity because the price rise makes you **effectively poorer** (your income now buys less). Sign depends on the good:
  - **normal good** → you buy **less** when poorer ⇒ income effect is **negative** (reinforces substitution);
  - **inferior good** → you buy **more** when poorer ⇒ income effect is **positive** (opposes substitution).

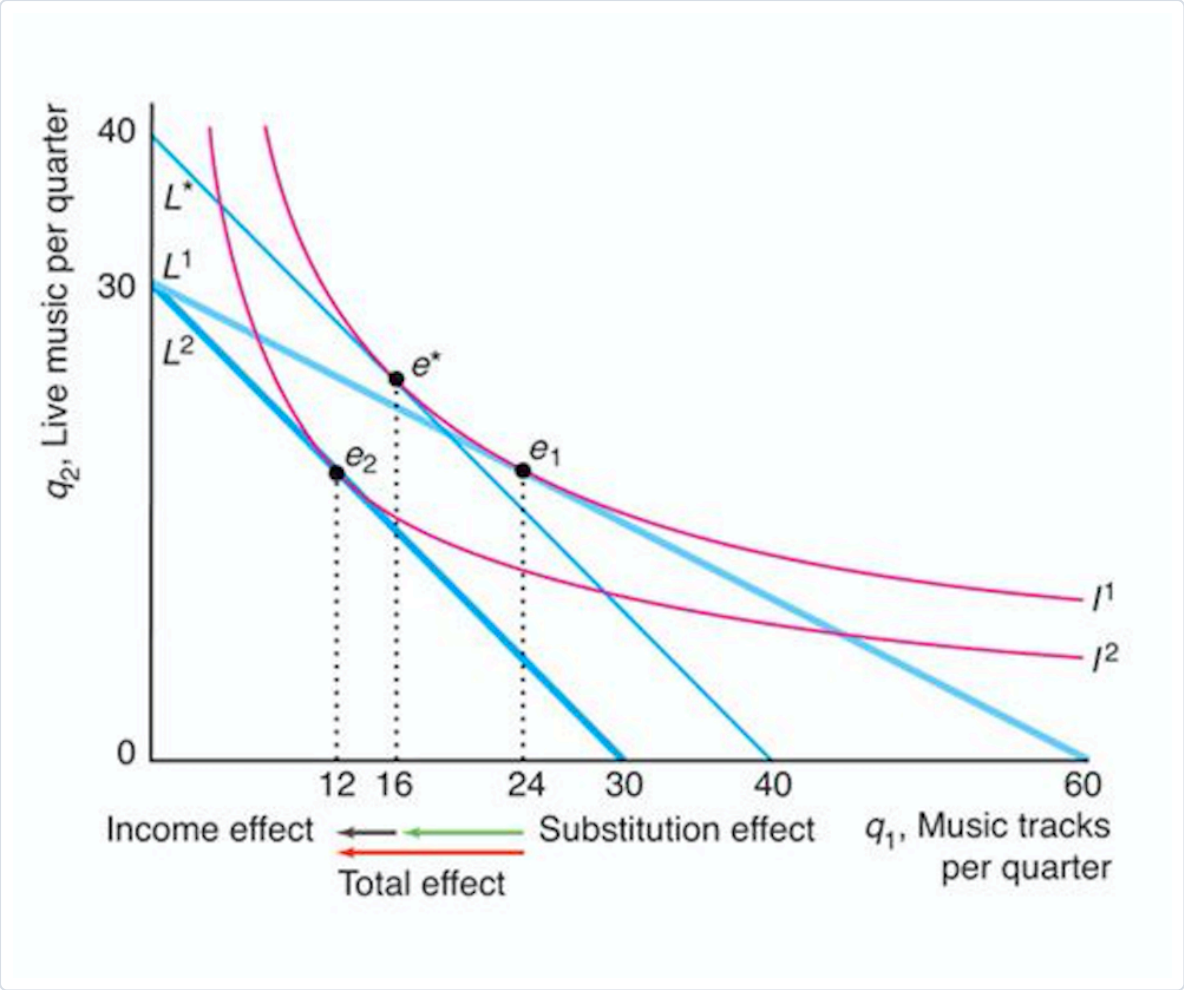
How the decomposition works (the three-step graph)

Core: compensate, then take the compensation away.

1. **Start** at optimum $e*$ on indifference curve I^1 with budget line L^1.
2. **Raise p_1** → budget line pivots inward and **steeper** (L^2) → new optimum e_2 on a lower curve I^2. The move $e* \to e_2$ is the **total effect.**
3. **Isolate substitution:** slide a budget line with the **new (steeper) slope** out until it is **tangent to the *original* curve I^1** (this "$L*$" imagines you were **compensated** to stay as happy as before). Its tangency e_1 gives the **substitution effect** $e* \to e_1$ (a slide *along* I^1).
4. **Income effect** is the leftover **parallel shift** $L* \to L^2$, i.e. $e_1 \to e_2$ — the pure loss of purchasing power.

A normal good — the two effects reinforce

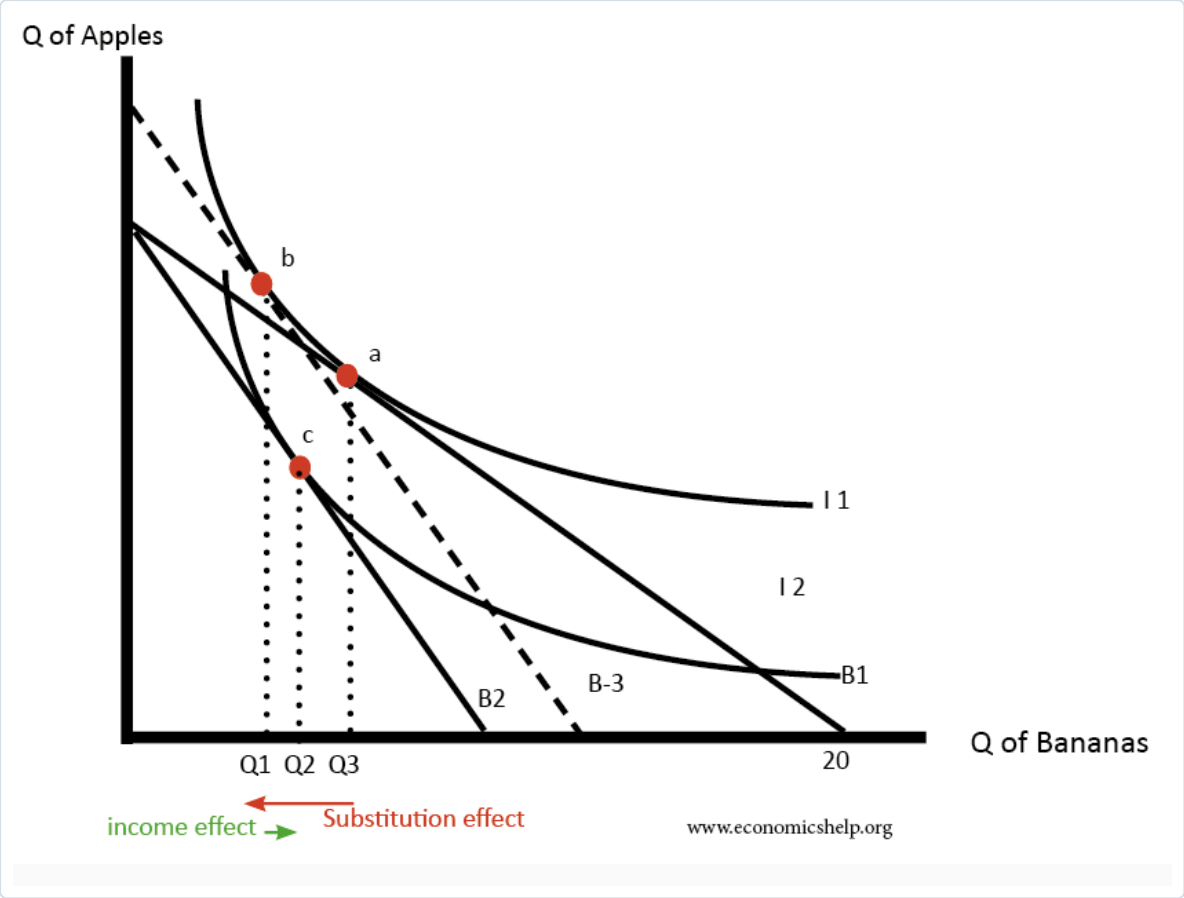
Core: for a **normal** good both effects push the **same way**, so demand falls **noticeably** when price rises.



How to read it. `e∗` is the original optimum. When the price of music tracks rises, the budget line pivots to `L²` and the optimum falls to `e₂`. The compensated line `L∗` (new slope, original indifference curve `I¹`) locates `e₁`: *substitution effect* = $e \to e_1^*$ (24→16), **income effect = $e_1 \to e_2$** (16→12). For a normal good the income arrow points the **same way** as the substitution arrow → the two **add up** to a large total fall (24→12).

An inferior good — the effects oppose

Core: for an **inferior** good the positive income effect **partly cancels** the substitution effect, so demand is **less responsive** to price.



How to read it. The substitution effect still pushes the good **down** (`a→b`), but because the good is **inferior**, being poorer makes you want **more** of it — the income arrow points **back the other way**, offsetting part of the substitution move. **Net:** demand still falls, but by **less** than for a normal good. "*The more inferior the good (the more negative its income elasticity) and the fewer the substitutes, the less demand shrinks when price rises.*"

Giffen goods — when demand slopes *up*

Core: the rare, dramatic case. A **Giffen good** is an **inferior** good whose **income effect is so strong it *dominates*** the substitution effect → **quantity demanded *rises* when price rises** → an **upward-sloping** demand curve, violating the law of demand.

- **The recipe for a Giffen good:** (1) strongly **inferior**; (2) takes up a **large share** of the budget; (3) has **few substitutes.** Then a price rise makes the household so much poorer that it **cuts the "luxury"** it used to afford and buys **even more** of the cheap staple.
- **The classic examples: potatoes in Ireland** in the 1840s (the origin of the name — still contested) and, more convincingly, **rice among poor households in China** (Jensen & Miller, 2008): when rice got dearer, families could no longer afford *any* meat, so they bought **more** rice to fill up. → "*They can only afford rice.*"
- **Why it's the exception, not the rule:** it needs a strong income effect (big budget share) *and* strong inferiority — rare in rich economies. The substitution effect usually wins, which is why demand almost always slopes down.

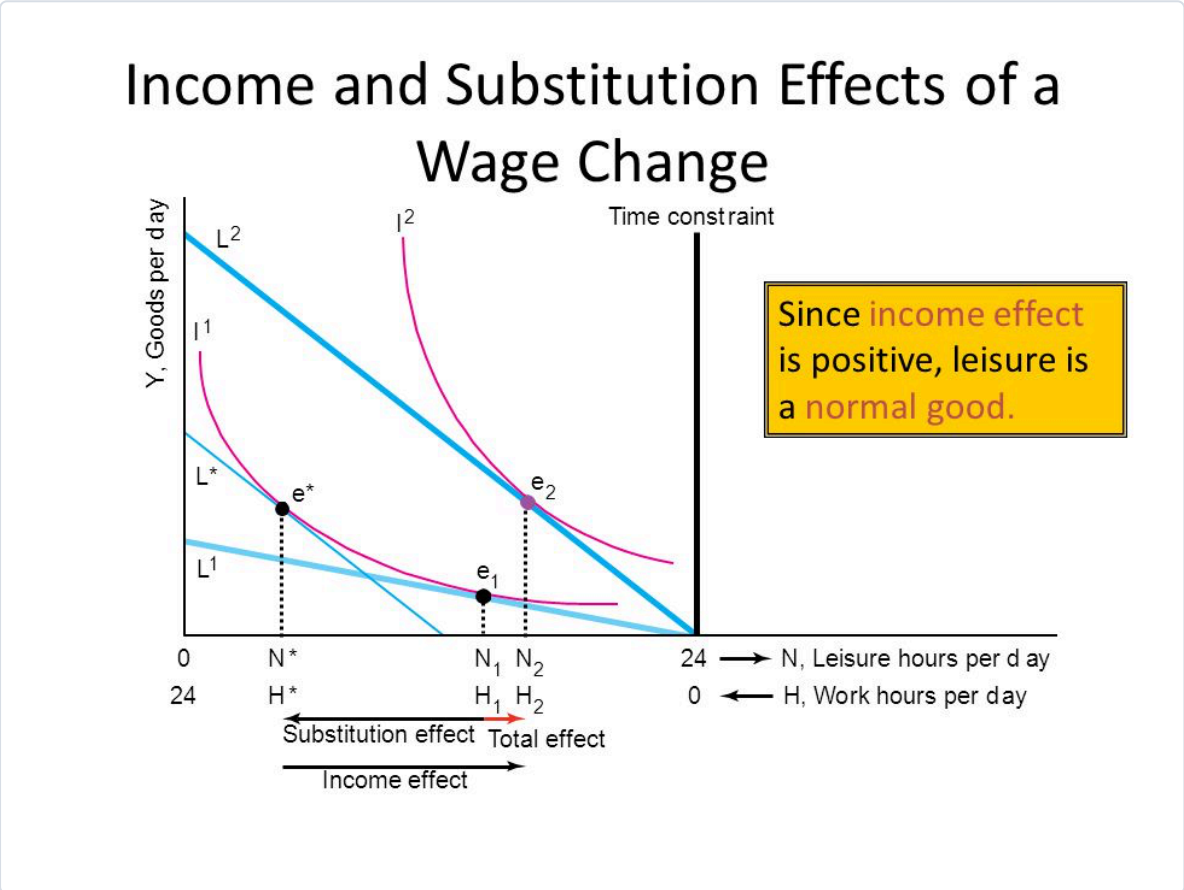
*(Aside — **Hicks vs Slutsky:** two ways to "compensate" for the substitution effect. **Hicks** keeps you on the **original indifference curve** (same utility); **Slutsky** keeps the **original basket just affordable** (rotate the budget line around*

the starting point). Hicks is the standard version; both give the same intuition.)

★ Application: labour supply (the wage puzzle)

Core: the decomposition cracks a real puzzle — *does a pay rise make you work more or less?* Model the worker choosing between **leisure** and **consumption (income)**. A wage is the **price of leisure** (an hour off costs you one hour's wage). So **a wage rise = leisure becomes more expensive** → apply income & substitution effects to *leisure*.

- **Substitution effect:** leisure is now dearer → consume **less** leisure → **work more.**
- **Income effect:** a higher wage makes you **richer**; if **leisure is a normal good**, you want **more** of it → **work less.**
- **Net effect is ambiguous** — it depends on which dominates.



How to read it. The horizontal axis is **leisure hours** `N` (read right-to-left, it's **work hours** `H`); the vertical axis is **goods/income**. A wage change tilts the budget line. Splitting `e*→e₂` at the compensated point `e₁` : the **substitution effect** cuts leisure (more work), the **income effect** adds leisure (less work — "*since the income effect is positive, leisure is a normal good*"). Because they pull **opposite ways**, the **backward-bending labour-supply curve** becomes possible: at **low** wages the substitution effect usually wins (higher wage → work more); at **high** wages the income effect can win (higher wage → work *less*).

- **The deck's puzzle made concrete:** a raise from **200 → 300 DKK/hr** probably makes you **work more** (substitution dominates); a raise from **200 → 4000 DKK/hr** might make you **work less** — you're now rich enough to "buy back" your time (income dominates).

Formula sheet (Lecture 5)

Tool	Formula
Tangency (optimum)	$MRS = MU_1/MU_2 = p_1/p_2$
Budget constraint	$p_1q_1 + p_2q_2 = Y$
Lagrangian	$\mathcal{L} = U(q_1,q_2) + \lambda(Y - p_1q_1 - p_2q_2)$
Lagrangian FOCs	$MU_1 = \lambda p_1$, $MU_2 = \lambda p_2$ $\rightarrow$ **$MU_1/p_1 = MU_2/p_2 = \lambda$**
λ interpretation	**marginal utility of income** (shadow price of the budget)
Expenditure function (dual)	$E(p_1,p_2,\bar{U})$ = min spending to reach utility $\bar{U}$
Slutsky decomposition	**total effect = substitution effect + income effect**
Substitution effect	**≤ 0 always** (holding utility constant)
Income effect (price ↑)	**normal:** negative · **inferior:** positive · **Giffen:** positive & dominant
Cobb–Douglas $U = q_1^a q_2^b$ demand	$q_1* = a/(a+b) \cdot Y/p_1$ (constant budget shares)

★ Practice problems (with solutions)

> Do these — Lecture 5 is where the calculus of consumer choice actually gets tested. Solutions below.

1. A consumer has $U(q_1,q_2) = q_1q_2$, income Y, prices p_1, p_2. Use the **substitution method** to find the demand for q_1.

2. Same consumer, now use the **Lagrangian** method to find q_1*, q_2*, and interpret **λ**.

3. Show that with **Cobb–Douglas** $U = q_1^a q_2^b$, the consumer spends a **constant share** of income on each good. What is the own-price elasticity of the demand $q_1* = [a/(a+b)]\cdot(Y/p_1)$?

4. The price of good 1 (a **normal** good) rises. On an indifference-curve diagram, identify the **substitution** and **income** effects, and say which direction each pushes q_1.

5. Classify the income effect (sign, for a **price rise**) and say what happens to total demand: (a) a normal good; (b) an inferior good; (c) a Giffen good.

6. Your hourly wage doubles. Using income & substitution effects, explain why you might **work more** *or* **less**. Which effect is the "substitution" one?

7. *(Concept.)* What does the **expenditure function** $E(p_1,p_2,\bar{U})$ measure, and how does it relate to the **substitution effect**?

Solutions

1. Budget: $q_2 = (Y - p_1q_1)/p_2$. Substitute: $U = q_1 \cdot (Y - p_1q_1)/p_2$. Differentiate wrt q_1 and set to 0: $dU/dq_1 = (Y - 2p_1q_1)/p_2 = 0 \rightarrow Y - 2p_1q_1 = 0 \rightarrow$

$$q1^* = Y / (2p1)$$

(and by symmetry `$q_2^* = Y/(2p_2)$`). The consumer spends **half** the budget on each good.

2. $\mathcal{L} = q_1q_2 + \lambda(Y - p_1q_1 - p_2q_2)$. FOCs: $q_2 = \lambda p_1$ and $q_1 = \lambda p_2$ and the budget. Divide the first two: $q_2/q_1 = p_1/p_2 \rightarrow p_1q_1 = p_2q_2$. Put into the budget $p_1q_1 + p_2q_2 = Y \rightarrow 2p_1q_1 = Y \rightarrow$ **$q_1^* = Y/(2p_1)$**, **$q_2^* = Y/(2p_2)$** (same as Q1 ✓). $\lambda = q_2/p_1 = Y/(2p_1p_2)$ — the **marginal utility of an extra euro** of income.

3. Lagrangian FOCs give $MU_1/MU_2 = (a\ q_1^{a-1}q_2^b)/(b\ q_1^a q_2^{b-1}) = (a/b)(q_2/q_1) = p_1/p_2 \rightarrow p_1q_1 = (a/b)p_2q_2$. Sub into the budget $\rightarrow p_1q_1 = [a/(a+b)] \cdot Y$. So **spending on good 1 is the fixed share $a/(a+b)$ of income** (independent of prices) $\rightarrow q_1^* = [a/(a+b)] \cdot (Y/p_1)$. **Own-price elasticity** $= (dq_1/dp_1)(p_1/q_1) = -[a/(a+b)]Y \cdot p_1^{-2} \cdot p_1/q_1 = -1$: Cobb–Douglas demand is **unit-elastic** (which is exactly why expenditure share is constant — recall Lecture 3: unit elasticity ⇒ spending unchanged).

4. Original optimum e^* on I^1. Price ↑ → new optimum e_2 on a lower I^2 (total effect, q_1 falls). Compensated line (new slope, tangent to I^1) gives e_1: **substitution effect $e^* \rightarrow e_1$** pushes q_1 **down** (dearer relative price); **income effect $e_1 \rightarrow e_2$** also pushes q_1 **down** (poorer, and it's normal). Both **negative** → they **reinforce**. *(This is the normal-good figure.)*

5. For a **price rise**: (a) **normal** → income effect **negative**; total demand **falls** (both effects down). (b) **inferior** → income effect **positive**, partly offsets substitution; total demand **falls, but less.** (c) **Giffen** → income effect **positive and dominant**; total demand **rises** (law of demand violated).

6. A wage rise makes leisure **more expensive** (its price = the wage). **Substitution effect:** leisure dearer → take less leisure → **work more. Income effect:** you're richer and leisure is normal → want more leisure → **work less.** The **substitution** effect is the "leisure-is-now-dearer" one; whether you net work more or less depends on which dominates (→ backward-bending supply).

7. $E(p_1,p_2,\bar{U})$ = the **minimum spending needed to reach utility $\bar{U}$** at given prices (the dual). Its quantities are the **Hicksian/compensated demands** — precisely what the **substitution effect** uses, since that effect asks how you re-optimise when **utility is held fixed** and only **relative prices** change.

★ Fun facts & memorable details (Lecture 5)

> Sticky bits from applying consumer theory.

- **The demand curve is a *result*, not an assumption.** Wiggle one price, re-solve the optimum over and over, and demand **draws itself.** Its whole shape is inherited from the curvature of your indifference curves.

- **λ has a name: the marginal utility of money.** The Lagrange multiplier isn't just algebra — it tells you **how much happier one extra euro would make you.**
- **Giffen goods really exist — barely.** For 150 years the **Irish-potato** story was economists' only (contested) example. Then **Jensen & Miller (2008)** found it for real: subsidise rice for poor Chinese families and they bought *less* — because they could finally afford some meat.
- "**They can only afford rice.**" The heart of the Giffen paradox: when your staple gets pricier, you're too poor to buy anything *but* the staple — so you buy **more** of it.
- **A pay rise can make you work less.** The **backward-bending labour supply curve**: past some wage, you're rich enough to "buy back" your own time. Lottery winners who quit their jobs are the income effect made visible.
- **Every price change is secretly two changes.** Splitting the substitution effect from the income effect is one of the most useful moves in all of microeconomics — it's why we can tell a normal good from an inferior one from a Giffen good just by signing an arrow.

Lecture 6: Firms and Production (Chapters 3–4)

Required reading: Friberg, **Ch. 3 — From technology to costs** (production functions, costs, isoquants & cost minimization) · **Ch. 4 — Supply by price-taking firms** (profit, MR = MC, shutdown, long-run supply). *Lecturer: L. Butera — deck "**Lecture 6: Production.**" (The lecture is titled "Chapter 4 / Firms & production"; the production-and-cost machinery is actually Friberg **Ch. 3**, and the profit/supply half is **Ch. 4** — this note covers both.)*

Résumé. Lectures 4–5 were the **consumer**; now it's the **firm.** The firm is the mirror image of the consumer: where the consumer maximised **utility** subject to a **budget**, the firm **minimises cost** subject to a **production function**, and then **maximises profit.** The lecture runs the whole chain **technology → cost → supply**: build the **production function** Q = f(K, L); derive **short-run product** (marginal product, diminishing returns) and **cost curves** (MC, ATC, AVC, AFC); add **revenue** to get the **profit-maximising rule MR = MC = p** and the firm's **supply curve**; add the **shutdown rule**; let **free entry** drive the **long run** to **zero profit**; and finally solve the **optimal input mix** (L*, K*) with **isoquants and isocosts** (the exact analogue of the consumer's indifference-curve optimum).

> ★ **The heart of the lecture — the firm as a two-stage optimiser.**
> • **Stage 1 — cost minimisation (how to produce):** for any target output, pick the input mix where the **isoquant is tangent to the lowest isocost** — **MRTS = w/r**, i.e. the **last-dollar rule** $MP_L/w = MP_K/r$. This gives the **cost function C(q).**
> • **Stage 2 — profit maximisation (how much to produce):** produce where **marginal revenue = marginal cost.** A **price-taking** firm has MR = p, so the rule is **p = MC** — and the firm's **supply curve is its MC curve** (above min AVC).
> • **The time dimension:** in the **short run** capital is **fixed** (so there are fixed costs, and a shutdown decision); in the **long run** everything is variable and **free entry drives economic profit to zero** (p = MC = min ATC).

Key concepts / "modes" to use: *perfect competition & the price taker · production function $Q = f(K,L)$; Cobb-Douglas $Q = \sqrt{(KL)}$ · short run vs long run (K fixed vs all variable) · marginal product $MP_L = \partial Q/\partial L$; law of diminishing marginal returns · returns to scale (CRS · IRS · DRS); the exponent-sum shortcut · FC · VC · TC; ATC · AVC · AFC;* ***MC = dTC/dq****; MC cuts ATC & AVC at their minima · $TR = pq$; $\pi = TR - TC$; $MR = dTR/dq$ · the output rule* ***MR = MC = p****; the firm's supply curve · the shutdown rule ($p <$ min AVC) · free entry → long-run zero economic profit; opportunity cost & economic rent · isoquants · $MRTS = MP_L/MP_K$ · isocost $wL + rK = C$ · the tangency / last-dollar rule.*

The firm, and the perfectly-competitive benchmark

Core: the whole lecture lives in **perfect competition**, whose assumptions make each firm a **price taker** — it can sell as much as it likes at the market price **p**, but nothing above it.

- **The five assumptions: (1)** a very large number of firms (no one firm moves the price); **(2) identical (homogeneous)** products; **(3) complete, symmetric information**; **(4) negligible transaction costs**; **(5) free entry and exit** (in the long run).
- **The consequence — the firm's demand curve is *horizontal* at p.** Raise your price a cent and you sell **zero** (buyers switch to an identical rival); so the firm **takes p as given.** *(Why nearly flat and not perfectly flat? The firm faces the market's residual demand; with many rivals that residual is hugely elastic — soybean example: market demand is only moderately elastic, yet one farm faces a "super-elastic," almost flat demand.)*
- **The firm** = the decision-maker that chooses **what** and **how** to produce; modelled as a single actor with one objective — **maximise profit** (and, as a sub-step, **minimise cost**).

Inputs and the production function

Core: production turns **inputs** into **output.** We simplify the firm's many inputs to two: **capital K** and **labour L.**

- **Factors of production: labour** (human time/effort), **capital** (things *used but not used up* — buildings, machines, servers; includes **intangible** capital like patents), plus **land** and **raw/intermediate materials.**
- **The production function** gives the maximum output from given inputs:

  Q = f(K, L).

  The lecture's running form is **Cobb-Douglas**, e.g. **Q = √(KL) = K^½ L^½.** It can also be written as a **table** (output for each K-row × L-column).

Short run vs long run (timing matters)

Core: the distinction is about **what can be varied**, not the calendar.

	Short run (SR)	Long run (LR)
Inputs	**at least one is fixed** — we assume **capital K is fixed**	**all inputs variable**
Costs	**fixed costs exist** (rent, machines)	**all costs variable**
Example	a sandwich shop can hire an extra pair of hands fast, but **can't add an oven** overnight	it can build a whole new sandwich station

In plain terms: the SR is "stuck with your factory"; the LR is "you can rebuild the factory (or leave the industry)."

★ Short-run production: marginal product & diminishing returns

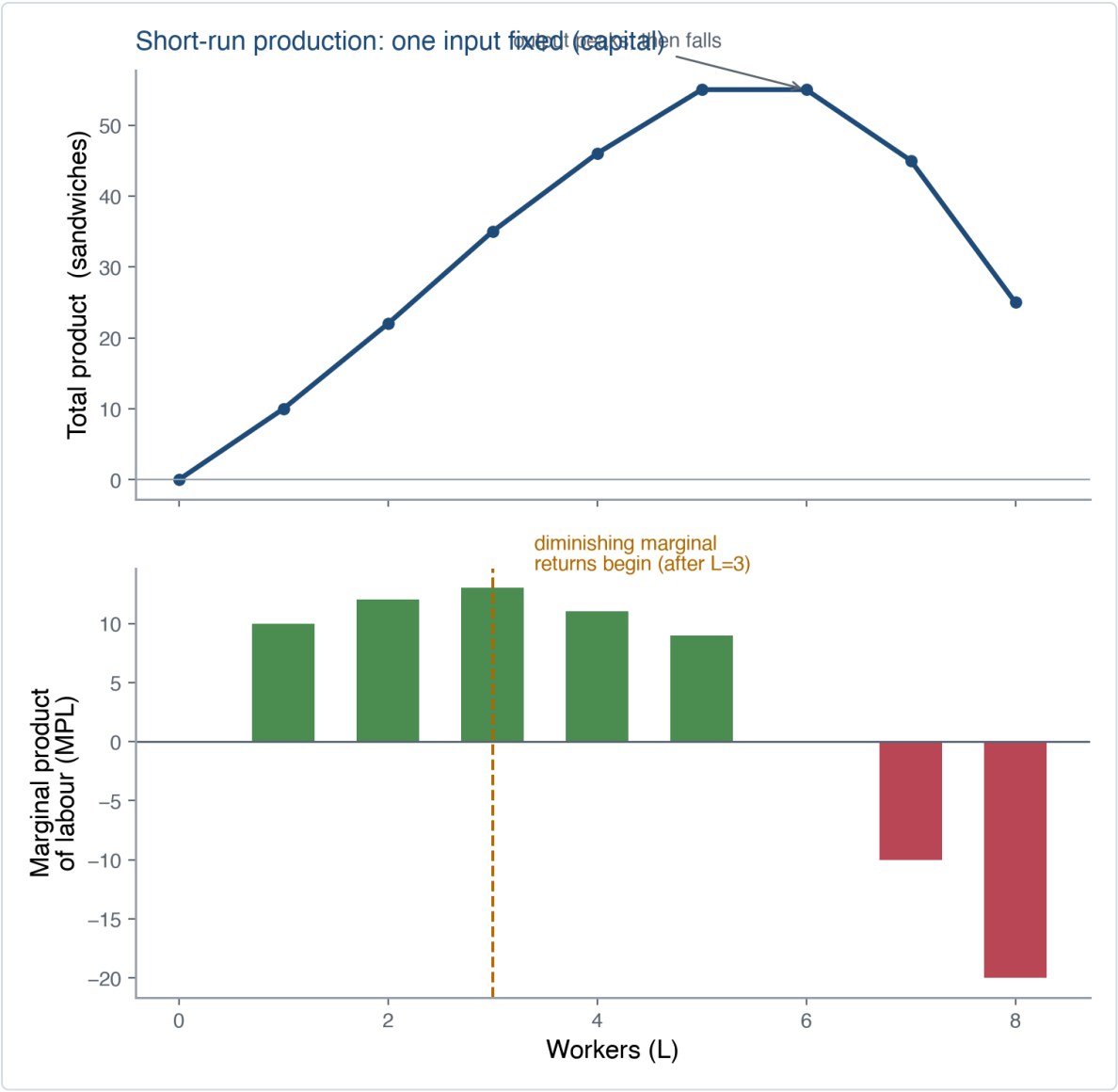
Core: with **K fixed**, keep adding **labour** and output rises — but each extra worker eventually adds **less** than the last.

- **Marginal product of labour (MP_L)** = the extra output from one more unit of labour, holding K fixed = the **partial derivative**:

  $MP_L = \partial Q/\partial L$. For $Q = \sqrt{(KL)}$: **$MP_L = \frac{1}{2}\sqrt{(K/L)}$.**

- **The law of diminishing marginal returns:** beyond some point, adding more of a **variable** input to **fixed** inputs makes **MP_L fall.** (Too many cooks crowd the one kitchen.)

- **Crucial distinction — diminishing *marginal* product ≠ *falling* output.** Total output is still **rising**; it is the **increments** that shrink (and only once MP_L goes **negative** does total output actually fall).



How to read it. **Top:** total product climbs steeply at first, then flattens and finally **turns down** once extra workers get in each other's way. **Bottom:** the **marginal** product (one worker's *addition*) **rises** through the first three workers, then **diminishes** (the amber line), and goes **negative** past the point where total product peaks. *The onset of diminishing marginal returns is where the marginal-product bars start shrinking — here, after the 3rd worker.*

Returns to scale (a long-run, all-inputs question)

Core: scale **all** inputs up by the same factor **α** and ask what happens to **output.**

	Definition (double all inputs ⇒ …)	Formula	Why
Constant returns (CRS)	output **doubles**	f(αK, αL) = α·f(K,L)	replication
Increasing returns (IRS)	output **more than doubles**	f(αK, αL) > α·f(K,L)	specialisation; one big plant beats two small
Decreasing returns (DRS)	output **less than doubles**	f(αK, αL) < α·f(K,L)	coordination gets harder as the firm grows

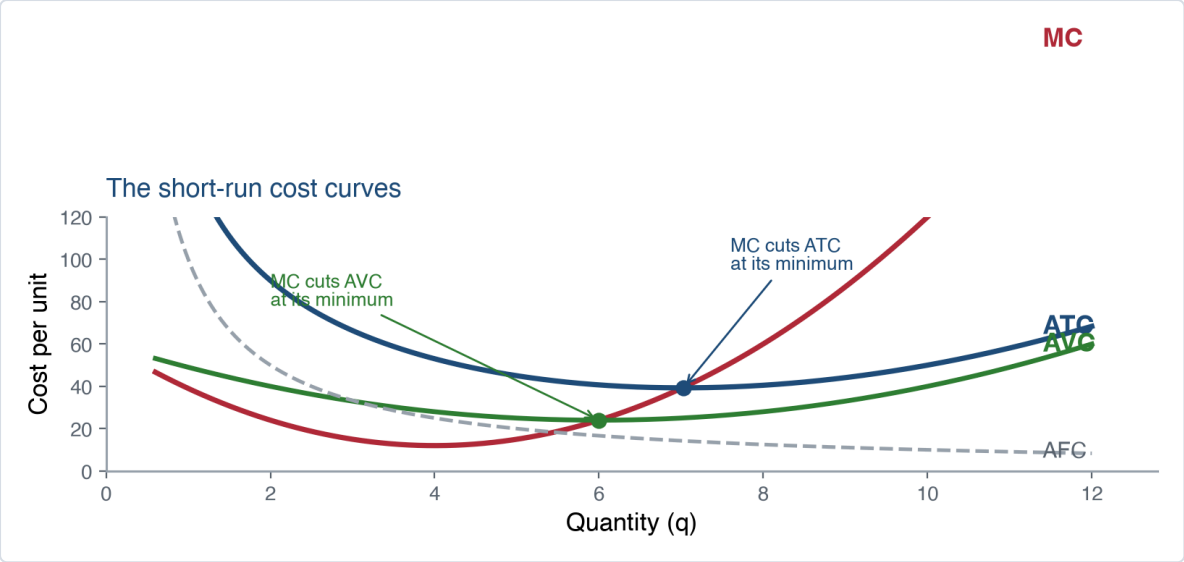
- **Worked checks: $Q = \sqrt{(KL)} \rightarrow f(\alpha K,\alpha L) = (\alpha K)\,\hat{}\,½(\alpha L)\,\hat{}\,½ = \alpha\cdot\sqrt{(KL)} \rightarrow$ CRS. $Q = K^2L^2 \rightarrow (\alpha K)^2(\alpha L)^2 = \alpha^4\cdot K^2L^2 \rightarrow$ IRS.**
- **Cobb-Douglas shortcut: sum the exponents.** Sum **= 1 → CRS; > 1 → IRS; < 1 → DRS.**
- *(Don't confuse with **economies of scale**, a **cost** idea: falling **long-run ATC** as output grows — linked to IRS but not identical.)*

★ The short-run cost curves

Core: total cost splits into a part that **varies** with output and a part that **doesn't**, and dividing by Q gives the per-unit curves.

- **Total cost TC = VC + FC** — **variable cost** (labour, materials, electricity: rises with Q) + **fixed cost** (rent, equipment: paid regardless in the SR).
- **Per-unit costs** (divide by Q): **ATC = AVC + AFC**, where **ATC = TC/Q, AVC = VC/Q, AFC = FC/Q.**
- **Marginal cost** — the cost of **one more unit:**

  MC = ΔTC/ΔQ = dTC/dq (the derivative of total cost).
- **The shapes (and the one fact that always shows up on exams):**
  - **AFC** falls continuously (a fixed cost spread over more and more units).
  - **ATC and AVC are U-shaped:** spreading fixed cost pulls ATC **down**, rising MC pulls it **up** → a U.
  - **★ MC cuts ATC and AVC at their *minimum* points.** *Intuition (the height analogy):* a new unit **cheaper** than the current average pulls the average **down**; **dearer** than the average pulls it **up** — so marginal crosses average exactly at the average's lowest point.



How to read it (illustrative diagram). **MC** (red) slopes up. **AVC** (green) and **ATC** (blue) are **U-shaped**, with ATC sitting above AVC by the falling **AFC** (grey dashed) gap. The two marked dots are the key fact: **MC passes exactly through the bottom of AVC and the bottom of ATC.** Left of each minimum, MC is **below** the average and drags it down; right of it, MC is **above** and pushes it up.

The rewards of doing business: revenue & profit

Core: the other half of the firm's problem — what it **earns.**

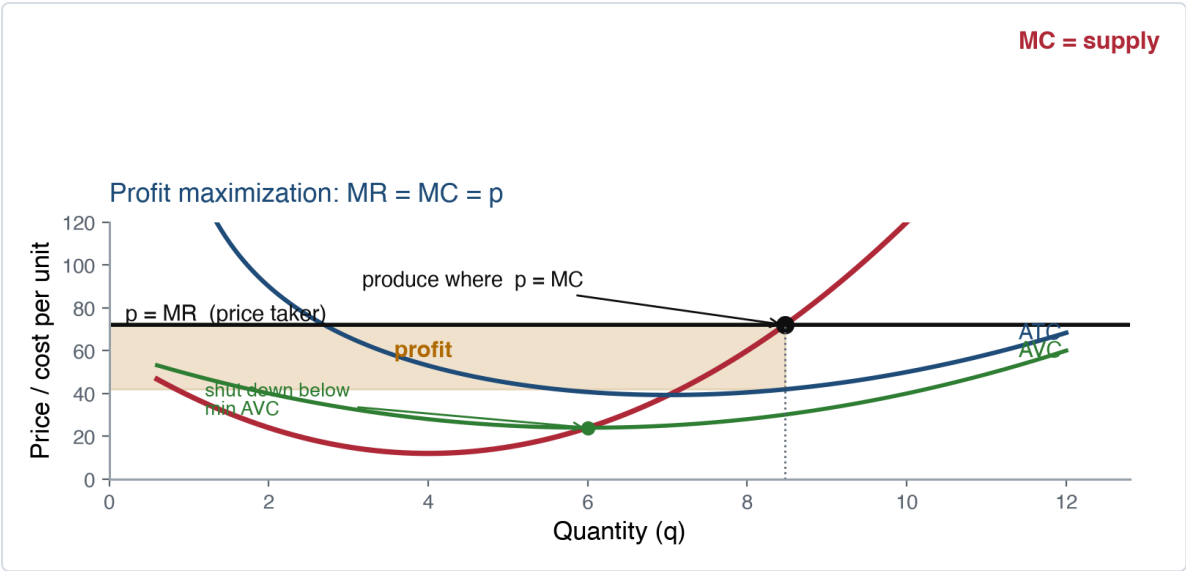
- **Total revenue TR = p × Q.**
- **Profit π = TR − TC.**
- **Marginal revenue** — the revenue from **one more unit sold: MR = ΔTR/ΔQ = dTR/dq.** For a **price taker**, every extra unit sells at the same market price, so **MR = p.**

★ Profit maximisation: MR = MC = p

Core: keep producing an extra unit as long as it **brings in more than it costs** — stop when the last unit's **revenue equals its cost.**

- **The output rule** (three equivalent statements): **(1)** profit is maximised; **(2)** marginal profit is zero; **(3) MR = MC.** Formally, maximise $\pi(q) = TR(q) - TC(q)$: set $d\pi/dq = MR - MC = 0 \rightarrow$ **MR = MC.**
- **For a price-taking firm MR = p**, so the rule collapses to

  p = MC → and therefore **the firm's supply curve *is* its marginal-cost curve** (the upward-sloping part, above min AVC).
- **Worked example (Friberg):** $C(q) = 10 + q^2$, price p = 12. Maximise $12q - (10 + q^2)$: FOC $12 - 2q = 0 \rightarrow$ **$q^* = 6$.** Profit **$\pi = (p - ATC)\cdot q = (12 - 46/6)\cdot 6 = 26 > 0$.** (At p = 10 → $q^* = 5$; at p = 14 → $q^* = 7$ — tracing out the MC/supply curve.)



How to read it (illustrative diagram). The flat black line is **p = MR** (the price taker's "demand"). The firm slides out along its **MC** curve and stops where **p = MC** (the dot). **Profit** is the shaded box: height **(p − ATC)** times width **q***. If the price line sits **below min AVC** (the green dot), the firm can't even cover its variable costs and **shuts down.**

The shutdown rule (short run)

Core: a firm losing money should keep going **only if producing loses *less* than closing.**

- In the SR, **fixed costs are unavoidable (sunk)** — you pay the rent whether you open or not. So the firm compares **revenue to *variable* cost**, not total cost.
- **Shutdown rule: shut down if price < minimum AVC** (i.e. if pq < VC — revenue can't even cover variable costs). If **p ≥ min AVC**, keep producing: the excess **pq − VC** ("variable profit") at least **reduces** the loss on fixed costs.
- So the firm's SR **supply curve = MC above min AVC**, and **zero** below it. (Shutdown is a **SR** decision; in the LR all costs are avoidable, so a loss-maker simply **exits.**)

Long-run competition: free entry drives profit to zero

Core: the same rule **MR = MC** still picks output, but now **free entry and exit** pin down the **price.**

- In the LR **all costs are variable** (no fixed cost; ATC = AVC) and **firms can enter or exit.** Positive economic profit **attracts entry**; entry **shifts market supply out** and **pushes the price down.** Entry stops only when **profit = 0.**
- **Long-run equilibrium condition:** price is driven to the **minimum of ATC**, where

  p = MR = MC = min ATC, and **economic profit = 0.**

  The **LR market supply curve** is therefore **flat at min ATC** (with identical firms, free entry and constant input prices).
- **"Zero profit" means zero *economic* profit. Economic cost = accounting cost + opportunity cost**; at the zero-profit point every input (including the owner's time and capital) earns exactly its **next-**

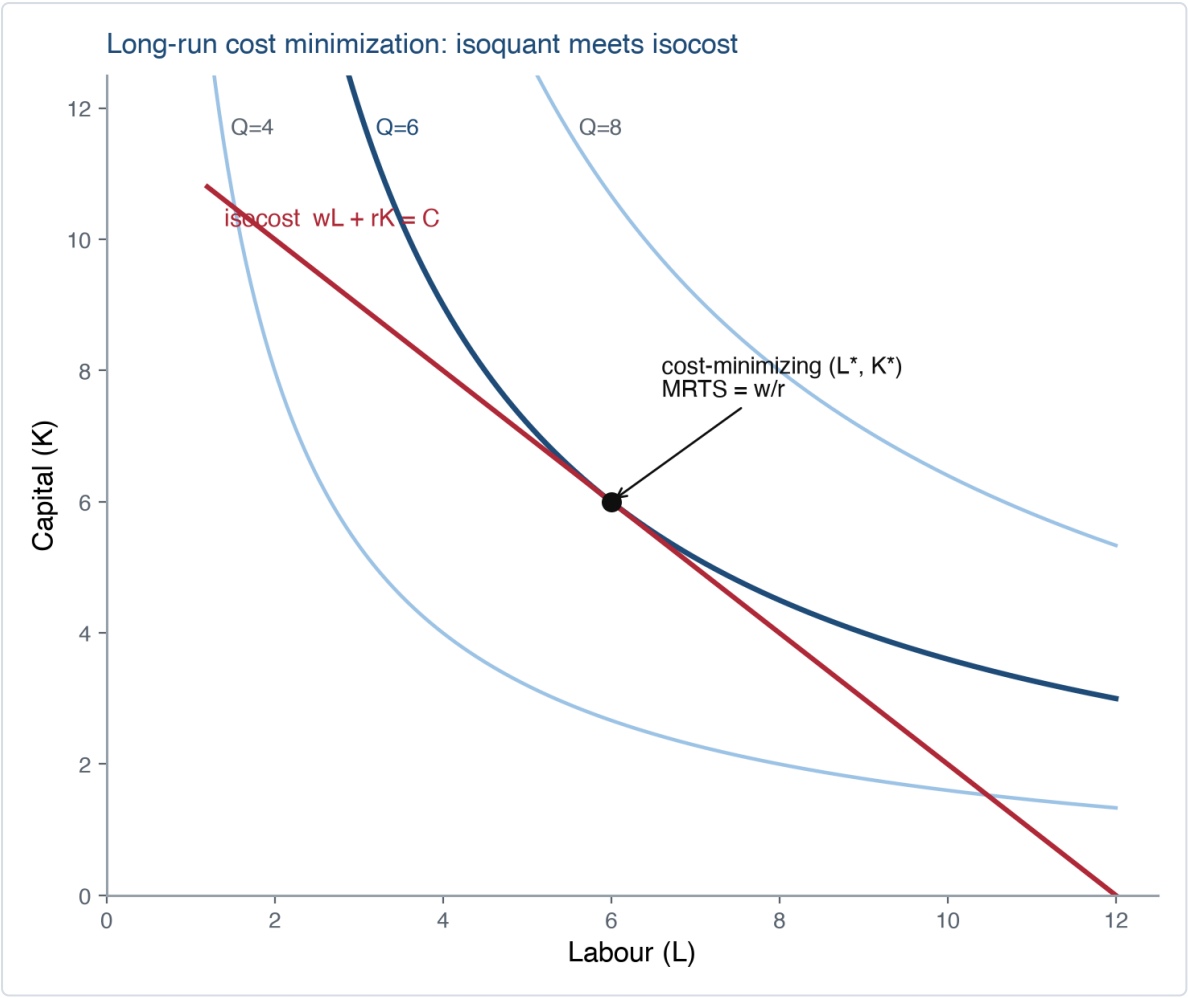
best return — the firm is doing fine, just not *better* than its alternatives. A return to owning a **scarce** input (prime Champagne land) is an **economic rent**, and can make the LR supply curve slope **up.** *(Learning-by-doing — chips, aircraft — can even make it slope **down**.)*

★ Optimal input choice: isoquants, isocosts & cost minimisation

Core: *how* to produce a given output most cheaply — the exact analogue of the consumer's optimum, with **isoquants** playing the role of indifference curves and **isocosts** the role of the budget line.

- **Isoquant** — all **(L, K) combinations that make the same output Q** ("iso" = equal). Shapes mirror the consumer's: **perfect substitutes** → straight line; **perfect complements** → L-shaped (rickshaws need exactly one driver each); the usual **imperfect substitutes** → **convex**, bowed toward the origin.
- **The slope of an isoquant = the marginal rate of technical substitution (MRTS)** — how much **K** you can drop when you add one **L**, holding output constant:

  MRTS = MP_L / MP_K (derived by totally differentiating Q = f(K,L) and setting dQ = 0).
- **Isocost line** — all input combinations costing the **same total, C = wL + rK** (w = wage, r = rental rate of capital). Rearranged: **K = C/r − (w/r)·L → intercept C/r, slope −w/r.**
- **Cost minimisation (three equivalent rules):**
  1. **Lowest-isocost rule:** pick the input bundle where the **lowest isocost** still touches the target isoquant.
  2. **Tangency rule:** where the **isoquant is tangent to the isocost** → slopes equal → **MRTS = w/r.**
  3. **Last-dollar rule: $MP_L/w = MP_K/r$** — the **last dollar** spent on each input buys the **same extra output.** *(The same algebra, via the Lagrangian: $MP_L/w = MP_K/r = \lambda$.)*
- **Mapping each target Q to its minimised cost gives the cost function C(q)** — closing the loop back to the cost curves above.



How to read it (illustrative diagram). Each blue curve is an **isoquant** — a fixed output level, more toward the top-right = more output. The red line is an **isocost** (all input mixes costing the same C). To make **Q = 6** as cheaply as possible, slide to the **lowest** isocost that still touches that isoquant — the **tangency** point **(L*, K*)**, where the isoquant's slope (**MRTS = MP_L/MP_K**) equals the isocost's slope (**w/r**).

- **Comparative statics — why factories differ by country.** Same technology, same r, but **Japan** has a **high wage** → a **steeper** isocost → the tangency uses **more capital, less labour** (automation). **Bangladesh** has a **low wage** → **flatter** isocost → **more labour, less capital** (labour-intensive garments). Relative **input prices** pick the technique.

Cases & examples

- **The CPH sandwich shop** — a new office opens next door: in the **SR** you hire hands (variable), but can't add ovens (fixed) → diminishing marginal product; in the **LR** you build a new station.
- **The Wisconsin Cheeseman** — "why not 500 boxes? why not 1500?" → the firm settles at **MR = MC.**
- **Japan vs Bangladesh** — same technology, different wages → capital-intensive vs labour-intensive input mix (comparative statics of cost minimisation).
- **The electricity supply curve** — the market supply stacks sources by MC ("merit order": wind/hydro cheapest, then nuclear, then fossil), so it is flat at low output and steep at the top.
- **Champagne land** — the forgone grape revenue is the opportunity cost of any other use; the premium for scarce land is **economic rent.**

- **The Black Death** — plague cut the workforce → each survivor had more land/capital → **MP_L and real wages rose** (marginal product drives wages).

Formula sheet

Concept	Formula
Production function	Q = f(K, L); Cobb-Douglas Q = K^a L^b
Marginal product of labour	$MP_L = \partial Q/\partial L$ (for $Q = \sqrt{(KL)}$: $\frac{1}{2}\sqrt{(K/L)}$)
Returns to scale	$f(\alpha K, \alpha L) = \alpha \cdot f \Rightarrow$ CRS; $> \alpha \cdot f \Rightarrow$ IRS; $< \alpha \cdot f \Rightarrow$ DRS (Cobb-Douglas: a + b vs 1)
Total / average costs	TC = VC + FC; ATC = TC/Q; AVC = VC/Q; AFC = FC/Q
Marginal cost	MC = dTC/dq
MC–AC relation	MC cuts ATC and AVC at their minima
Revenue, profit	TR = pQ; π = TR − TC; MR = dTR/dq
Output rule	MR = MC; price taker ⇒ **p = MC** (supply = MC above min AVC)
Shutdown (SR)	produce iff **p ≥ min AVC**
Long-run equilibrium	**p = MC = min ATC**, economic profit = 0
Isocost	C = wL + rK ⇒ K = C/r − (w/r)L
MRTS (isoquant slope)	$MRTS = MP_L/MP_K$
Cost-min (tangency / last dollar)	MRTS = w/r ⇔ **$MP_L/w = MP_K/r$**

★ Practice problems (with solutions)

Problem 1 — short-run firm (the lecture's example). A firm has **$TC = 2q^2 + 50q + 50$** and faces a market price **$p = 100$.** *(a)* Find FC, VC, AFC, AVC, ATC and MC. *(b)* How much does the firm produce, and what profit does it make?

> **Solution. (a) FC = 50; $VC = 2q^2 + 50q$; AFC = $50/q$; $AVC = 2q + 50$; $ATC = 2q + 50 + 50/q$; $MC = dTC/dq = 4q + 50$. (b)** Price taker ⇒ **p = MC**: $100 = 4q + 50 \rightarrow$ **$q^* = 12.5$.** Then TR = $100(12.5) = 1250$; TC = $2(12.5^2) + 50(12.5) + 50 = 312.5 + 625 + 50 = 987.5$ → **$\pi = 1250 - 987.5 = 262.5 > 0$.** (Since p = 100 > min AVC, the firm produces.)

Problem 2 — long run with free entry. Same cost function $TC = 2q^2 + 50q + 50$; market demand **$p = 210 - Q$.** Find the **long-run** price, each firm's output, the number of firms, and each firm's profit.

> **Solution.** Free entry drives price to **min ATC.** Minimise ATC = 2q + 50 + 50/q: dATC/dq = 2 − 50/q^2 = 0 → q^2 = 25 → **$q^* = 5$**, giving **min ATC = 2(5) + 50 + 50/5 = 70.** So the **LR price = 70.** Market quantity from demand: Q = 210 − 70 = **140.** Number of firms **$n = Q / q^* = 140 / 5 = 28$.** Each firm's profit **$\pi = (p - ATC) \cdot q = (70 - 70) \cdot 5 = 0$** — zero economic profit, as required. *(Entry has pushed the price down from 100 to 70 and wiped out the short-run profit.)*

Problem 3 — returns to scale & cost minimisation. *(a)* Does **Q = K^(1/3) L^(1/2)** show increasing, constant or decreasing returns to scale? *(b)* A firm produces with **Q = √(KL)**, wage **w = 4**, rental rate **r = 1.** In what proportion should it use L and K?

> **Solution. (a)** Sum of exponents = 1/3 + 1/2 = **5/6 < 1 → decreasing returns to scale. (b)** MP_L = ½√(K/L), MP_K = ½√(L/K) → MRTS = MP_L/MP_K = **K/L.** Tangency: MRTS = w/r → **K/L = 4/1 → K = 4L.** The firm uses **four units of capital per unit of labour** (capital-intensive, because labour is relatively dear).

★ Fun facts & memorable details (Lecture 6: Firms & Production)

> Sticky bits from the production lecture.

- **The firm is a consumer in a mirror.** Indifference curve ↔ **isoquant**; budget line ↔ **isocost**; MRS = price ratio ↔ **MRTS = w/r.** Learn one optimum, you've learned both.
- **Marginal cuts average at the bottom — always.** Whether it's cost curves or your grade-point average, the **marginal** value crosses the **average** exactly at the average's minimum. Add something below your average and it drops; above, it rises.
- **You can rationally run at a loss.** In the short run, as long as price covers **average variable cost**, staying open loses **less** than closing — because the rent is sunk either way. The shutdown line is **min AVC**, not min ATC.
- **"Zero profit" is a success, not a failure.** Long-run zero *economic* profit means every input — including the owner — is earning exactly its best alternative return. It only looks like zero because the **opportunity cost** is already counted.
- **The Black Death raised wages.** Kill a third of the workforce and each survivor works more land → higher **marginal product** → higher real wages. A grim demonstration that **wages track marginal product.**
- **Why Japanese factories are robotic and Bangladeshi ones aren't.** Same technology — different **wages.** High wages tilt the isocost toward **capital**; low wages toward **labour.** The technique follows the price of the input.
- **One big plant can beat two small ones.** That's **increasing returns to scale** (specialisation) — and, on the cost side, the **economies of scale** that let a few firms dominate aircraft, pharma and chips.

Lecture 7: Competition & Long-Run Production

(Chapter 4)

Required reading: Friberg, **Ch. 4 — Supply by price-taking firms** (the **long-run** half: output rules, long-run shutdown & entry, the long-run market supply curve). *Lecturer: L. Butera — deck "**Competition and long-run production.**" (Lecture 6 built the firm's costs and the short-run supply rule; this lecture pushes the same firm into the **long run**, where firms can **enter and exit** — and derives the **market** supply curve.)*

Résumé. Lecture 6 left the firm making a **short-run** profit at price p by setting **MR = MC**. This lecture asks the natural follow-up: *are those profits sustainable?* In **perfect competition** the answer is **no** — because **entry is free**, any economic profit attracts new firms, output rises, and the **price is bid down** until profit hits **zero**. That pins the **long-run price to the bottom of the average-cost curve** (min AC), tells us **how much each firm makes** and **how many firms** the market supports, and makes the **long-run market supply curve flat** at min AC. We also check *why* a competitive firm's demand is nearly horizontal, and the three cases where long-run supply **slopes** instead.

> ★ **The heart of the lecture — free entry drives price to the bottom of the AC curve.**
> • **Same output rule as the short run:** every firm still produces where **MR = MC**; for a price taker that is **p = MC**.
> • **What's new is entry/exit.** In the long run *all* costs are variable and firms can come and go. **Economic profit ⇒ entry** (more firms, price falls); **economic loss ⇒ exit** (fewer firms, price rises). The only resting point is **zero economic profit.**
> • **The long-run equilibrium condition** stacks four things on top of each other: **p = MR = MC = min AC**. Price sits at the **minimum of average cost**, so the typical firm makes **zero economic profit** and produces at **efficient scale**.
> • **Market supply** is therefore **flat at min AC** (with identical firms, free entry, constant input prices): the market delivers whatever quantity consumers want, all at the same minimum-cost price.

Red thread (two questions the whole lecture answers): *(1) Once we allow firms to enter and exit, what price and quantity does a competitive market settle at? (2) How many firms does it support — and when is the long-run supply curve flat vs sloped?*

Key concepts / "modes" to use: *market structure (the 3 determinants) · perfect competition & the price taker (→ Lecture 6) · the three equivalent **output rules** · marginal profit $\pi'(q) = 0$ · MR = MC, and p = MC for a price taker · the **long-run shutdown** rule (exit if economic loss) · free entry ⇒ **zero economic profit** · the **long-run price = min AC** (found by dAC/dq = 0) · efficient scale · the **long-run market supply curve** (horizontal sum; flat at min AC) · residual (firm-level) demand elasticity $\varepsilon_A = n\cdot\varepsilon - (n-1)\cdot\eta$ · increasing- vs decreasing-cost industries.*

Market structure & the perfectly-competitive benchmark

***Core:** market structure* is just **how a market is shaped**, and it predicts **how firms will behave.** Perfect competition is the extreme where every firm is a **price taker**.

What determines market structure?

Three features (from the deck): | Determinant | Question it asks | |---|---| | **Number of firms** | many tiny firms, or a few big ones? | | **Ease of entry/exit** | can new firms join (and leave) freely? | | **Product differentiation** | identical products, or can firms make theirs distinct? |

In plain terms: the more firms, the freer the entry, and the more identical the product, the closer you are to **perfect competition** (and the less power any one firm has over price).

Perfect competition — the 5 assumptions (recap from Lecture 6)

(1) a **large number** of firms · **(2) identical (homogeneous)** products · **(3) full information** (buyers know every price) · **(4) negligible transaction costs** · **(5) free entry and exit.** → The payoff of these assumptions: each firm is a **price taker** — its own demand curve is a **horizontal line at the market price**. Charge one cent more and you sell **zero** (buyers switch to an identical rival); you can already sell all you want at p, so you never charge less. *(Assumption (5), free entry/exit, is the one doing the real work **this** lecture — it's what makes the long run different from the short run.)*

★ The three output rules (how much to produce)

Core: "produce the profit-maximising quantity" can be stated **three equivalent ways** — they always give the **same** q*.

Profit is revenue minus cost:

PROFIT

$$\pi(q) = R(q) - C(q) = p{\cdot}q - C(q)$$

Reads: profit **π** = total revenue **R(q)** (for a price taker, price **p** × quantity **q**) minus total cost **C(q)**. · **p** = the market price the firm takes as given.

#	Output rule	What it says
1	**Profit is maximised**	pick the q at the **top** of the profit "hill" π(q).
2	**Marginal profit = 0**	at the top of the hill the **slope** is zero: π′(q) = 0.
3	**MR = MC**	the slope of profit is MR − MC, so π′(q) = 0 means **marginal revenue = marginal cost**.

Why #2 and #3 are the same thing

Differentiate profit and set it to zero (that's rule #2):

RULE #2 ⇒ RULE #3

$$\pi'(q) = R'(q) - C'(q) = MR - MC = 0 \Rightarrow \mathbf{MR = MC}$$

Reads: the slope of the profit curve is **marginal revenue minus marginal cost**; setting it to zero gives the familiar **MR = MC.** · **R'(q) = MR** = extra revenue from one more unit · **C'(q) = MC** = extra cost of one more unit.

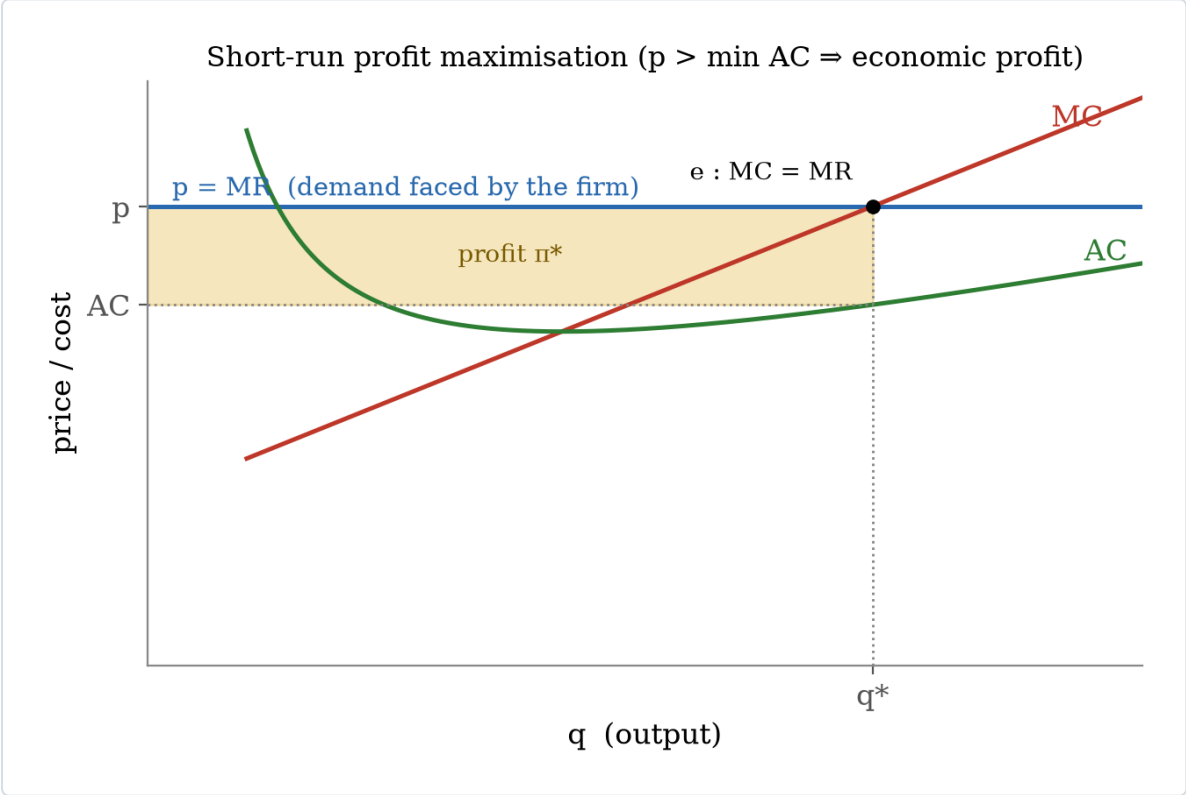
For a price taker, MR = p

Because R = p·q with *p* fixed, **MR = dR/dq = p**. So the three rules collapse to the one you already know from Lecture 6:

PRICE-TAKER'S OUTPUT RULE

$$p = MR = MC$$

Reads: a competitive firm produces the quantity where **price equals marginal cost** — that point *is* its profit-maximising output.



How to read it *(illustrative diagram)*.

- **Axes:** price/cost (y) vs output q (x).
- **p = MR** (blue) is **horizontal** — the price-taking firm's demand. **MC** (red) rises; **AC** (green) is U-shaped.
- **q*** is read off where **MC crosses p = MR** (rule #3). Rules #1 and #2 give the *same* q* at the top of the profit curve.

- The **shaded rectangle** = profit = **(p − AC)** × **q***. Here **p > AC**, so the firm makes an **economic profit** — which is exactly what the rest of the lecture says **can't last**.

Are short-run profits sustainable? (the hinge of the lecture)

Core: **no** — in the short run firms **can't** enter, so profits can persist; but the moment we allow the **long run**, free entry competes those profits away.

- In the **short run** the number of firms is **fixed** (no entry/exit), so a firm earning the shaded profit above just... keeps earning it.
- In the **long run**, that profit is a **signal**: outsiders see easy money and **enter**. More firms → more market output → the **market price falls**. Entry continues **as long as there are economic profits to be made** — i.e. until profit is **zero**.
- The symmetric case: if firms are making an **economic loss**, some **exit**, market output falls, and the price **rises** back up — again until profit is **zero**.

→ So the long-run resting point is **zero economic profit**, reached by **entry/exit moving the price**, not by any single firm changing its behaviour. *(Recall from Lecture 6: "zero economic profit" is a success — every input, including the owner's time, is earning its **opportunity cost**.)*

Long-run output & shutdown decisions

Core: the output rule is **unchanged** (MR = MC); what changes in the long run is that **all costs are variable**, which simplifies both the cost curves and the shutdown rule.

Output: same rule as the short run

The firm still picks q where **MC = MR**. Nothing new here — the rule doesn't care about the time horizon.

All costs are variable ⇒ AC = AVC

In the long run there are **no fixed costs** (you can sell the factory, not renew the lease). So **average total cost and average variable cost are the same curve**:

> LONG RUN: NO FIXED COSTS
>
> $$\text{AC} = \text{AVC} \text{ (because FC} = 0\text{, so AFC} = 0)$$

Reads: with every cost variable, total = variable cost, so **average cost (AC) equals average variable cost (AVC)** — the two SR curves merge. · *AFC = average fixed cost, which vanishes in the LR.*

Long-run shutdown: exit if you'd make a loss

Because nothing is sunk, the rule is simpler than the short run's "min AVC":

LONG-RUN SHUTDOWN RULE

$$\text{exit the market if } p < \min AC \text{ (an economic loss)}$$

Reads: in the long run a firm **leaves** the market entirely if the price is below its **minimum average cost** — because, with no sunk fixed costs, continuing would just pile up losses. · *(Contrast the **short-run** rule, min AVC: there, fixed costs are sunk, so you stay open as long as price beats **variable** cost.)*

★ The long-run price = the minimum of average cost

Core: competition drives the price **down to the lowest point of the AC curve** — so to find the long-run price, **minimise AC** (take its derivative and set it to zero).

The chain: free entry pushes the price down; it **stops** only when there's no profit left, i.e. when the price just equals the lowest possible average cost (any lower and everyone makes a loss and exits). So:

FIND THE LONG-RUN PRICE

$$\text{minimise } AC(q)\text{: } dAC/dq = 0 \Rightarrow p_{LR} = \min AC$$

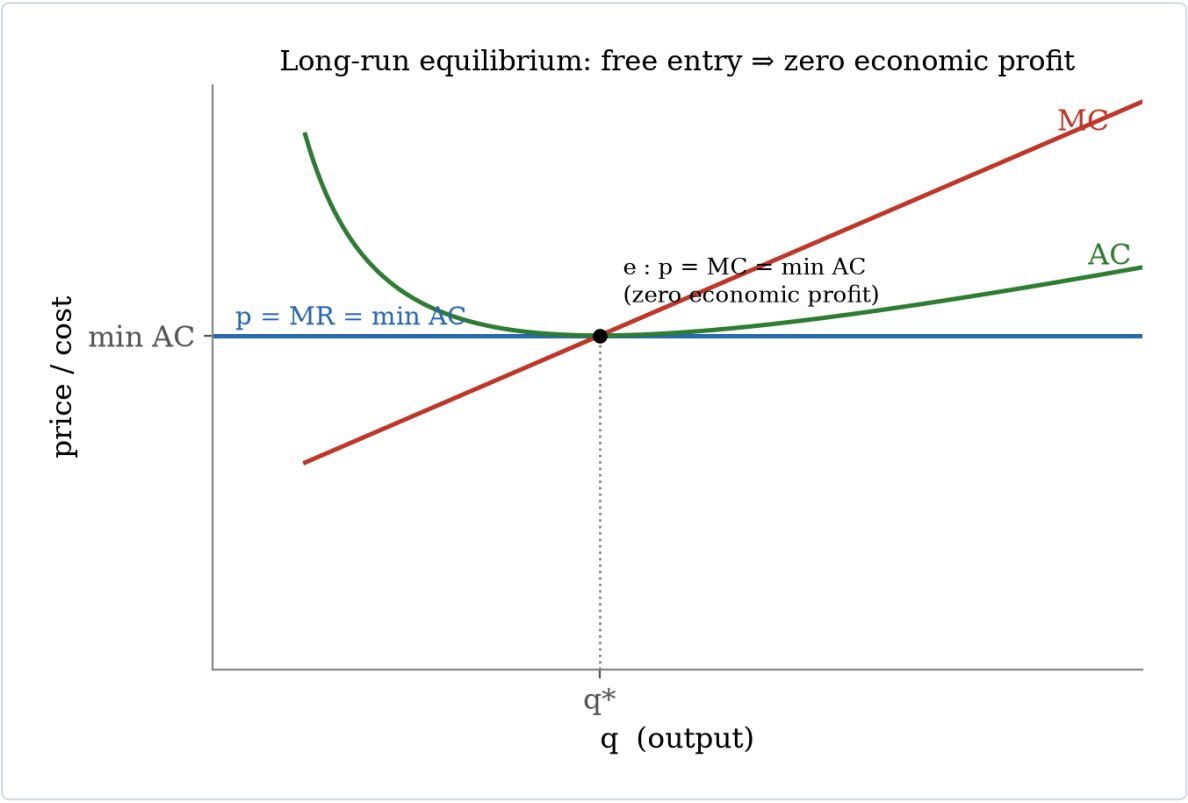
Reads: the long-run price equals the **minimum of the average-cost curve**; you find the minimising quantity by setting the **derivative of AC to zero**. · *At that minimum, **MC = AC** (marginal cuts average at its lowest point — the fact from Lecture 6).*

So the full **long-run equilibrium** stacks everything together:

LONG-RUN COMPETITIVE EQUILIBRIUM

$$p = MR = MC = \min AC \Rightarrow \pi = 0$$

Reads: in long-run equilibrium the price equals marginal revenue, marginal cost **and** the minimum of average cost all at once — so economic **profit is zero** and each firm produces at its **efficient scale** (the output that minimises average cost).



How to read it *(illustrative diagram)*.

- Compare with the previous figure: entry has pushed **p = MR** (blue) **down** until it just **touches the bottom of AC** (green).
- At that point **MC** (red) also passes through — so **p = MC = min AC**.
- The profit rectangle has **shrunk to nothing**: **zero economic profit.** The firm produces at **efficient scale** (the q that minimises AC).

★ Worked example — entry, the long-run price, and how many firms

Core: the exam-style question: given one firm's costs and the market demand, find **each firm's output**, the **long-run price**, and the **number of firms**. *(This is the deck's own example, extending Lecture 6's short-run version.)*

Given: each firm has **TC = 2q² + 50q + 50**; the short-run price was **p = 100**; **market demand is p = 210 − Q** (Q = total market quantity).

Step 1 — build MC and AC.

MARGINAL & AVERAGE COST

$$\mathrm{MC} = \mathrm{dTC/dq} = 4\mathrm{q} + 50 \quad \mathrm{AC} = \mathrm{TC/q} = 2\mathrm{q} + 50 + 50/\mathrm{q}$$

Reads: **MC** is the derivative of total cost; **AC** is total cost per unit. · the **50/q** term is the fixed cost (50) spread over output — it's what makes AC U-shaped.

Step 2 — find the long-run price = min AC (set dAC/dq = 0):

MINIMISE AC

$$dAC/dq = 2 - 50/q^2 = 0 \Rightarrow q^2 = 25 \Rightarrow \mathbf{q^* = 5}$$

Reads: differentiate AC, set to zero, solve for q. Each firm's **efficient-scale output is q* = 5.** · (Second check: AC'' = $100/q^3 > 0$, so it's a minimum.)

Step 3 — the long-run price (plug $q^* = 5$ into AC):

LONG-RUN PRICE

$$p_{LR} = \min AC = 2(5) + 50 + 50/5 = \mathbf{70} \text{ (check: } MC = 4(5)+50 = 70 = AC \checkmark)$$

Reads: the long-run price is **70** — and MC equals AC there, confirming it's the minimum. · *(The SR price of 100 > 70 meant SR economic profit → that's exactly what drew firms in and dragged the price down to 70.)*

Step 4 — market quantity (from demand at p = 70):

TOTAL MARKET OUTPUT

$$p = 210 - Q \Rightarrow Q = 210 - 70 = \mathbf{140}$$

Reads: at the long-run price of 70, consumers buy **Q = 140** in total across the whole market.

Step 5 — how many firms? (market quantity ÷ each firm's output):

NUMBER OF FIRMS

$$n = Q \,/\, q^* = 140 \,/\, 5 = \mathbf{28 \text{ firms}}$$

Reads: with each firm producing 5 and the market absorbing 140, the market supports **n = 28** identical firms in the long run. · **This is the number that free entry delivers** — exactly enough firms to meet demand at zero profit.

★ The long-run market supply curve

Core: the **market** supply curve is the **horizontal sum** of all the firms' supply curves — but in the long run the **number of firms isn't fixed**, so with identical firms it comes out **flat at min AC**.

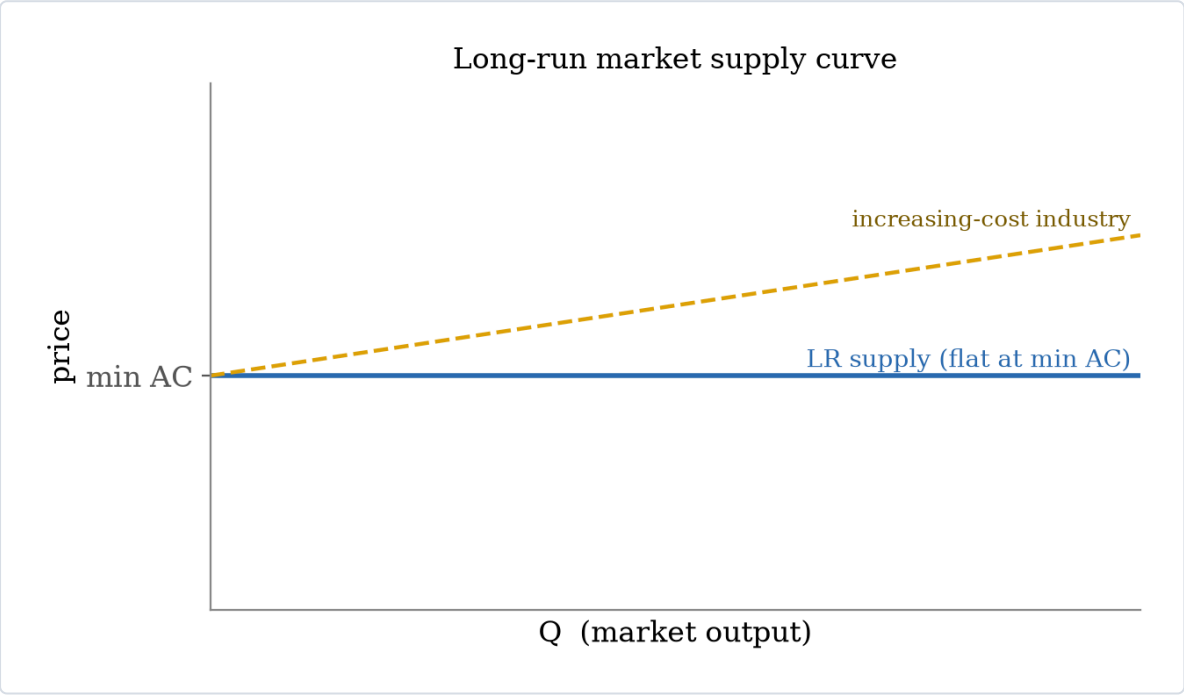
- As in the short run, market supply = **add up quantities across firms at each price** (horizontal sum).
- The twist: firms **enter** whenever a long-run profit is available and **exit** to avoid a long-run loss. So the market can supply **any** quantity at the price **min AC** — just add or subtract firms.

- With **identical firms + free entry + constant input prices**, the result is a **horizontal (flat) long-run supply curve at min LR average cost**: as new firms enter, profits fall to zero, and the price can't stay above (entry) or below (exit) **min AC**.

LONG-RUN MARKET EQUILIBRIUM (IDENTICAL FIRMS)

$$P = MR = MC = AC \text{ (at min AC)} \Rightarrow \text{long-run profit} = 0$$

Reads: the whole market settles where price equals every firm's marginal **and** minimum average cost, so **aggregate long-run profit is zero** — the market delivers consumers' desired quantity all at the one minimum-cost price.



How to read it *(illustrative diagram)*.

- **Axes:** price (y) vs **market** quantity Q (x).
- The solid **blue line is flat at min AC**: whatever quantity demand calls for, the market supplies it at that one price — because **entry/exit** adjusts the number of firms rather than the price.
- The dashed amber line previews the next section: if input prices rise with industry output (an **increasing-cost industry**), long-run supply **slopes up** instead.

Is the firm's demand *really* flat?

Core: an individual competitive firm faces a demand curve **far more elastic** (flatter) than the **market's** — flat enough to treat as horizontal — and we can show *how much* flatter.

The firm's own (**residual**) demand elasticity depends on the market elasticity, the number of firms, and how elastically the **other** firms supply:

FIRM-LEVEL (RESIDUAL) DEMAND ELASTICITY

$$\varepsilon_A = n \cdot \varepsilon - (n-1) \cdot \eta_{\text{other firms}}$$

Reads: the elasticity of demand faced by **firm A** = the **number of firms** n times the **market demand elasticity** ε, minus $(n-1)$ times the **supply elasticity of the other firms** η. · **The more firms there are, and the more elastically they supply, the more elastic (flatter) firm A's demand.**

Worked numbers (soybean, from the deck): market demand elasticity **$\varepsilon = -1.1$**, other-firm supply elasticity **$\eta = 3.1$**, **$n = 78$** firms:

SOYBEAN EXAMPLE

$$\varepsilon_A = 78 \cdot (-1.1) - 77 \cdot (3.1) = -85.8 - 238.7 \approx \mathbf{-324.5}$$

Reads: although the **whole soybean market** is only **moderately** elastic (-1.1), a **single farm** faces a **super-elastic** demand (≈ -324.5) — in effect a **flat** line. · *This is what justifies the "price taker / horizontal demand" assumption: it's an approximation, but a very good one when firms are many.*

When the long-run supply curve is *not* flat

Core: the flat LR supply curve relies on identical firms + free entry + constant input prices. **Break any of those** and the curve **slopes**.

Scenario	What breaks	LR supply shape
Entry is limited	government caps the number of firms, a **scarce resource** is needed, or **entry is costly**	**upward-sloping**
Firms differ	low-min-AC firms enter at lower prices than high-cost ones	**upward-sloping** (cheap firms first, then dearer)
Input prices vary with output — increasing-cost industry	expanding the industry **bids up** input prices	**upward-sloping**
Input prices vary with output — decreasing-cost industry	expanding the industry **lowers** input prices (e.g. supplier economies of scale)	**downward-sloping**

In plain terms: the flat curve is the special "textbook" case; the moment some firms are cheaper than others, or growing the industry changes what inputs cost, the long-run supply curve tilts.

Formula sheet (Lecture 7)

Concept	Formula	In words
Profit	$\pi(q) = p \cdot q - C(q)$	revenue minus cost
Output rule (general)	MR = MC	produce where marginal revenue = marginal cost
Output rule (price taker)	**p = MC**	…and MR = p, so price = marginal cost
Long run: no fixed costs	AC = AVC	fixed costs vanish
Long-run shutdown	exit if **p < min AC**	leave if you'd make an economic loss
Long-run price	**p = min AC** (dAC/dq = 0)	entry drives price to the bottom of AC
LR equilibrium	**p = MR = MC = min AC**, $\pi = 0$	zero profit, efficient scale
Number of firms	$n = Q / q^*$	market quantity ÷ each firm's output
Firm demand elasticity	$\varepsilon_A = n \cdot \varepsilon - (n-1) \cdot \eta$	why one firm's demand is near-flat

★ Fun facts & memorable details (Lecture 7: Competition & Long-Run Production)

> The sticky bits from the long-run lecture.

- **Profit is a self-destruct button (in competition).** The very thing every firm wants — economic profit — is the signal that **summons the competitors who compete it away.** In perfect competition, visible profit can't survive the long run.
- **"How many firms?" is just division.** Once you've found each firm's efficient output (min AC) and the market quantity at that price, the number of firms is literally **Q ÷ q***. In the worked example: 140 ÷ 5 = **28**.
- **A single soybean farm faces demand ~300× more elastic than the market.** Market elasticity −1.1, but farm A's ≈ **−324.5**. That gap is *why* "price taker" is a fair approximation even though no demand curve is *perfectly* flat.
- **The long run erases fixed costs — and simplifies the shutdown rule.** Short run: stay open above **min AVC** (rent is sunk). Long run: nothing is sunk, so you leave below **min AC**. Same firm, different rule, because of what you can still change.
- **Competition is secretly an efficiency machine.** Driving price to **min AC** means the good is produced at the **lowest possible average cost** and sold at that cost — consumers capture all the gains, firms make zero economic profit. That's the welfare punchline behind "perfect competition."